

2. Optical Properties

2.1 Refractive Index and Dispersion

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2.1.1 Introduction

Optical glass is traditionally used to transmit visible light, including its neighbouring spectral regions, for a variety of purposes (e.g., optical windows, lenses). Optical imaging systems, whether individual lenses, photographic lenses, microscopes, telescopes, or other instruments, utilize glass in order to deflect rays of light emitted from a single point in the object plane in such a way that most of them are focused to a single point in the image plane. This deflection in transmission optics is caused by *refraction*. The refraction is different for different glasses and the refractive index depends on the wavelength of the light. This phenomenon is called *dispersion*. Both the refractive index and dispersion are considered in the following.

2.1.2 Law of Refraction

If light enters a non-absorbing homogeneous material, *reflection* (discussed in Sect. 2.3) and *refraction* occur at the boundary surface. The *refractive index* n is given by the ratio of the velocity of light in the vacuum c to that in the medium v :

$$n = \frac{c}{v} . \quad (2.1)$$

The ratio of the index in medium 1 to that in medium 2 is called the *relative index* of refraction. An important special case is the refractive index n_{rel} defined by the equation

$$n_{\text{rel}} = \frac{n_{\text{glass}}}{n_{\text{air}}} . \quad (2.2)$$

The refractive index data quoted in this book and in the catalogues of glass manufacturers are relative to air rather than to vacuum.

At a surface an incident ray is refracted; this phenomenon is described by *Snell's law* [2.1] (sometimes called the law of Descartes [2.2, 3]), which is given here in its vectorial form [2.4]:

$$\mathbf{s} \times \mathbf{o} = \mathbf{s}' \times \mathbf{o} . \quad (2.3)$$

$\mathbf{s}$ and $\mathbf{s}'$ are vectors with the direction cosines of the incident and of the refracted rays and with lengths corresponding to the refractive index in the media 1 and 2, respectively (cf. Fig. 2.1). The vector $\mathbf{o}$ designates a unit vector in the direction normal to the surface. The formulation using the vector cross product automatically takes into account that the incident ray, the refracted ray, and the normal vector of the surface are coplanar.

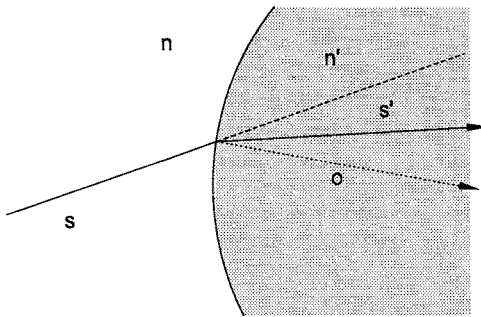


Fig. 2.1. Refraction at a surface between two optical media

In the following, we will consider optical glasses as *homogeneous* and *isotropic* optical materials (i.e., the refractive index varies neither spatially nor directionally) with a high light transmission in the wavelength range used. Deviations from this assumption will be discussed in Sect. 2.4. and in Chap. 3.

2.1.3 Dispersion and the Various Dispersion Formulae

Basic Dispersion Theory

In general the refractive index and therefore the phase velocity in the medium depend on the wavelength used. This phenomenon is called *dispersion*.

The dispersion is called *normal* if the refractive index *increases* with shorter wavelengths. Otherwise *anomalous* dispersion occurs. Normal dispersion is observed far away from the absorption bands of the glass, whereas anomalous dispersion occurs at or near the absorption bands (see Fig. 2.2).

In principle the quantum-mechanical dispersion theory should be considered, but the classical electromagnetic theory is sufficient for our purpose. The variation of the refractive index n with the wavelength λ or the corresponding frequency ν can be explained by the electromagnetic theory applied to the molecular structure of matter. If an electromagnetic wave impinges on an atom or a molecule, the bound charges vibrate at the frequency of the incident wave.

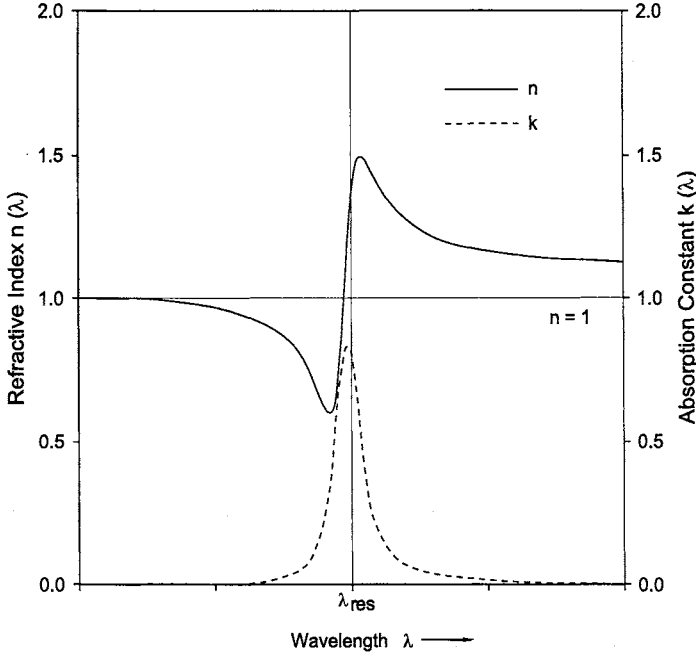


Fig. 2.2. Absorption and dispersion of light in a dielectric medium at the resonance wavelength λ_{res}

The bound charges have eigenfrequencies or *resonances* at the wavelengths λ_j (cf. *Born and Wolf* [2.5]). Then one can obtain the following dispersion relation, which applies to media of *low density*:

$$\hat{u}^2 - 1 = (n - ik)^2 = \frac{Ne^2}{2\pi c\epsilon_0 m} \sum_j \frac{f_j \lambda_j^2 \lambda^2}{2\pi c(\lambda^2 - \lambda_j^2) + i\gamma_j \lambda_j^2 \lambda}. \quad (2.4)$$

In (2.4) $\hat{u}$ represents the complex refractive index in vacuum, where n is the real and k the imaginary part, N is the number of molecules per unit volume, f_j the oscillator strength of the resonance with wavelength λ_j and damping constant γ_j , and m and e are the mass and the charge of the electron. The constants f_j determine the strength of the absorption, and the constants γ_j may be considered as a measure of the frictional force at the resonance wavelength. The pair n and k are called the optical constants.

In the regions of negligible absorption, far from resonances where $k = 0$, (2.4) reduces to

$$n^2 - 1 = \frac{Ne^2}{4\pi^2 c^2 \epsilon_0 m} \sum_j \frac{f_j \lambda_j^2 \lambda^2}{\lambda^2 - \lambda_j^2}. \quad (2.5)$$

With the abbreviation $A_j = Ne^2 f_j \lambda_j^2 / 4\pi^2 c^2 \epsilon_0 m$ we can write (2.5) in a simplified form:

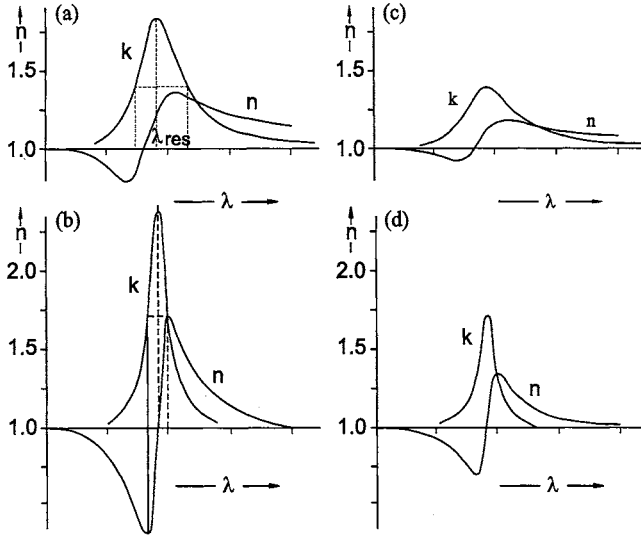


Fig. 2.3a–d. Dispersion curves for a single resonance with different amounts of absorption and friction ($k = n\kappa$): (a) strong absorption, strong friction; (b) strong absorption, weak friction; (c) weak absorption, strong friction; (d) weak absorption, weak friction [2.6]

$$n^2 - 1 = \sum_j \frac{A_j \lambda^2}{\lambda^2 - \lambda_j^2} . \quad (2.6)$$

In *dense media* where an interaction of atoms with the effective electric field cannot be neglected, the term $n^2 - 1$ in (2.5) and (2.6) should, according to *Lorentz* [2.7] and *Lorenz* [2.8], be replaced by $(n^2 - 1)/(n^2 + 2)$:

$$\frac{n^2 - 1}{n^2 + 2} = \sum_j \frac{A_j^2 \lambda^2}{\lambda^2 - \lambda_j^2} . \quad (2.7)$$

Kramers and *Kronig* [2.9] showed that under very general conditions a relationship between the real part, Re , and the imaginary part, Im , of a complex function,

$$F(\tilde{\nu}) = \text{Re}(\tilde{\nu}) - i \text{Im}(\tilde{\nu}) ,$$

can be derived, where $\tilde{\nu} = \omega/c = 2\pi/\lambda$ designates the free-space wave number.

$$\text{Re}(\tilde{\nu}) = \frac{2}{\pi} \int_0^\infty \frac{\tilde{\nu}' \text{Im}(\tilde{\nu}')}{\tilde{\nu}'^2 - \tilde{\nu}^2} d\nu' . \quad (2.8)$$

Functions that fulfil this relation are

$$F(\tilde{\nu}) = N(\tilde{\nu}) - 1 = \underbrace{n(\tilde{\nu}) - 1}_{\text{Re}(\tilde{\nu})} - i \underbrace{k(\tilde{\nu})}_{\text{Im}(\tilde{\nu})} \quad (2.9)$$

and

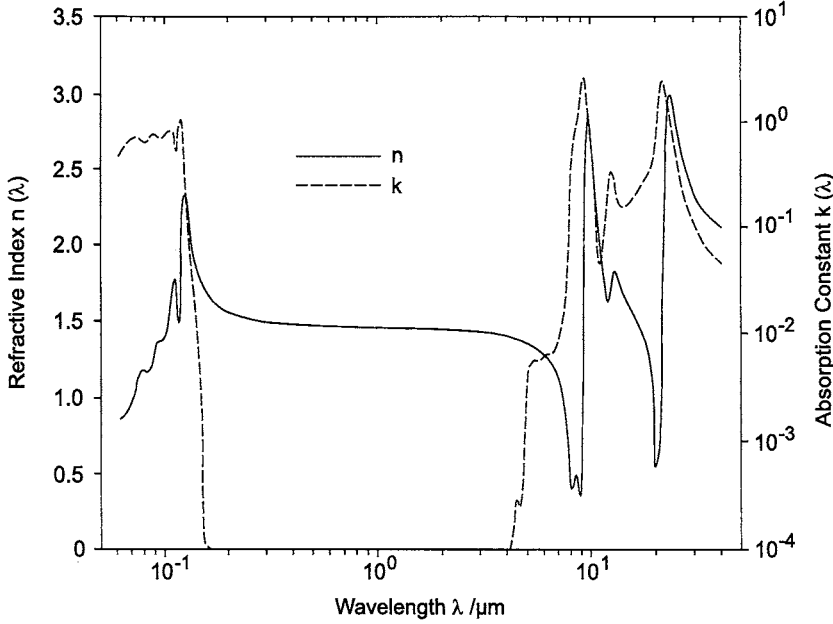


Fig. 2.4. Measured optical constants of fused silica (SiO_2 glass) [2.10]

$$F(\tilde{\nu}) = \varepsilon - 1 = N^2(\tilde{\nu}) - 1 = \underbrace{n^2(\tilde{\nu}) - k^2(\tilde{\nu}) - 1}_{\text{Re}(\tilde{\nu})} - i \underbrace{2n(\tilde{\nu})k(\tilde{\nu})}_{\text{Im}(\tilde{\nu})}, \quad (2.10)$$

where ε denotes the complex dielectric function and N the complex refractive index. Further possible expressions for $F(\tilde{\nu})$ exist but they will not be considered here.

In the literature the form (2.10) is preferred. However, the form (2.9) is just as valid as (2.10) and is more convenient for various analytic considerations. Therefore we can write (2.6) in a simplified form:

$$n - 1 = \sum_j \frac{\tilde{A}_j \lambda^2}{\lambda^2 - \tilde{\lambda}_j^2}. \quad (2.11)$$

Real Dispersion Curves of Glasses

The dispersion curves of Figs. 2.2 and 2.3 indicate the variation of n and k for the case of a single-resonator model. In reality the resonance phenomena are much more complex. This is demonstrated in Fig. 2.4, where the measured values of the optical constants for fused silica are shown (data from *Philipp* [2.10]). Even this simple compound has three main absorption features (one in the UV at $0.1 \mu\text{m}$ and two in the IR at $9 \mu\text{m}$ and $21 \mu\text{m}$) with a complicated fine structure. If one wants to express the variation of n with the wavelength in terms of a dispersion formula, it is useful to define an *effective* resonance wavelength corresponding to the centre of gravity of the corresponding absorption band.

The dispersion curves of optical glasses are fixed by the number, position, and strengths of the absorption bands of the glass components and their structure. Within certain limits the designer of an optical glass can vary these properties in order to obtain a desired dispersion curve (Chap. 3).

Requirements for Dispersion Formulae

Often one would like to know the refractive index for unmeasured wavelengths. An analytical representation of the variation of the refractive index n with the wavelength λ is also necessary for computational purposes during the design process of optical systems. Many scientists in the fields of optics and glass science have therefore been trying to derive a *dispersion formula* in order to calculate the refractive index with a few constants derived from measurements. The formula should cover the whole useful spectral range of a glass (from the UV-absorption edge to IR) and should be as accurate as necessary (4 decimal places for analytical purposes during optical design, 5 or 6 decimal places for high-precision interpolation of measured data points) and as simple as possible. For analytical purposes it would be advantageous if n could be expressed in the form of $n = \sum a_j \lambda^j$.

Consequently, there exist a lot of dispersion formulae that fulfil these requirements with varying precision. In general, they are only applicable in a spectral range where absorption is negligible.

Warning: It should be noted here that extrapolation of refractive index data outside the measured wavelengths by using a dispersion formula can lead to serious errors!

Cauchy Dispersion Formula

Cauchy [2.11] was the first to succeed in describing normal dispersion algebraically in 1830. His equation, which can easily be derived from (2.6), reads:

$$n = a + \frac{b}{\lambda^2} + \frac{c}{\lambda^4} . \quad (2.12)$$

The accuracy within the visible spectrum is of the order of 10^{-4} .

Hartmann Dispersion Formula

In the visible range the dispersion formula of *Hartmann*

$$n = n_0 + \frac{A}{(\lambda - \lambda_0)^B} \quad (2.13)$$

is valid. The constants A , B , n_0 , and λ_0 must be derived empirically. The exponent B varies between 0.5 and 2, but can be replaced by unity in most cases. The accuracy is 3 to 4 decimal places.

Sellmeier Dispersion Formula

This dispersion formula was derived by *Sellmeier* in 1871 [2.12] and can be written in the following form:

$$n^2 = 1 + \sum_{j=1}^N \frac{a_j \lambda^2}{\lambda^2 - \lambda_j^2} . \quad (2.14)$$

The λ_j designate the centre of gravity of the j th absorption band in the relevant spectral range (called the *effective resonance wavelength*). It can be shown that virtually all dispersion curves of optical glasses can be expressed by using two effective resonance wavelengths in the UV (typically in the region 50 nm to 150 nm) and one in the IR (roughly at 7 to 10 μ m).

The main advantage of the Sellmeier dispersion formula is that it is based on physical reasoning and therefore has excellent fitting properties in the data reduction process of measured refractive indices. Furthermore, interpolation errors are minor and even the extrapolation errors are significantly smaller than the Laurent series expansion formula (2.16).

In 1992 a Sellmeier formula containing three effective resonance wavelengths was adopted as the standard dispersion formula for the Schott *Optical Glass Catalogue* [2.13]. The accuracy in the refractive index is given below:

$$\begin{array}{llll} 404 \text{ nm} & \leq \lambda \leq & 706 \text{ nm} & \pm 3 \times 10^{-6} \\ 365 \text{ nm} & \leq \lambda \leq & 404 \text{ nm} & \text{and} \\ 706 \text{ nm} & \leq \lambda \leq & 1014 \text{ nm} & \pm 5 \times 10^{-6} \\ 1014 \text{ nm} & \leq \lambda \leq & 2.3 \text{ } \mu\text{m} & \pm 1 \times 10^{-5} \end{array}$$

Helmholtz–Ketteler–Drude Dispersion Formula

The Helmholtz–Ketteler–Drude [2.14, 15] dispersion formula is a modification of the Sellmeier formula (for a derivation see *Born* and *Wolf* [2.5]):

$$n^2 = a_0 + \sum_{j=1}^N \frac{a_j}{\lambda^2 - \lambda_j^2} . \quad (2.15)$$

The λ_j are the effective resonance wavelengths of the oscillators. The accuracy is expected to be comparable to that of the Sellmeier formula.

Schott Dispersion Formula

This dispersion formula, which gives an accuracy of 5 or 6 decimal places, was proposed by Schott in 1966 [2.16] as well suited for optical glasses and was accepted by most glass manufacturers. It can be derived as a Laurent series expansion of *finite* order of the Sellmeier dispersion formula with an *arbitrary* number of effective resonance wavelengths (see e.g. *Born* and *Wolf* [2.5]). The

λ^2 term comes from the IR-absorption band and the λ^{-j} terms model the UV-absorption bands.

$$n^2 = A_0 + A_1\lambda^2 + A_2\lambda^{-2} + A_3\lambda^{-4} + A_4\lambda^{-6} + A_5\lambda^{-8} . \quad (2.16)$$

The constants A_i are determined by a least squares fit to the measured refractive index data. The accuracy is given below:

$$400 \text{ nm} \leq \lambda \leq 750 \text{ nm} \quad \pm 3 \times 10^{-6}$$

$$\begin{aligned} 355 \text{ nm} &\leq \lambda \leq 400 \text{ nm} \quad \text{and} \\ 750 \text{ nm} &\leq \lambda \leq 1014 \text{ nm} \quad \pm 6 \times 10^{-6}. \end{aligned}$$

Since this formula takes into account an arbitrary number of effective resonance wavelengths, the interpolation accuracy between the measured refractive indices is somewhat poorer than that of the Sellmeier formula. The validity of this dispersion formula can be extended further into the UV by adding a λ^{-10} term and into the IR by adding a λ^4 term and higher terms.

Herzberger Dispersion Formula

Based on an analytical study, *Herzberger* [2.17] came to the conclusion that in the visible range the manifold of optical glasses can be explained by four parameters

$$n = A_0 + A_1\lambda^2 + \frac{A_2}{\lambda^2 - \lambda_0^2} + \frac{A_3}{(\lambda^2 - \lambda_0^2)^2} , \quad (2.17)$$

with $\lambda_0 = 168 \text{ nm}$. This dispersion formula is accurate to four decimal places in the visible spectral range.

Geffcken Dispersion Formula

Geffcken [2.18–20] sought to express the dependence of the refractive index on the wavelength λ as a combination of one function describing the spectral behaviour of the so-called normal glasses (cf. below in Sect. 2.1.5) and a δ function which accounts for the abnormal dispersion of a specific glass. By definition, the δ function becomes zero for all normal glasses.

$$n(\lambda) = 1 + [1 - D(\lambda)] [n(\lambda_1) - 1] + B(\lambda) [n(\lambda_2) - n(\lambda_1)] + D(\lambda)\delta(\lambda) . \quad (2.18)$$

In this equation λ_1 and λ_2 are standard wavelengths (e.g. $\lambda_1 = 644 \text{ nm}$, $\lambda_2 = 480 \text{ nm}$), n_1 and n_2 the corresponding refractive indices, $B(\lambda)$ and $D(\lambda)$ tabulated functions for a fixed pair of λ_1 and λ_2 (independent of the glass type), and finally δ a tabulated function specific to a given glass. The δ function describes the deviation of the relative partial dispersion from the behaviour of “normal glasses”.

For the wavelength range from 365 nm to 1014 nm an accuracy of about 10^{-6} can be achieved.

Buchdahl Dispersion Formula

In terms of analytical algebraic expressions it is very desirable to have a dispersion model in the form $n = \sum a_j \omega^j$ for the design of highly colour-corrected lens systems. The a_j should be independent of the wavelength and the variable ω should be a function of the wavelength, $\omega = \omega(\lambda)$. None of the dispersion formulae mentioned so far fulfils this requirement. *Buchdahl* [2.21] introduced a change in variables from the wavelength λ to a chromatic coordinate ω defined as

$$\omega(\lambda) = \frac{\lambda - \lambda_0}{1 + \alpha(\lambda - \lambda_0)} . \quad (2.19)$$

λ_0 is a reference (or base) wavelength and α is a constant for all glasses and has a value of 2.5. For applications in the visible spectrum, λ_0 is chosen as 587.6 nm.

The refractive index n can now be written as an implicit function of the wavelength in terms of the chromatic coordinate ω :

$$n(\omega) = n_0 + \nu_1 \omega + \nu_2 \omega^2 + \nu_3 \omega^3 + \dots + \nu_n \omega^n . \quad (2.20)$$

n_0 is the index of refraction at λ_0 , and the ν_j correspond to the dispersion of the glass. This series expansion converges very quickly so that only a few terms are necessary. The accuracy for an expansion with powers in ω of $j = 2$ and $j = 3$ is given below (cf. *Robb* and *Mercado* [2.22]):

$$\begin{array}{llll} j = 2 : 400 \text{ nm} & \leq \lambda \leq & 700 \text{ nm} & \pm 1 \times 10^{-4} \\ j = 3 : 355 \text{ nm} & \leq \lambda \leq & 1014 \text{ nm} & \pm 2 \times 10^{-4} . \end{array}$$

This model was used by *Robb* [2.23] for the selection of glass doublets and triplets having excellent colour correction.

2.1.4 Measurement and Limitations of the Accuracy of Refractive Index Data

General Remarks

There are a lot of measuring techniques deriving the optical constants of glasses (cf. *Bell* [2.24] and *Ward* [2.25] and references therein). Usually the refractive index is measured by determining the angle of the minimum deviation of light in a prism with a spectrometer (accurate to $\pm 1 \times 10^{-6}$) or by determination of the critical angle with the help of a Pulfrich refractometer (accurate to $\pm 1 \times 10^{-5}$) for routine work. Despite great efforts, the accuracy of the measured index data can barely exceed $\pm 1 \times 10^{-6}$. There are several reasons for these limitations, which we shall discuss further. Interferometric techniques can be applied to obtain relative measurements such as for checking homogeneity (spatial index variation). In this case the relative accuracy may reach $\pm 1 \times 10^{-7}$.

The refractive indices given in catalogues and handbooks are obtained by measuring pieces of glass samples in air under controlled environmental conditions (i.e., temperature, air pressure, humidity, etc.). The differences in the

refractive index are up to $\pm 1 \times 10^{-3}$ from melt to melt. Single selected pieces of glass (for a few glasses up to 1 m in diameter) can be manufactured with a relative index variation of about $\pm 1 \times 10^{-6}$.

Preparation of Samples

In order to measure the refractive index with high accuracy, the preparation of the samples to be measured is crucial. One needs highly homogeneous and striae-free pieces of glass. Much greater efforts are necessary to achieve a surface flatness compatible with the desired index accuracy. If an accuracy of $\pm 1 \times 10^{-6}$ is required, the variation, due to surface imperfections, of the angle of incidence must not exceed a few tenths of an arcsecond. The influence of irregular deviations is reduced by integration over the prism aperture. On the other hand, if the angular deviation of a prism is measured in a spectral range where the transmission is low, the influence of the surface curvature is increased. Because of the different lengths of light paths across the aperture of the prism, the highest transmission is near the thin edge, which may easily have surface imperfections (e.g. turned down edge). Therefore a systematic error may occur.

Refractive Index of Air

Since the refractive indices are relative to air, it is necessary to know the refractive index of air very accurately [2.26–28]. More details will be given in Sect. 2.4 below.

As an example, at the wavelength of sodium ($\lambda_0 = 589.46$ nm, temperature $T = 20^\circ$ C, air pressure $p = 760$ Torr, partial pressure of water vapour $w = 10$ Torr) one obtains the following expressions for the differential variation of the refractive index of air [2.26]:

$$\begin{aligned}\frac{dn_{\text{air}}}{dT} &= -\alpha \frac{n_{\text{air}} - 1}{1 + \alpha T} = -1.071 \times 10^{-6} \text{ K}^{-1}, \\ \frac{dn_{\text{air}}}{dp} &= \frac{n_{\text{air}} - 1}{p} = +0.268 \times 10^{-6} \text{ hPa}^{-1}, \\ \frac{dn_{\text{air}}}{dw} &= -\frac{41 \times 10^{-9} \text{ hPa}^{-1}}{1 + \alpha T} = -0.039 \times 10^{-6} \text{ hPa}^{-1},\end{aligned}\tag{2.21}$$

where α designates the thermal expansion of air ($\alpha = 0.00367 \text{ K}^{-1}$). Consequently variations of the temperature of $dT = \pm 5 \text{ K}$ and the pressure of $dp = \pm 20 \text{ hPa}$ have a noticeable effect on the refractive index of air.

Wavelengths of Spectral Lines for the Determination of the Refractive Index

For the determination of refractive indices one needs high-intensity monochromatic light sources. It is convenient to use *low-pressure* spectral discharge lamps. The implicit assumption is that the wavelengths of the spectral lines are fixed and known as accurately as necessary; this is not correct.

In general, a spectral line consists of several multiplet components (due to fine and hyperfine structure) with various intensities which depend on pressure, isotope composition, and other conditions in a discharge lamp. Therefore, if it is measured by a spectrometer with limited resolution, the centre of gravity of such a multiplet may noticeably shift in wavelength.

Additionally, the *absolute* accuracy of the wavelength of a spectral line may be questioned. It is true that high-resolution spectroscopy (e.g. laser spectroscopy) allows a highly accurate measurement of the multiplet structure of a spectral line. However, the data obtained are only *relative* to a given reference wavelength. In general, most existing data on absolute wavelengths are measured *relative to air* and afterwards reduced to those *in vacuum*. These data are often quite old and not as accurate as actually needed today; this status is not expected to change quickly since the task of measuring absolute wavelengths is extremely laborious.

Table 2.1 shows the spectral lines that are generally used to determine the refractive indices of glasses.

Table 2.1. Wavelength of spectral lines for the determination of refractive indices

Wavelength nm	Designation	Spectral Line	Element
2325.4		IR mercury line	Hg
1970.1		IR mercury line	Hg
1529.6		IR mercury line	Hg
1060.0		neodymium glass laser	Nd
1013.98	<i>t</i>	IR hydrogen line	Hg
852.1101	<i>s</i>	IR caesium line	Cs
706.5188	<i>r</i>	red helium line	He
656.2725	<i>C</i>	red hydrogen line	H
643.8469	<i>C'</i>	red cadmium line	Cd
632.8		helium-neon gas laser	He-Ne
589.2938	<i>D</i>	yellow sodium line	Na
		center of doublet	
587.5618	<i>d</i>	yellow helium line	He
546.0740	<i>e</i>	green mercury line	Hg
486.1327	<i>F</i>	blue hydrogen line	H
479.9914	<i>F'</i>	blue cadmium line	Cd
435.8343	<i>g</i>	blue mercury line	Hg
404.6561	<i>h</i>	violet mercury line	Hg
365.0146	<i>i</i>	UV mercury line	Hg

2.1.5 Definition of the Characteristic Properties of Optical Glasses

Characteristic Values of Glass Types

For the design of optical systems, a range of optical glasses is indispensable. Consequently, a classification of a specific glass according to its optical properties, *refraction* and *dispersion*, is necessary. The following numbers and values

characterize these properties for optical applications in the visible spectral range: the main refractive index n , the main dispersion $n_{\lambda_1} - n_{\lambda_2}$, the Abbe number ν [2.29], and the relative partial dispersion P .

(a) The *main refractive index* n_d or n_e is the index for the helium line at $\lambda = 587.56$ nm or mercury line at $\lambda = 546.07$ nm. This value varies between 1.4 and 2.3 for oxide glasses.

(b) The main dispersion $n_F - n_C$ or $n_{F'} - n_{C'}$ describes the variation of the refractive index between the hydrogen lines F ($\lambda = 486.13$ nm) and C ($\lambda = 656.27$ nm) or the cadmium lines F' (479.99 nm) and C' (643.85 nm). In the case of optical glasses, n_F is always larger than n_C (normal dispersion).

(c) The *reciprocal dispersive power* or *Abbe number* ν is basically the ratio between the refractive power and the main dispersion of a glass. It is defined by

$$\begin{aligned}\nu_d &= \frac{n_d - 1}{n_F - n_C}, \\ \nu_e &= \frac{n_e - 1}{n_{F'} - n_{C'}}.\end{aligned}\quad (2.22)$$

In most cases the refractive index at the d line n_d or e line n_e is used as a reference wavelength; the corresponding Abbe numbers are designated as ν_d or ν_e . A small number corresponds to large dispersion and vice versa. A generalized differential version of (2.22), covering a continuous wavelength range, can be written as

$$\nu^*(\lambda) = \frac{1 - n(\lambda)}{dn/d\lambda}. \quad (2.23)$$

The unit is a length.

(d) The *relative partial dispersion* P or P' is needed to control the colour correction in more sophisticated optical systems (correction of the secondary spectrum). It can be considered as a number describing the similarity in the dispersion of two glasses.

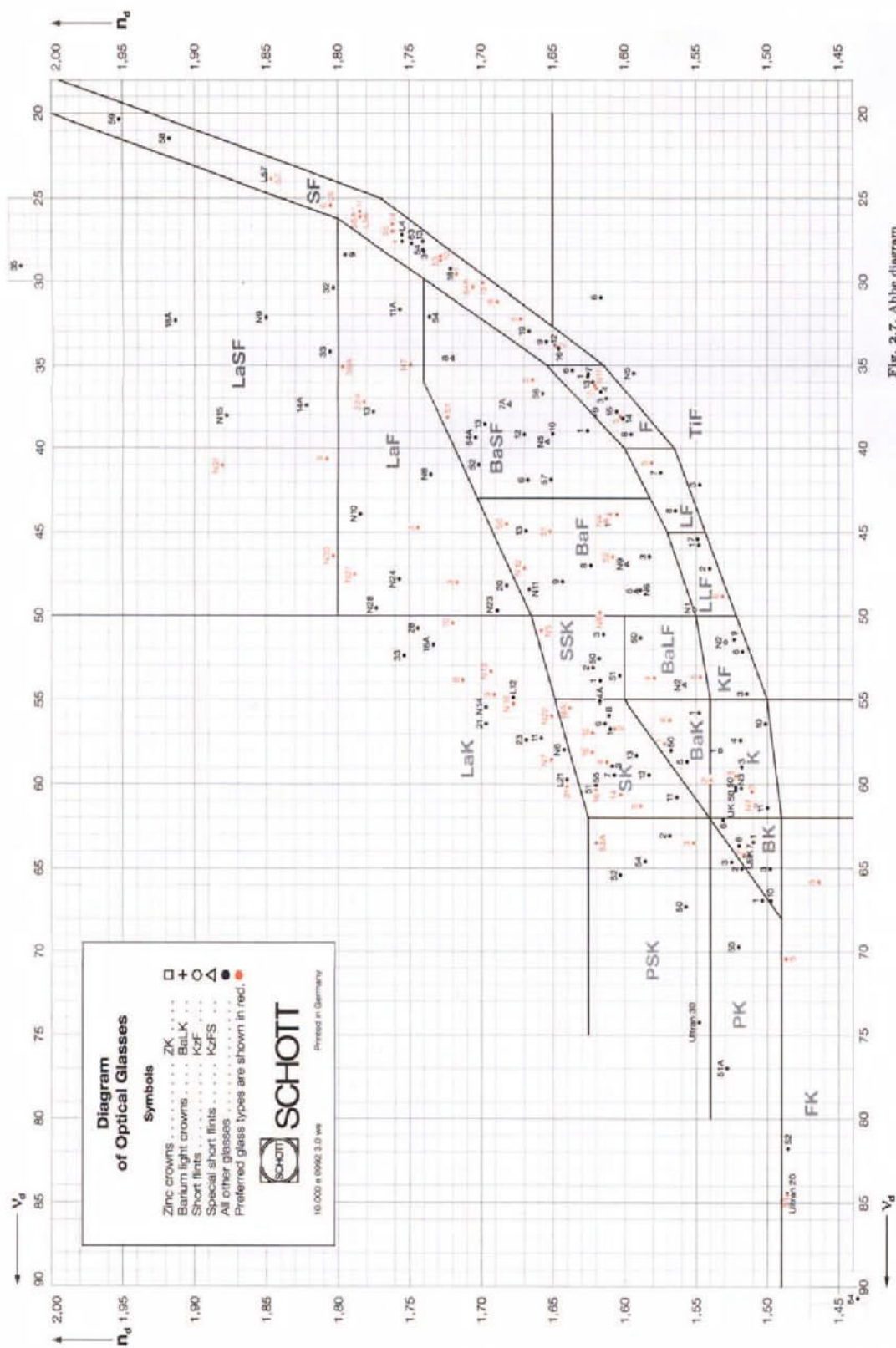
$$\begin{aligned}P_{\lambda_1, \lambda_2} &= \frac{n_{\lambda_1} - n_{\lambda_2}}{n_F - n_C} \\ P'_{\lambda_1, \lambda_2} &= \frac{n_{\lambda_1} - n_{\lambda_2}}{n_{F'} - n_{C'}}.\end{aligned}\quad (2.24)$$

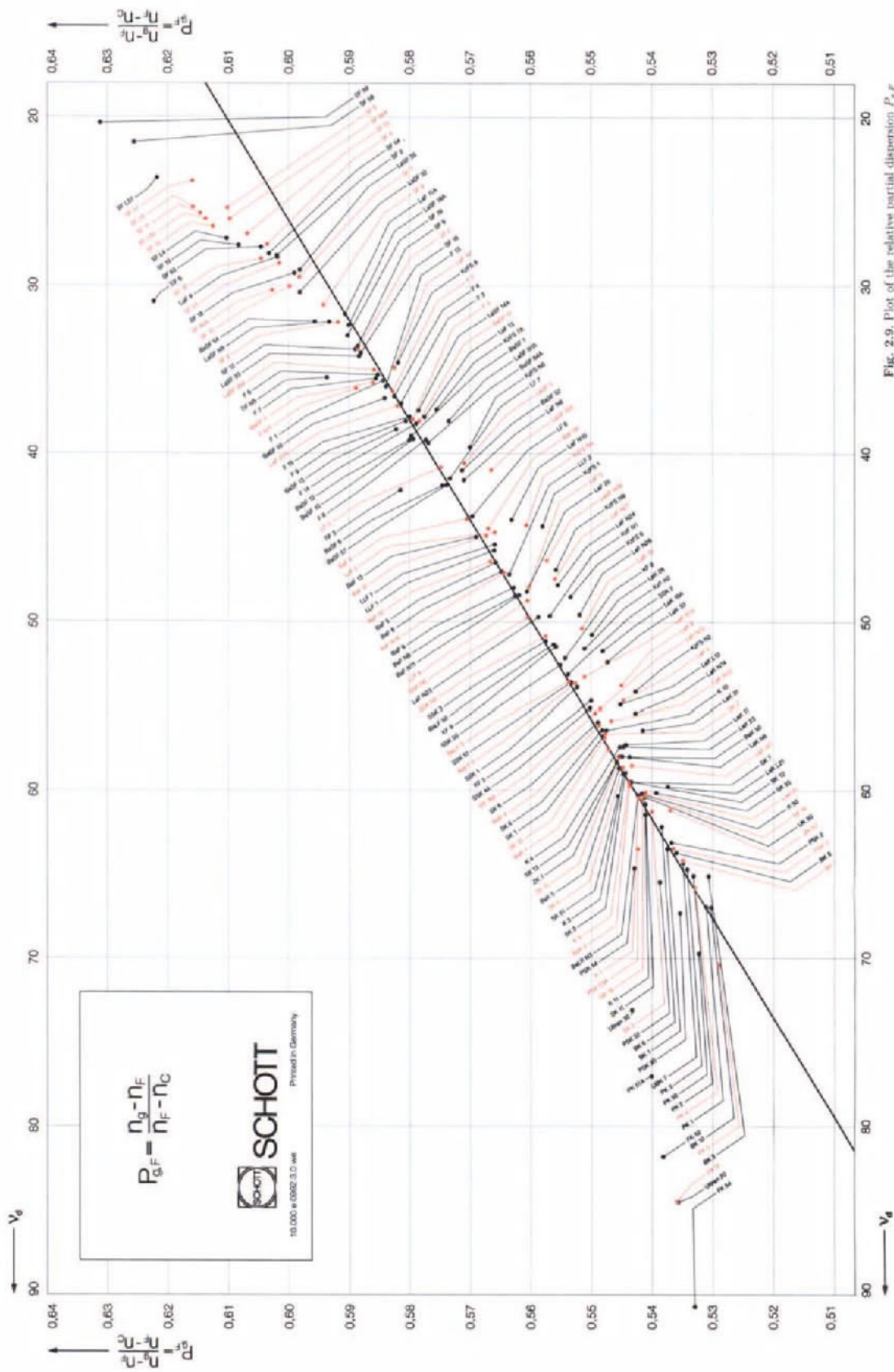
Basically it is a measure of the rate of refractive index change dependent on the wavelength (i.e. the second derivative).

Traditionally optical glasses in the range of

$$\begin{array}{llll}n_d < 1.60, & \nu_d > 55 & n_e < 1.6028, & \nu_e > 54.7 \\ n_d > 1.60, & \nu_d > 50 & n_e > 1.6028, & \nu_e > 49.7\end{array}$$

are designated as *crown* glasses, the other ones as *flint* glasses. Although these names have no other significance today they roughly divide the Abbe diagram (see fold-out diagram Fig. 2.7) into glasses with relatively low and those with relatively high dispersive powers.





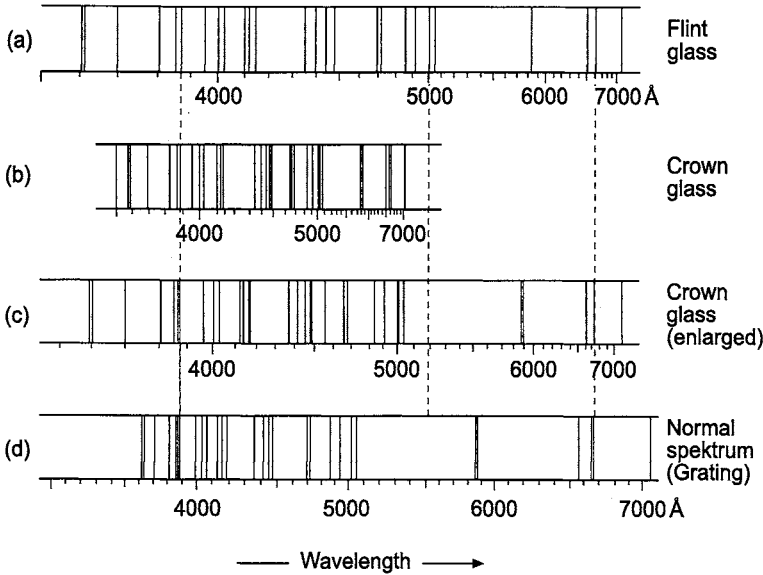


Fig. 2.5. Comparison of the helium spectra produced by flint glass and crown glass prisms with a normal spectrum [2.6]

Figure 2.5 shows helium spectra produced by a crown prism and a flint prism in comparison to a grating (normal) spectrum. It can be seen that the dispersions of the two glasses and especially the relative spacings of the lines are different even if the crown glass spectrum is so enlarged that the two selected lines coincide with those in the flint glass spectrum. Obviously the relative partial dispersions of both glasses are not equal. We will discuss this topic later in greater detail.

In Fig. 2.6 the variation of the refractive index n with the wavelength λ is shown for glasses with similar values of n_d but different ν_d . Additionally, the first derivative $dn/d\lambda$ is plotted. Clearly, at the wavelength of the d line (587 nm) the refractive indices of the glasses coincide, whereas the glass with the *lower*

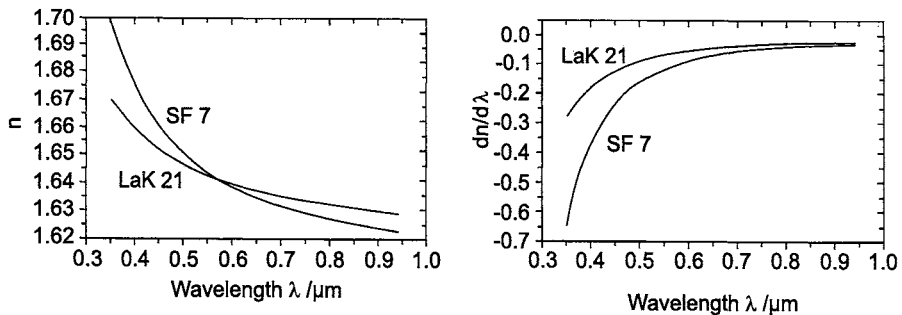


Fig. 2.6. Comparison of dispersion curves $n(\lambda)$ for glasses with similar n_d but different ν_d and the corresponding first derivative $dn/d\lambda$

Abbe number has a *higher* refraction at shorter wavelengths and vice versa. The corresponding plot of the dispersion $dn/d\lambda$ stresses the different behaviour of both glasses.

A specific glass type has a designation based on its glass composition and a running number (e.g., boron crown 7 as BK 7), see Table 2.2. A lot of different types exist, as can be seen in the Schott nomenclature system.

Some newer glasses with similar optical properties differ significantly in composition (e.g. lightweight glasses).

In order to give a glass type a unique designation a 6-digit code number was introduced:

$$1000(n - 1) + 10\nu ,$$

where n was rounded to 3 decimal places and ν to 1 decimal place. For example, the glass type BK 7 with $n_d = 1.5168$ and $\nu_d = 64.17$ was designated BK 7-517642. In Table 2.3, optical glasses of the four major manufacturers (i.e., Corning, Hoya, Ohara, Schott) are listed. They are sorted according to increasing code numbers.

Glass Diagrams

In order to provide an overview of the range of optical properties for the large (> 200) number of optical glasses, their positions are plotted in so-called *glass diagrams*. They are useful for designing optical systems.

The n - ν diagram (often called the Abbe diagram [2.29]) is a plot of the refractive index against the Abbe number for the various glass types (see fold-out diagram Fig. 2.7).

Occasionally a plot of n against $1/\nu$ is preferred in order to stress the importance of $1/\nu$ for the colour correction of optical systems (see Zimmer [2.31]). The reciprocal relative dispersion ν has been introduced only for practical reasons – handier numbers – but already Abbe preferred the representation $1/\nu$. Therefore one can introduce a *dispersion number* m according to

$$m_d = 1831 \frac{n_F - n_C}{n_d - 1} = \frac{1831}{\nu_d} , \quad (2.25)$$

where $1831 = 1/0.000546$ corresponds to the reciprocal wavelength of the e line, expressed in millimetres.

Figure 2.8 shows the resulting diagram, where the deviations of the relative partial dispersions from the line of “normal glasses” in the blue $\Delta P_{g,F}$ and red $\Delta P_{C,s}$ are designated as vectors.

The majority of glasses (the so-called normal glasses) show a more or less linear relationship between the Abbe number and the relative partial dispersion, as already found by Abbe:

$$P_{x,y} \approx a_{xy} + b_{xy}\nu_d = \tilde{P}_{x,y} . \quad (2.26)$$

The glasses K 7 and F 2 are often chosen as reference glasses, defining the line (cf. fold-out diagram Fig. 2.9). A somewhat better representation, which takes

Table 2.2. Designations of Schott glasses [2.30]

Glass type	Name	Boundary	Main components
FK	fluorine crown	$n < 1.49$ $\nu > 62$	phosphate or borosilicate glass with fairly high fluorine content
PK	phosphorus crown	$n 1.49 - 1.54$ $\nu > 68 - 62$	phosphate or borosilicate glass, partially with fluorine
PSK	dense phosphorus crown	$n 1.54 - 1.625$ $\nu > 62$	borosilicate or borophosphate glasses
BK	boron crown	$n \sim 1.51^*$ $\nu \sim 64$	borosilicate glasses
BaLK	barium light crown	$n \sim 1.52$ $\nu \sim 60$	borosilicate glasses
K	crown	$n 1.495 - 1.54$ $\nu 55 - 62$	alkaline-silicate glasses, partially with boron
ZK	zinc crown	$n 1.51 - 1.53$ $\nu 57 - 61$	silicate or borosilicate glasses with $> 10\%$ wt. ZnO
BaK	barium crown	$n 1.55 - 1.60^*$ $\nu 55 - 62$	borosilicate glasses with $> 10\%$ wt. BaO
SK	dense crown	$n 1.55 - 1.60^*$ $\nu 55 - 62$	alumino-borosilicate glasses with $> 15\%$ wt. BaO
KF	crown flint	$n 1.55 - 1.545^*$ $\nu 50 - 55$	alkaline-lead-silicate glasses
BaLF	barium light flint	$n 1.55 - 1.59^*$ $\nu 51 - 54$	alkaline-lead-silicate glasses with $> 10\%$ wt. BaO
SSK	extra-dense crown	$n 1.60 - 1.66$ $\nu 50 - 55$	alkaline-borosilicate glasses with $> 29\%$ wt. BaO
LaK	lanthanum crown	$n > 1.64^*$ $\nu > 50$	silico-borate glasses with RE oxides
LLF	extra-light flint	$n 1.53 - 1.57^*$ $\nu 45 - 50$	alkaline-silicate glasses with 20-27% wt. PbO
BaF	barium flint	$n 1.58 - 1.69^*$ $\nu 43 - 50$	alkaline-silicate glasses with $\sum$ PbO and BaO $< 40\%$ wt.
LF	light flint	$n 1.56 - 1.59^*$ $\nu 40 - 45$	alkaline-silicate glasses with 28-35% wt. PbO
F	flint	$n 1.595 - 1.64^*$ $\nu 35 - 40$	alkaline-silicate glasses with 39-47% wt. PbO
BaSF	dense barium flint	$n 1.60 - 1.74^*$ $\nu 32 - 43$	alkaline-silicate with $\sum$ PbO and BaO $> 40\%$ wt.
LaF	lanthanum flint	$n 1.63 - 1.80^*$ $\nu 28 - 50$	borosilicate or silico-borate glasses with RE oxides
LaSF	dense lanthanum flint	$n > 1.80^*$ $\nu 27 - 50$	RE-oxide-borate or RE-oxide alkaline-earth-silicate glasses
SF	dense flint	$n > 1.63^*$ $\nu < 35$	alkaline-silicate glasses with $> 47\%$ wt. PbO
TiK	low crown	$n < 1.495$ $\nu 55 - 62$	alkaline-alumino-borosilicate glass
TiF	low flint	$n 1.50 - 1.65^*$ $\nu < 33$ @ 1.65 $\nu < 55$ @ 1.50	titanium-alkaline-alumino-boro-silicate or phosphate glass

Table 2.2. (Cont.)

Glass type	Name	Boundary	Main components
TiSF	dense low flint	$n > 1.65^*$ $\nu < 30$	titanium-alkaline-silicate glass
KzF	short flint	$n \sim 1.54$ $\nu \sim 50$	antimony-borosilicate glass
KzFS	short flint special	$n \ 1.59 - 1.72$ $\nu \ 34 - 54$	alumino-lead-borate glass

* only approximate due to non-rectangular boundaries in the $n_d - \nu_d$ diagram

the small curvature of the line of “normal glasses” into account, was given by *Izumitani* and *Nakagawa* [2.32]

$$P_x = a_x + b_x \nu_d + c_x \nu_d^2 . \quad (2.27)$$

In order to correct the secondary spectrum (i.e. colour correction for more than two wavelengths) glasses are required which do *not* fulfil relation (2.26). A plot of the relative partial dispersion in a specific wavelength range (e.g. $P_{g,F}$ in the blue, $P_{C,s}$ in the red) against the Abbe number ν facilitates the selection of interesting glasses (see fold-out diagram Fig. 2.9). For a doublet in the thin-lens approximation, both glasses should have the same relative partial dispersions P (possibly in a wide spectral range), but the difference between the $1/\nu$ values should be as large as possible to keep the individual refractive powers of both lenses as low as possible.

If a triplet lens is required to bring four colours to a common focus (e.g. red, green, blue, and violet), the three glasses must additionally lie on a straight line in a plot of the red P_r against the blue P_b as shown by *Herzberger* [2.17].

Colour Correction and Relative Partial Dispersions

Generally, a nearly perfect correction of polychromatic aberrations is a laborious task and it essentially depends on the available glass types with abnormal relative partial dispersions. From the thin-lens approximation one can derive the following condition for perfect colour correction for a lens doublet with vanishing separation d

$$n_2(\lambda) = \alpha + \beta n_1(\lambda) . \quad (2.28)$$

In other words, the dispersion curves of the two glasses must be related by a *linear* transformation. In general, however, this cannot be attained for an arbitrarily wide wavelength range as will now be shown. If we express both dispersion curves by Sellmeier formulae, we obtain, assuming a fixed number N of resonance wavelengths,

$$n_1(\lambda) = 1 + \sum_{j=1}^N a_j \frac{\lambda^2}{\lambda^2 - \lambda_{1j}^2} ,$$

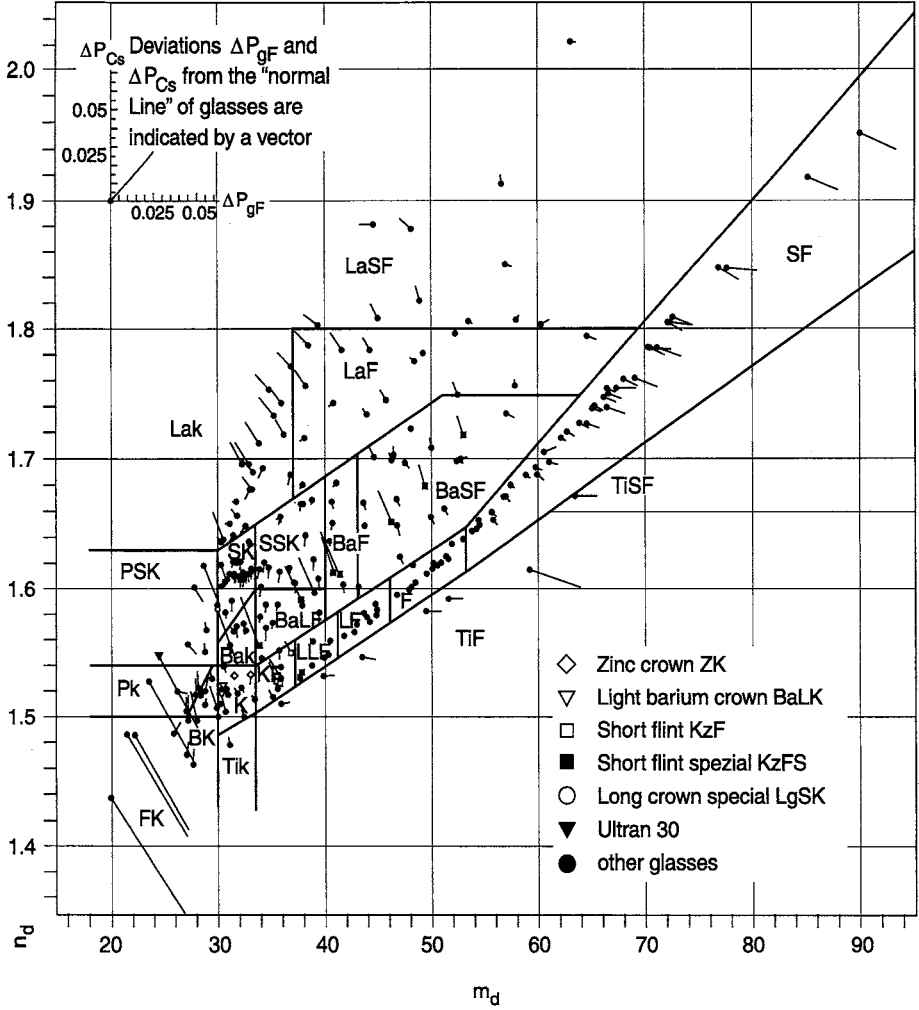


Fig. 2.8. Refractive index plotted against dispersion number [2.31] and indication of the partial dispersion

$$n_2(\lambda) = 1 + \sum_{j=1}^N b_j \frac{\lambda^2}{\lambda^2 - \lambda_{2j}^2} \quad (2.29)$$

If we expand $n_2(\lambda)$ using (2.28), we can write

$$n_2(\lambda) = \alpha + \beta + \beta \sum_{j=1}^N a_j \frac{\lambda^2}{\lambda^2 - \lambda_{1j}^2} \quad (2.30)$$

Then the following conclusions can be drawn.

(a) The resonance wavelengths of both glasses must be the **same**. With some effort this can be achieved in practice.

(b) All oscillator strengths of glass 2 must be decreased/increased by the **same** factor β . This is virtually impossible.

(c) The constants α and β must fulfil the condition $\alpha + \beta = 1$ or $\alpha = 1 - \beta$. The generalized Abbe numbers for both glasses are given by

$$\begin{aligned}\nu_1(\lambda) &= \frac{1 - n_1(\lambda)}{dn_1/d\lambda} , \\ \nu_2(\lambda) &= \frac{1 - n_2(\lambda)}{dn_2/d\lambda} .\end{aligned}\tag{2.31}$$

If ν_2 is expanded by using (2.28), we obtain

$$\nu_2(\lambda) = \frac{1 - \alpha - \beta n_1(\lambda)}{d(\alpha + \beta n_1(\lambda))/d\lambda} = \frac{1 - \alpha - \beta n_1(\lambda)}{\beta dn_1/d\lambda} .\tag{2.32}$$

With use of (c) this results in $\nu_1 = \nu_2$! As is well known, this condition does not allow achromatization.

A viable way to solve this problem could be the introduction of additional resonance wavelengths; a method which makes an approximate solution for a limited spectral range possible. If we add a single resonance λ_{2r} , only the following condition must be fulfilled:

$$1 + b_r \frac{\lambda^2}{\lambda^2 - \lambda_{2r}^2} = \alpha + \beta .\tag{2.33}$$

If $\lambda \gg \lambda_{2r}$, we can approximate (2.33) by

$$1 + b_r \approx \alpha + \beta ,\tag{2.34}$$

which is independent of λ . For visible applications the new resonance wavelength must be as far in the UV as possible.

On the other hand, if $\lambda_{2r} \gg \lambda$, i.e., the new resonance is in the IR, one obtains

$$1 - b_r \frac{\lambda^2}{\lambda_{2r}^2} = \alpha + \beta ,\tag{2.35}$$

which depends quadratically on λ . This explains why it is much easier to develop glasses with abnormal partial dispersions such as the fluorine crown glasses having resonances in the far UV.

A key idea for the development of new glasses could be the design of glass pairs (doublets) with relative partial dispersions adapted to each other. If we form the first and second derivative of (2.28) with respect to the wavelength, we can create a differential partial dispersion

$$P^*(\lambda) = \frac{n'(\lambda)}{n''(\lambda)} .\tag{2.36}$$

In the case of adapted glasses the following condition must be fulfilled:

$$\frac{n'_2(\lambda)}{n''_2(\lambda)} = \frac{n'_1(\lambda)}{n''_1(\lambda)} .\tag{2.37}$$

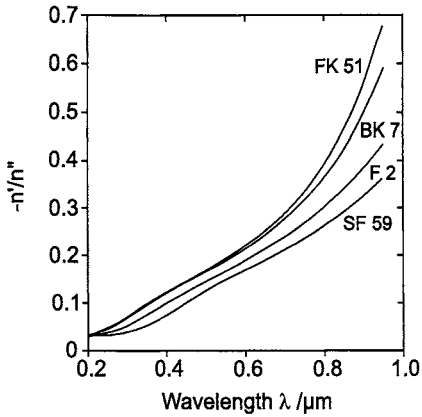


Fig. 2.10. Spectral variation of the differential partial dispersions of several glasses

A plot of $P^*(\lambda)$ has the advantage of being independent of a chosen reference wavelength system (e.g. F and C lines) so that the spectral variation of the relative partial dispersions and their similarity between glasses can easily be checked.

In Fig. 2.10 a plot of several optical glasses is shown. Obviously BK 7 and FK 51 have similar relative partial dispersions in the visible and near-UV, whereas in the near-IR both glasses do not go well together. On the other hand, the relative partial dispersions of F 2 and SF 59 differ significantly from the two other glasses.

If one were to succeed in developing optical glasses that fulfil (2.37), at least approximately over a limited spectral range, a lot would be gained.

Acknowledgement

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Table 2.3. Optical glass cross-reference by index (producers: C, Corning France; H, Hoya; O, Ohara; S, Schott Glas and Schott Glass Technologies)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
437-907	FK 54	1.43700	90.70	0.0418	-0.1249	S
438-950	FPL 53	1.43875	94.97	0.0480	-0.1520	O
456-903	FPL 52	1.45600	90.31	0.0390	-0.1310	O
457-903	FCD10	1.45650	90.77	0.0504	-0.1343	H
464-658	FK 3	1.46450	65.77	-0.0003	0.0207	S
465-657	FC A63-65	1.46450	65.70			C
465-658	FC3	1.46450	65.77	0.0055	0.0271	H
465-659	FSL 3	1.46450	65.94	-0.0020	0.0260	O
471-673	FC1	1.47069	67.31	0.0052	0.0207	H
471-674	FSL 1	1.47069	67.39	0.0010	0.0170	O
486-817	PFC A86-82	1.48605	81.70			C
486-818	FK 52	1.48605	81.80	0.0319	-0.1114	S
487-702	FSL 5	1.48749	70.21	0.0020	0.0140	O
487-704	FC A87-70	1.48725	70.40			C
487-704	FC5	1.48749	70.44	0.0091	0.0238	H
487-704	FK 5	1.48749	70.41	0.0036	0.0202	S
487-845	FK 51	1.48656	84.47	0.0342	-0.1112	S
487-845	Ultran [®] 20	1.48656	84.47	0.0340	-0.1097	S
488-658	FC A88-66	1.48750	65.80			C
497-816	FCD1	1.49700	81.61	0.0374	-0.0967	H
497-816	FPL 51	1.49700	81.61	0.0280	-0.0990	O
498-650	BSL 3	1.49831	65.03	-0.0040	0.0336	O
498-651	BSC3	1.49831	65.13	-0.0021	0.0427	H
498-651	BK 3	1.49831	65.06	-0.0036	0.0394	S
498-668	BSL 10	1.49782	66.83	-0.0040	0.0300	O
498-670	BK 10	1.49782	66.95	-0.0008	0.0314	S
500-614	K 11	1.50013	61.44	0.0006	0.0130	S
500-660	BSC4	1.50048	66.00	0.0016	0.0384	H
500-660	BSL 4	1.50048	65.99	-0.0031	0.0319	O
501-563	C10	1.50137	56.30	0.0038	0.0109	H
501-564	FTL 10	1.50137	56.40	0.0035	-0.0019	O
501-564	K 10	1.50137	56.41	-0.0015	0.0094	S
504-668	BSL 21	1.50378	66.81	-0.0030	0.0220	O
504-669	PC1	1.50378	66.89	0.0041	0.0244	H
504-669	PK 1	1.50378	66.92	-0.0001	0.0258	S
508-608	ZSL 7	1.50847	60.83	-0.0063	0.0263	O
508-612	ZKN 7	1.50847	61.19	-0.0039	0.0266	S
508-613	ZnC7	1.50847	61.27	-0.0017	0.0283	H
510-621	NSL 1	1.50977	62.10	-0.0016	-0.0021	O

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
510-634	BSC1	1.51009	63.43	0.0028	0.0136	H
510-635	BSC B10-63	1.51000	63.50			C
510-635	BK 1	1.51009	63.46	0.0004	0.0101	S
510-636	BSL 1	1.51009	63.63	-0.0001	0.0112	O
511-510	FF1	1.51118	51.03	0.0072	0.0092	H
511-510	FTL 8	1.51118	51.02	0.0058	0.0035	O
511-604	C B11-60	1.51100	60.40			C
511-604	K 7	1.51112	60.41	0.0000	0.0000	S
511-605	C7	1.51112	60.49	0.0000	0.0000	H
511-605	NSL 7	1.51112	60.49	0.0000	0.0000	O
512-581	ZSL 4	1.51190	58.14	-0.0020	-0.0079	O
515-546	CF3	1.51454	54.63	0.0014	0.0057	H
515-547	NSL 33	1.51454	54.71	-0.0021	-0.0014	O
515-547	KF 3	1.51454	54.70	-0.0018	0.0039	S
516-568	C2	1.51602	56.77	0.0016	0.0058	H
516-568	NSL 2	1.51602	56.80	-0.0033	0.0053	O
516-642	BSL 7	1.51633	64.15	-0.0026	0.0199	O
517-522	CF6	1.51742	52.15	0.0005	0.0084	H
517-522	KF 6	1.51742	52.20	-0.0017	0.0073	S
517-524	NSL 36	1.51742	52.41	-0.0020	0.0076	O
517-642	BSC B16-64	1.51680	64.20			C
517-642	BSC7	1.51680	64.20	0.0015	0.0223	H
517-642	BK 7	1.51680	64.17	-0.0009	0.0216	S
517-643	UBK 7	1.51680	64.29	-0.0008	0.0189	S
517-696	APL 1	1.51728	69.56	0.0034	-0.0017	O
517-697	PCS1	1.51728	69.68	0.0084	-0.0040	H
518-590	C3	1.51823	58.96	0.0035	0.0010	H
518-590	NSL 3	1.51823	58.96	-0.0004	-0.0045	O
518-590	K 3	1.51823	58.98	0.0001	-0.0017	S
518-603	C B18-60	1.51820	60.30			C
518-603	NSL 23	1.51835	60.27	-0.0019	0.0025	O
518-603	BaLK N3	1.51849	60.26	-0.0001	0.0037	S
518-604	BaCL3	1.51835	60.40	0.0021	-0.0068	H
518-650	BSL 22	1.51821	65.04	-0.0036	0.0284	O
518-651	BSC B18-65	1.51820	65.10			C
518-651	PK 2	1.51821	65.05	-0.0012	0.0255	S
518-652	PC2	1.51821	65.16	0.0025	0.0249	H
519-574	K 4	1.51895	57.40	-0.0020	0.0086	S
519-645	BSL 12	1.51874	64.48	-0.0032	0.0188	O
520-637	BK 8	1.52015	63.69	-0.0007	0.0152	S

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
521-526	SSL 5	1.52130	52.55	-0.0072	0.0302	O
521-528	SbF5	1.52130	52.78	-0.0053	0.0337	H
521-697	PK 50	1.52054	69.71	0.0058	0.0009	S
522-595	C5	1.52249	59.45	0.0033	-0.0035	H
522-595	K 5	1.52249	59.48	0.0000	-0.0025	S
522-598	NSL 5	1.52249	59.79	0.0003	-0.0020	O
523-508	NSL 35	1.52310	50.84	-0.0024	0.0054	O
523-510	CF5	1.52310	50.95	0.0003	0.0088	H
523-515	KF 9	1.52341	51.49	-0.0014	0.0064	S
523-585	NSL 51	1.52307	58.49	-0.0005	0.0014	O
523-586	C12	1.52307	58.64	0.0030	0.0005	H
523-594	C B23-59	1.52300	59.40			C
523-600	NSL 50	1.52257	60.03	-0.0001	0.0054	O
523-602	K 50	1.52257	60.18	-0.0009	0.0068	S
523-604	C40	1.52300	60.40	0.0014	0.0073	H
523-604	UK 50	1.52257	60.38	-0.0010	0.0064	S
525-646	PC3	1.52542	64.62	0.0027	0.0230	H
525-646	BSL 23	1.52542	64.55	-0.0014	0.0242	O
525-647	PK 3	1.52542	64.66	-0.0009	0.0218	S
526-511	CF2	1.52630	51.05	-0.0019	0.0023	H
526-512	NSL 32	1.52630	51.17	-0.0018	0.0021	O
526-600	NSL 21	1.52642	59.99	-0.0024	-0.0021	O
526-601	BaCL1	1.52642	60.11	0.0048	-0.0043	H
527-511	SSL 6	1.52682	51.13	-0.0075	0.0283	O
529-516	SbF2	1.52944	51.64	-0.0036	0.0321	H
529-516	KzF N2	1.52944	51.63	-0.0058	0.0304	S
529-517	SSL 2	1.52944	51.72	-0.0067	0.0297	O
529-518	CHD B29-52	1.52900	51.80			C
529-770	PK 51A	1.52855	76.98	0.0258	-0.0991	S
531-621	BSC6	1.53113	62.07	0.0017	0.0057	H
531-622	BK 6	1.53113	62.15	-0.0009	0.0101	S
531-625	BSL 6	1.53113	62.45	-0.0025	0.0085	O
532-488	FEL6	1.53172	48.84	-0.0030	0.0020	H
532-488	LLF 6	1.53172	48.76	-0.0013	0.0054	S
532-489	PBL 6	1.53172	48.91	-0.0027	-0.0013	O
533-459	FF2	1.53256	45.94	0.0095	0.0116	H
533-459	FTM 8	1.53256	45.91	0.0077	0.0044	O
533-580	ZSL 1	1.53315	57.98	-0.0022	-0.0048	O
533-580	ZK 1	1.53315	57.98	-0.0008	-0.0009	S
533-581	ZnCl	1.53315	58.06	-0.0006	0.0019	H

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
534-515	CF4	1.53358	51.54	0.0012	0.0027	H
534-516	NSL 34	1.53358	51.56	-0.0025	-0.0068	O
534-555	ZSL 5	1.53375	55.53	-0.0026	-0.0174	O
540-510	NSL 31	1.54041	51.00	-0.0010	-0.0050	O
540-595	BAL 12	1.53996	59.45	-0.0010	-0.0098	O
540-597	BCL B39-59	1.53950	59.70			C
540-597	BaC2	1.53996	59.72	0.0039	-0.0051	H
540-597	BaK 2	1.53996	59.71	0.0004	-0.0089	S
541-472	FEL2	1.54072	47.20	-0.0004	0.0043	H
541-472	PBL 2	1.54072	47.23	-0.0019	0.0019	O
541-472	LLF 2	1.54072	47.17	-0.0010	0.0032	S
541-476	FeL B41-48	1.54099	47.62			C
542-734	PFC B42-73	1.54220	73.37			C
547-536	BAL 5	1.54739	53.55	-0.0006	-0.0022	O
547-536	BaLF 5	1.54739	53.63	0.0001	-0.0122	S
548-422	TiF 3	1.54765	42.20	0.0087	0.0052	S
548-457	FeL B48-46	1.54810	45.70			C
548-458	FEL1	1.54814	45.82	-0.0016	0.0014	H
548-458	PBL 1	1.54814	45.78	-0.0020	0.0006	O
548-458	LLF 1	1.54814	45.75	-0.0009	0.0025	S
548-535	FBL B48-53	1.54775	53.50			C
548-628	BAL 21	1.54771	62.84	-0.0045	0.0121	O
548-742	Ultran [®] 30	1.54830	74.24	0.0224	-0.0901	S
549-454	FEL7	1.54883	45.44	-0.0015	0.0025	H
549-454	LLF 7	1.54883	45.41	-0.0016	0.0054	S
549-456	PBL 7	1.54869	45.56	-0.0027	0.0022	O
551-495	SbF1	1.55115	49.52	-0.0012	0.0186	H
551-496	KzF N1	1.55115	49.64	-0.0035	0.0153	S
552-634	PCD3	1.55232	63.42	0.0042	0.0100	H
552-635	PSK 3	1.55232	63.46	-0.0005	0.0118	S
552-638	BAL 23	1.55232	63.75	-0.0036	0.0184	O
557-486	BAM 1	1.55690	48.55	0.0001	-0.0102	O
557-586	BaC5	1.55671	58.56	-0.0006	-0.0093	H
557-587	BAL 15	1.55671	58.69	-0.0023	0.0016	O
557-587	BaK 5	1.55671	58.65	0.0003	-0.0127	S
558-525	FBL5 B58-5	1.55750	52.50			C
558-542	KzFS N2	1.55836	54.16	-0.0100	0.0601	S
558-673	PSK 50	1.55753	67.28	0.0048	-0.0052	S
560-470	PBL 3	1.56013	46.99	-0.0011	-0.0030	O
560-471	FEL3	1.56013	47.09	0.0009	0.0018	H

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
560-612	BAL 50	1.55963	61.17	-0.0033	0.0058	O
561-452	FEL4	1.56138	45.23	0.0001	0.0024	H
561-452	PBL 4	1.56138	45.19	-0.0007	0.0011	O
564-438	PBL 28	1.56444	43.79	-0.0012	-0.0008	O
564-438	LF 8	1.56444	43.75	-0.0006	0.0000	S
564-607	BAL 41	1.56384	60.70	-0.0029	0.0045	O
564-608	BaCD11	1.56384	60.83	0.0020	-0.0024	H
564-608	SK 11	1.56384	60.80	-0.0004	-0.0024	S
564-609	BCD B64-61	1.56400	60.90			C
565-530	ADF1	1.56500	52.96	-0.0084	0.0458	H
567-428	FL6	1.56732	42.84	-0.0004	0.0067	H
567-428	PBL 26	1.56732	42.83	-0.0007	-0.0012	O
568-580	BaK 50	1.56774	57.99	-0.0026	0.0025	S
569-560	BCL B69-56	1.56880	56.00			C
569-560	BaC4	1.56883	56.04	0.0011	0.0003	H
569-561	BaK 4	1.56883	56.13	-0.0007	-0.0068	S
569-563	BAL 14	1.56883	56.34	-0.0025	-0.0047	O
569-631	PCD2	1.56873	63.10	0.0014	0.0145	H
569-631	PSK 2	1.56873	63.08	-0.0009	0.0119	S
569-632	BAL 22	1.56873	63.16	-0.0029	0.0120	O
570-481	SSL 4	1.57041	48.13	-0.0030	0.0108	O
570-493	BAM 2	1.56965	49.33	-0.0013	-0.0141	O
570-494	BaF2	1.56965	49.39	0.0030	-0.0165	H
571-508	BAL 2	1.57099	50.80	-0.0007	-0.0129	O
571-509	BaFL2	1.57099	50.86	0.0015	-0.0139	H
571-530	BaFL3	1.57135	52.98	0.0002	-0.0150	H
571-530	BAL 3	1.57135	52.97	-0.0007	-0.0207	O
573-426	FL1	1.57309	42.59	0.0010	0.0019	H
573-426	PBL 21	1.57309	42.58	-0.0007	-0.0021	O
573-575	BCL B72-57	1.57250	57.50			C
573-575	BaC1	1.57250	57.49	0.0043	-0.0110	H
573-576	BaK 1	1.57250	57.55	0.0002	-0.0167	S
573-578	BAL 11	1.57250	57.76	-0.0018	-0.0021	O
574-564	BaC6	1.57444	56.36	0.0026	-0.0029	H
574-565	BAL 16	1.57444	56.48	-0.0024	-0.0083	O
575-415	FL7	1.57501	41.51	-0.0009	0.0001	H
575-415	PBL 27	1.57501	41.49	-0.0015	0.0011	O
575-415	LF 7	1.57501	41.49	-0.0007	0.0024	S
576-414	FL57	1.57616	41.39	-0.0002	0.0034	H
578-415	PBL 24	1.57845	41.52	0.0001	0.0006	O

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
578-417	FL4	1.57845	41.71	-0.0024	0.0064	H
580-537	BaFL4	1.57957	53.71	0.0000	-0.0113	H
580-537	BAL 4	1.57957	53.71	-0.0014	-0.0100	O
580-537	BaLF 4	1.57957	53.71	-0.0009	-0.0075	S
581-408	FL B81-41	1.58140	40.80			C
581-408	PBL 25	1.58144	40.75	-0.0005	0.0029	O
581-409	FL5	1.58144	40.89	-0.0002	0.0014	H
581-409	LF 5	1.58144	40.85	-0.0003	-0.0006	S
582-420	FL3	1.58215	42.03	0.0007	0.0027	H
582-421	PBL 23	1.58215	42.09	-0.0012	0.0024	O
583-464	BAM 3	1.58267	46.39	-0.0004	-0.0066	O
583-465	BaF3	1.58267	46.46	-0.0032	-0.0160	H
583-465	BaF 3	1.58267	46.47	0.0001	-0.0095	S
583-594	BAL 42	1.58313	59.38	-0.0022	-0.0063	O
583-595	BaCD12	1.58313	59.46	-0.0002	-0.0018	H
583-595	SK 12	1.58313	59.45	-0.0003	-0.0078	S
586-646	PSK 54	1.58599	64.60	0.0077	-0.0440	S
589-410	FL2	1.58921	40.96	-0.0008	0.0032	H
589-411	PBL 22	1.58921	41.08	-0.0005	-0.0026	O
589-485	BaF6	1.58900	48.54	0.0024	-0.0063	H
589-485	BaF N6	1.58900	48.45	0.0002	-0.0015	S
589-486	BAM 6	1.58900	48.59	-0.0010	-0.0115	O
589-512	BAL 7	1.58875	51.18	-0.0012	-0.0180	O
589-514	BaLF 50	1.58893	51.37	-0.0011	-0.0051	S
589-529	BaFL6	1.58904	52.93	-0.0034	-0.0084	H
589-532	BAL 6	1.58904	53.20	-0.0016	-0.0135	O
589-612	BCD B89-61	1.58900	61.20			C
589-612	BAL 35	1.58913	61.18	-0.0021	0.0017	O
589-613	BaCD5	1.58913	61.25	0.0023	0.0046	H
589-613	SK 5	1.58913	61.27	-0.0007	0.0008	S
592-485	KzFS 6	1.59196	48.51	-0.0088	0.0460	S
592-583	BaCD13	1.59181	58.31	0.0026	-0.0104	H
592-583	SK 13	1.59181	58.30	0.0001	-0.0140	S
593-353	FTM 16	1.59270	35.30	0.0090	0.0061	O
593-355	FF5	1.59270	35.45	0.0081	0.0078	H
594-355	TiF N5	1.59355	35.51	0.0096	0.0069	S
596-392	F8	1.59551	39.22	-0.0002	0.0024	H
596-392	PBM 8	1.59551	39.21	-0.0009	-0.0007	O
596-392	F 8	1.59551	39.18	-0.0006	0.0022	S
597-553	ADC2	1.59700	55.29	0.0077	-0.0020	H

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
599-469	KzFS N9	1.59856	46.90	-0.0091	0.0452	S
600-425	BaFD16	1.60000	42.46	0.0026	0.0174	H
601-382	F 14	1.60140	38.23	-0.0006	0.0022	S
603-380	F5	1.60342	38.01	-0.0010	-0.0010	H
603-380	PBM 5	1.60342	38.01	-0.0004	-0.0007	O
603-380	F 5	1.60342	38.03	-0.0003	0.0017	S
603-423	BAM 25	1.60323	42.32	0.0000	-0.0081	O
603-424	BaFD5	1.60323	42.39	0.0014	-0.0026	H
603-606	SK 14	1.60311	60.60	-0.0003	-0.0033	S
603-607	BaCD14	1.60311	60.69	0.0020	0.0024	H
603-607	BSM 14	1.60311	60.70	-0.0024	-0.0024	O
603-654	PSK 52	1.60310	65.41	0.0049	-0.0183	S
603-655	PHM 53	1.60300	65.48	0.0046	-0.0273	O
604-381	FD C04-38	1.60350	38.10			C
604-536	SSK 51	1.60361	53.63	-0.0003	-0.0132	S
604-640	PCDS C04-6	1.60350	64.00			C
606-378	F 15	1.60565	37.83	-0.0002	0.0013	S
606-379	F15	1.60565	37.90	-0.0007	0.0022	H
606-437	BAM 4	1.60562	43.72	0.0004	-0.0128	O
606-439	FB C06-44	1.60560	43.90			C
606-439	BaF4	1.60562	43.88	0.0014	-0.0083	H
606-439	BaF 4	1.60562	43.93	0.0006	-0.0118	S
607-403	BAM 23	1.60717	40.26	0.0008	-0.0040	O
607-404	BaFD3	1.60717	40.36	0.0010	-0.0059	H
607-492	BAM 5	1.60729	49.19	-0.0010	-0.0112	O
607-493	BaF5	1.60729	49.34	0.0032	-0.0105	H
607-550	BaCD50	1.60700	54.98	0.0000	0.0013	H
607-567	BaCD2	1.60738	56.71	0.0019	-0.0132	H
607-567	SK 2	1.60738	56.65	-0.0008	-0.0162	S
607-568	BSM 2	1.60738	56.81	-0.0010	-0.0179	O
607-594	BSM 7	1.60729	59.38	-0.0017	-0.0101	O
607-595	BaCD7	1.60729	59.47	0.0018	-0.0083	H
607-595	SK 7	1.60729	59.46	-0.0004	-0.0087	S
608-462	BaF7	1.60801	46.21	0.0015	-0.0085	H
608-566	BCD C07-57	1.60750	56.60			C
609-464	BaF 52	1.60859	46.44	0.0009	-0.0122	S
609-589	BaCD3	1.60881	58.86	0.0025	-0.0098	H
609-589	SK 3	1.60881	58.92	-0.0003	-0.0085	S
609-590	BCD C09-59	1.60875	59.00			C
609-590	BSM 3	1.60881	58.95	-0.0021	-0.0098	O

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
610-565	BSM 1	1.61025	56.51	-0.0021	-0.0174	O
610-567	BaCD1	1.61025	56.65	0.0009	-0.0149	H
610-567	SK 1	1.61025	56.71	-0.0005	-0.0145	S
611-558	BaCD8	1.61117	55.77	0.0005	-0.0104	H
611-559	BSM 8	1.61117	55.92	-0.0021	-0.0135	O
611-559	SK 8	1.61117	55.92	-0.0008	-0.0134	S
613-370	F3	1.61293	36.96	-0.0001	0.0021	H
613-370	PBM 3	1.61293	37.01	0.0000	0.0005	O
613-370	F 3	1.61293	37.04	0.0000	0.0012	S
613-438	BPM 4	1.61340	43.84	-0.0080	0.0352	O
613-442	BPM 51	1.61340	44.24	-0.0084	0.0376	O
613-443	FSB C13-44	1.61340	44.30			C
613-443	ADF40	1.61340	44.29	-0.0079	0.0421	H
613-443	KzFS 1	1.61310	44.34	-0.0112	0.0546	S
613-443	KzFS N4	1.61340	44.29	-0.0087	0.0406	S
613-444	ADF10	1.61310	44.36	-0.0100	0.0564	H
613-449	ADF4	1.61250	44.87	-0.0056	0.0446	H
613-574	BSM 19	1.61342	57.41	-0.0012	-0.0094	O
613-585	BCD C13-58	1.61270	58.50			C
613-586	BaCD4	1.61272	58.58	0.0019	-0.0078	H
613-586	SK 4	1.61272	58.63	-0.0004	-0.0082	S
613-588	BSM 4	1.61272	58.75	-0.0014	-0.0130	O
614-550	BSM 9	1.61405	54.95	-0.0020	-0.0145	O
614-551	BaCD9	1.61405	55.12	0.0014	-0.0101	H
614-564	BCD C13-56	1.61350	56.40			C
614-564	BaCD6	1.61375	56.38	0.0002	-0.0141	H
614-564	BSM 6	1.61375	56.36	-0.0021	-0.0154	O
614-564	SK 6	1.61375	56.40	-0.0007	-0.0169	S
615-512	BaCED3	1.61484	51.15	0.0004	-0.0152	H
615-512	BSM 23	1.61484	51.17	-0.0012	-0.0120	O
615-512	SSK 3	1.61484	51.16	-0.0002	-0.0142	S
616-444	FSB C16-44	1.61600	44.40			C
617-310	TiF 6	1.61650	30.97	0.0306	-0.0081	S
617-366	F4	1.61659	36.61	0.0001	0.0011	H
617-366	PBM 4	1.61659	36.63	0.0005	-0.0005	O
617-366	F 4	1.61659	36.63	0.0003	0.0004	S
617-539	BaCED1	1.61720	53.94	0.0011	-0.0131	H
617-539	SSK 1	1.61720	53.91	-0.0008	-0.0128	S
617-540	BSM 21	1.61720	54.04	-0.0026	-0.0163	O
617-628	PCD5	1.61700	62.83	0.0059	-0.0104	H

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
617-628	PHM 51	1.61700	62.80	0.0031	-0.0127	O
618-498	BSM 28	1.61772	49.83	0.0005	-0.0028	O
618-498	SSK N8	1.61772	49.77	0.0004	-0.0040	S
618-526	SSK 50	1.61795	52.61	-0.0002	-0.0139	S
618-551	BCD C17-55	1.61760	55.10			C
618-551	BSM 24	1.61765	55.05	-0.0021	-0.0135	O
618-551	SSK 4A	1.61765	55.14	-0.0008	-0.0149	S
618-552	BaCED4	1.61765	55.16	0.0005	-0.0115	H
618-634	PCD4	1.61800	63.39	0.0059	-0.0154	H
618-634	PHM 52	1.61800	63.39	0.0052	-0.0346	O
620-363	FD C20-36	1.62000	36.30			C
620-363	F2	1.62004	36.30	0.0000	0.0000	H
620-363	PBM 2	1.62004	36.26	0.0000	0.0000	O
620-364	F 2	1.62004	36.37	0.0002	0.0007	S
620-381	F9	1.62045	38.09	0.0001	0.0081	H
620-381	PBM 9	1.62045	38.12	-0.0010	0.0044	O
620-381	F 9	1.62045	38.08	0.0009	0.0035	S
620-496	BSM 29	1.62012	49.55	-0.0005	-0.0080	O
620-498	BaCED9	1.62012	49.82	0.0015	-0.0040	H
620-601	SK 55	1.62041	60.12	-0.0033	0.0162	S
620-603	BCD C20-60	1.62040	60.30			C
620-603	BaCD16	1.62041	60.34	-0.0003	0.0105	H
620-603	BSM 16	1.62041	60.28	-0.0012	-0.0022	O
620-603	BSM 16C	1.62041	60.26	-0.0014	-0.0017	O
620-603	SK 16	1.62041	60.32	-0.0011	0.0016	S
620-622	ADC1	1.62000	62.19	0.0088	-0.0394	H
620-622	ATC1	1.62000	62.19	0.0088	-0.0394	H
620-635	PCD53	1.62014	63.52	0.0059	-0.0265	H
620-635	PSK 53A	1.62014	63.48	0.0053	-0.0274	S
621-359	PBM 11	1.62096	35.88	0.0058	0.0136	O
621-360	F11	1.62096	35.95	0.0054	0.0141	H
621-362	F N11	1.62096	36.18	0.0060	0.0139	S
621-603	SK 51	1.62090	60.31	0.0032	-0.0205	S
622-360	F 13	1.62237	36.04	0.0008	-0.0006	S
622-531	BaCED2	1.62230	53.11	0.0005	-0.0105	H
622-532	BSM 22	1.62230	53.20	-0.0018	-0.0157	O
622-532	SSK 2	1.62230	53.15	-0.0006	-0.0135	S
623-531	BCDD C23-5	1.62250	53.10			C
623-569	BCD C23-57	1.62280	56.90			C
623-569	BaCD10	1.62280	56.91	0.0008	-0.0144	H

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
623-569	SK 10	1.62280	56.90	-0.0003	-0.0153	S
623-571	BSM 10	1.62280	57.06	-0.0025	-0.0105	O
623-581	BCD C23-58	1.62300	58.10			C
623-581	BaCD15	1.62299	58.12	0.0008	-0.0068	H
623-581	SK 15	1.62299	58.06	-0.0008	-0.0090	S
623-582	BSM 15	1.62299	58.15	-0.0017	-0.0094	O
624-365	PBM 10	1.62364	36.54	0.0002	0.0011	O
624-469	FB C24-47	1.62400	46.90			C
624-470	BaF 8	1.62374	47.00	0.0001	-0.0130	S
624-471	BaF8	1.62374	47.05	0.0005	-0.0109	H
624-471	BAM 8	1.62374	47.10	-0.0019	-0.0041	O
625-356	F7	1.62536	35.58	0.0007	-0.0022	H
625-356	F 7	1.62536	35.56	0.0016	-0.0031	S
626-356	FD C26-36	1.62590	35.60			C
626-357	F1	1.62588	35.74	-0.0016	0.0017	H
626-357	PBM 1	1.62588	35.70	0.0005	0.0001	O
626-357	F 1	1.62588	35.70	0.0007	-0.0001	S
626-390	BaSF 1	1.62606	38.96	0.0014	-0.0077	S
626-391	BaFD1	1.62606	39.09	0.0000	-0.0135	H
626-392	BAM 21	1.62606	39.21	0.0001	-0.0008	O
636-353	F6	1.63636	35.34	-0.0008	-0.0013	H
636-353	F 6	1.63636	35.34	0.0009	-0.0013	S
636-354	PBM 6	1.63636	35.38	0.0009	-0.0012	O
637-353	FD C37-35	1.63650	35.30			C
639-449	BAM 12	1.63930	44.88	-0.0001	-0.0125	O
639-451	BaF12	1.63930	45.07	0.0016	-0.0108	H
639-554	BSM 18	1.63854	55.38	-0.0019	-0.0048	O
639-554	SK 18A	1.63854	55.42	-0.0013	-0.0047	S
639-555	BCD C39-56	1.63850	55.50			C
639-555	BaCD18	1.63854	55.45	-0.0011	-0.0057	H
640-345	PBM 27	1.63980	34.48	0.0018	-0.0012	O
640-346	FD7	1.63980	34.57	0.0009	0.0020	H
640-598	LaK L21	1.64048	59.75	-0.0059	0.0365	S
640-601	BSM 81	1.64000	60.09	-0.0077	0.0348	O
640-601	LaK 21	1.64049	60.10	-0.0017	0.0052	S
640-602	LaCL6	1.64000	60.15	-0.0023	0.0142	H
640-602	LaCL60	1.64000	60.20	-0.0039	0.0334	H
641-568	LaCL1	1.64085	56.83	-0.0028	0.0141	H
641-569	BSM 93	1.64100	56.93	-0.0024	-0.0015	O
641-601	BCS C41-60	1.64050	60.10			C

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
642-580	LaK N6	1.64250	57.96	-0.0014	-0.0078	S
643-479	BaF9	1.64328	47.94	0.0007	-0.0074	H
643-479	BAM 9	1.64328	47.85	0.0000	-0.0122	O
643-480	BaF 9	1.64328	47.96	-0.0003	-0.0129	S
643-580	BCS C42-58	1.64250	58.04			C
643-580	LaC6	1.64250	57.96	-0.0030	0.0144	H
643-584	BSM 36	1.64250	58.37	-0.0055	0.0148	O
645-408	ADF355	1.64450	40.76	-0.0053	0.0363	H
645-408	BPH 35	1.64450	40.82	-0.0067	0.0292	O
646-341	SF 16	1.64611	34.05	0.0016	-0.0008	S
648-338	FD2	1.64769	33.84	0.0014	0.0006	H
648-338	FD12	1.64831	33.79	0.0016	0.0008	H
648-338	PBM 32	1.64831	33.83	0.0017	-0.0013	O
648-338	PBM 22	1.64769	33.80	0.0015	-0.0022	O
648-338	SF 12	1.64831	33.84	0.0021	0.0011	S
648-339	FDD C48-34	1.64800	33.90			C
648-339	SF 2	1.64769	33.85	0.0017	-0.0010	S
648-462	FB C48-46	1.64775	46.20			C
649-530	BaCED20	1.64850	53.03	0.0012	-0.0123	H
649-530	BSM 71	1.64850	53.03	-0.0018	-0.0112	O
650-392	FBD C51-39	1.65020	39.20			C
650-392	BaSF 10	1.65016	39.15	0.0015	-0.0114	S
650-393	BaFD10	1.65016	39.34	0.0005	-0.0094	H
650-394	BAH 30	1.65016	39.39	0.0015	-0.0141	O
650-557	LaCL2	1.65020	55.74	-0.0009	-0.0036	H
651-380	ATF2	1.65052	38.04	0.0193	-0.0214	H
651-383	BaFD4	1.65128	38.31	0.0013	-0.0055	H
651-383	BAH 24	1.65128	38.25	0.0024	-0.0100	O
651-419	BaSF 57	1.65147	41.90	0.0006	-0.0027	S
651-559	BCS C51-56	1.65113	55.90			C
651-559	LaK N22	1.65113	55.89	-0.0031	-0.0058	S
651-562	LAL 54	1.65100	56.15	-0.0023	-0.0058	O
652-449	BaF23	1.65224	44.92	0.0009	-0.0069	H
652-449	BaF 51	1.65224	44.93	-0.0009	-0.0080	S
652-584	LaC7	1.65160	58.40	-0.0035	0.0075	H
652-585	BCS C52-58	1.65160	58.50			C
652-585	LAL 7	1.65160	58.52	-0.0041	0.0052	O
652-585	LaK N7	1.65160	58.52	-0.0021	0.0010	S
654-336	PBM 29	1.65446	33.62	0.0014	-0.0006	O
654-337	FD9	1.65446	33.72	0.0004	0.0002	H

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
654-337	SF 9	1.65446	33.65	0.0014	0.0003	S
654-396	ADF50	1.65412	39.62	-0.0071	0.0384	H
654-396	KzFS N5	1.65412	39.63	-0.0071	0.0371	S
654-397	BPH 5	1.65412	39.70	-0.0069	0.0345	O
654-447	ATF4	1.65376	44.72	0.0041	-0.0201	H
656-401	FBDS C55-4	1.65550	40.10			C
657-367	BaSF 56	1.65715	36.74	0.0021	-0.0072	S
658-509	BaCED5	1.65844	50.85	0.0007	-0.0093	H
658-509	BSM 25	1.65844	50.86	-0.0038	-0.0063	O
658-509	SSK N5	1.65844	50.88	-0.0007	-0.0090	S
658-534	LAL 55	1.65830	53.44	-0.0022	-0.0094	O
658-572	BCS C57-57	1.65750	57.20			C
658-573	LaC11	1.65830	57.26	-0.0036	0.0113	H
658-573	LAL 11	1.65830	57.33	-0.0037	-0.0049	O
658-573	LaK 11	1.65830	57.26	-0.0032	0.0041	S
659-510	BCDD C58-5	1.65850	51.00			C
664-358	BAH 22	1.66446	35.81	0.0013	-0.0054	O
664-358	BaSF 2	1.66446	35.83	0.0026	-0.0071	S
664-359	BaFD2	1.66446	35.89	0.0015	-0.0102	H
664-489	BaF21	1.66422	48.94	0.0027	-0.0193	H
665-534	LaCL3	1.66480	53.40	-0.0011	-0.0008	H
667-330	SF 19	1.66680	33.01	0.0020	0.0004	S
667-331	FDD C67-33	1.66700	33.10			C
667-331	FD19	1.66680	33.06	0.0013	-0.0038	H
667-331	PBM 39	1.66680	33.05	0.0014	0.0011	O
667-483	BaF11	1.66672	48.30	0.0021	-0.0119	H
667-483	BAH 11	1.66672	48.32	-0.0013	-0.0128	O
667-484	FB C67-48	1.66670	48.40			C
667-484	BaF N11	1.66672	48.42	-0.0005	-0.0080	S
668-419	BaFD6	1.66755	41.93	0.0021	-0.0076	H
668-419	BAH 26	1.66755	41.93	0.0011	-0.0105	O
668-419	BaSF 6	1.66755	41.93	0.0013	-0.0095	S
669-449	BaF13	1.66892	44.91	0.0022	-0.0050	H
669-450	BAH 13	1.66892	44.99	0.0005	-0.0093	O
669-450	BaF 13	1.66892	44.96	0.0008	-0.0093	S
669-574	LaK 23	1.66882	57.38	-0.0026	-0.0002	S
670-392	BaSF 12	1.66998	39.20	0.0022	-0.0083	S
670-393	BAH 32	1.66998	39.28	-0.0004	-0.0071	O
670-471	FB C70-47	1.67000	47.10			C
670-471	BaF N10	1.67003	47.11	-0.0001	-0.0071	S

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
670-472	BaF10	1.67003	47.20	0.0007	0.0035	H
670-473	BAH 10	1.67003	47.25	-0.0006	-0.0095	O
670-516	LAL 53	1.67000	51.63	-0.0017	-0.0127	O
670-517	LaCL4	1.66960	51.66	-0.0047	-0.0002	H
670-573	LaCL7	1.67000	57.31	-0.0049	0.0206	H
670-573	LAL 52	1.67000	57.33	-0.0049	0.0044	O
673-321	PBM 25	1.67270	32.10	0.0027	-0.0025	O
673-322	FD5	1.67270	32.17	0.0014	-0.0019	H
673-322	SF 5	1.67270	32.21	0.0023	-0.0010	S
673-323	FeD C72-32	1.67270	32.30			C
677-375	ADF405	1.67650	37.48	-0.0050	0.0251	H
677-375	BPH 40	1.67650	37.54	-0.0045	0.0207	O
678-506	LaCL9	1.67790	50.55	-0.0018	-0.0115	H
678-507	LAL 56	1.67790	50.72	-0.0028	-0.0136	O
678-534	LaCL8	1.67790	53.42	-0.0009	-0.0053	H
678-534	LAL 57	1.67790	53.36	-0.0049	-0.0100	O
678-549	LaK L12	1.67790	54.92	-0.0062	0.0143	S
678-552	LaK N12	1.67790	55.20	-0.0024	-0.0126	S
678-553	LAL 12	1.67790	55.33	-0.0055	-0.0063	O
678-555	BCS C78-56	1.67800	55.50			C
678-555	LaC12	1.67790	55.52	-0.0032	-0.0021	H
681-374	KzFS 7A	1.68064	37.39	-0.0054	0.0300	S
682-482	LaFL20	1.68248	48.20	-0.0012	-0.0121	H
682-482	LaF 20	1.68248	48.20	-0.0021	-0.0096	S
683-445	BaF 50	1.68273	44.50	-0.0020	-0.0075	S
683-447	BaF22	1.68250	44.67	0.0040	-0.0011	H
683-447	BAH 51	1.68250	44.65	-0.0005	-0.0048	O
686-439	BaF20	1.68578	43.90	-0.0002	-0.0065	H
686-440	BAH 53	1.68578	44.00	0.0010	-0.0034	O
686-492	LAM 56	1.68600	49.16	-0.0023	-0.0116	O
686-494	LaFL1	1.68600	49.35	0.0008	-0.0215	H
687-429	ADF8	1.68650	42.86	-0.0055	0.0300	H
689-311	PBM 28	1.68893	31.08	0.0031	-0.0014	O
689-312	FeD C89-31	1.68900	31.20			C
689-312	FD8	1.68893	31.16	0.0022	-0.0005	H
689-312	SF 8	1.68893	31.18	0.0029	-0.0013	S
689-495	LaFL23	1.68900	49.48	-0.0015	-0.0103	H
689-496	FBS C90-50	1.68900	49.60			C
689-497	LaF N23	1.68900	49.71	-0.0015	-0.0152	S
691-547	LaC9	1.69100	54.70	-0.0079	0.0225	H

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
691-547	LaK 9	1.69100	54.71	-0.0071	0.0223	S
691-548	BCS C90-55	1.69100	54.80			C
691-548	LAL 9	1.69100	54.84	-0.0077	0.0204	O
694-508	LaCL5	1.69350	50.79	-0.0049	-0.0152	H
694-508	LAL 58	1.69350	50.81	-0.0042	-0.0119	O
694-532	LAL 13	1.69350	53.23	-0.0082	0.0157	O
694-533	LaC13	1.69350	53.34	-0.0067	0.0206	H
694-533	LaK N13	1.69350	53.33	-0.0030	-0.0139	S
695-422	BAH 54	1.69500	42.16	-0.0039	0.0068	O
696-365	FBDS C97-3	1.69600	36.50			C
697-485	LaFL2	1.69700	48.51	-0.0023	-0.0121	H
697-485	LAM 59	1.69700	48.51	-0.0025	-0.0110	O
697-554	BCS C97-55	1.69700	55.40			C
697-554	LaK N14	1.69680	55.41	-0.0079	0.0273	S
697-555	LaC14	1.69680	55.46	-0.0059	0.0282	H
697-555	LAL 14	1.69680	55.53	-0.0085	0.0263	O
697-561	LaC15	1.69680	56.11	-0.0057	0.0271	H
697-562	BCS C97-56	1.69700	56.20			C
697-564	LaK 31	1.69673	56.42	-0.0074	0.0287	S
697-565	LAL 64	1.69680	56.47	-0.0085	0.0279	O
698-386	BaSF 13	1.69761	38.57	0.0033	-0.0045	S
699-301	FeD C99-30	1.69900	30.10			C
699-301	FD15	1.69895	30.05	0.0052	0.0007	H
699-301	PBM 35	1.69895	30.12	0.0063	-0.0023	O
699-301	SF 15	1.69895	30.07	0.0066	-0.0014	S
700-349	BaFD14	1.69968	34.92	0.0035	-0.0019	H
700-478	LaFL3	1.70030	47.84	-0.0028	-0.0119	H
700-481	LAM 51	1.70000	48.08	-0.0016	-0.0062	O
702-401	BAH 71	1.70200	40.10	0.0003	-0.0101	O
702-402	BaFD15	1.70200	40.20	-0.0015	0.0097	H
702-410	BaSF 52	1.70181	41.01	-0.0035	0.0164	S
702-411	FBD D01-41	1.70180	41.10			C
702-412	BaFD7	1.70154	41.15	0.0027	-0.0092	H
702-412	BAH 27	1.70154	41.24	0.0024	-0.0047	O
702-412	BAH 77	1.70154	41.21	0.0003	-0.0032	O
704-394	BaSF 64A	1.70400	39.38	-0.0006	0.0069	S
706-303	SF 64A	1.70585	30.30	0.0099	0.0066	S
713-432	LAM 57	1.71285	43.19	-0.0005	-0.0048	O
713-433	LaFL4	1.71270	43.29	-0.0046	0.0212	H
713-538	BCS D13-54	1.71300	53.80			C

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
713-538	LAL 8	1.71300	53.84	-0.0089	0.0190	O
713-538	LaK 8	1.71300	53.83	-0.0083	0.0266	S
713-539	LaC8	1.71300	53.94	-0.0070	0.0228	H
717-295	FeD D17-29	1.71740	29.50			C
717-295	FD1	1.71736	29.50	0.0034	0.0003	H
717-295	PBH 1	1.71736	29.51	0.0038	-0.0027	O
717-295	SF 1	1.71736	29.51	0.0042	-0.0018	S
717-479	LAM 3	1.71700	47.94	-0.0035	-0.0111	O
717-480	FBS D17-48	1.71700	48.00			C
717-480	LaF30	1.71700	47.98	-0.0041	0.0029	H
717-480	LaF3	1.71700	47.98	-0.0062	-0.0085	H
717-480	LaF 3	1.71700	47.96	-0.0028	-0.0054	S
719-335	ADF455	1.71852	33.46	-0.0030	0.0144	H
719-335	BPH 45	1.71850	33.52	-0.0009	0.0145	O
720-293	FD20	1.72022	29.30	0.0053	0.0005	H
720-293	PBH 51	1.72022	29.31	0.0045	-0.0038	O
720-346	KzFS 8	1.72047	34.61	-0.0037	0.0217	S
720-347	BPH 8	1.72047	34.72	-0.0032	0.0200	O
720-420	LaFL6	1.72000	42.02	0.0005	-0.0027	H
720-420	LAM 58	1.72000	41.98	-0.0003	-0.0051	O
720-437	LAM 52	1.72000	43.71	-0.0008	-0.0086	O
720-439	LaFL5	1.72000	43.90	-0.0052	-0.0046	H
720-460	LAM 61	1.72000	46.03	-0.0024	-0.0179	O
720-503	BCS D20-50	1.72000	50.30			C
720-503	LaC10	1.72000	50.34	-0.0075	0.0061	H
720-503	LAL 10	1.72000	50.25	-0.0069	-0.0009	O
720-504	LaK 10	1.72000	50.41	-0.0075	0.0240	S
722-292	FD18	1.72151	29.24	0.0031	-0.0004	H
722-292	PBH 18	1.72151	29.24	0.0046	-0.0033	O
722-293	SF 18	1.72151	29.25	0.0045	-0.0021	S
723-380	BaFD8	1.72342	37.99	0.0020	-0.0033	H
723-380	BAH 28	1.72342	37.95	0.0035	-0.0025	O
723-380	BAH 78	1.72342	38.03	-0.0005	-0.0044	O
724-380	FBD D23-38	1.72350	38.00			C
724-381	BaSF 51	1.72373	38.11	-0.0013	0.0013	S
726-534	TaC1	1.72600	53.42	-0.0094	0.0182	H
726-536	LAL 60	1.72600	53.57	-0.0091	0.0245	O
728-283	FD10	1.72825	28.32	0.0073	-0.0001	H
728-284	FeD D28-28	1.72830	28.40			C
728-284	SF 10	1.72825	28.41	0.0085	-0.0012	S

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
728-285	PBH 10	1.72825	28.46	0.0088	-0.0020	O
728-287	FDS2	1.72830	28.67	0.0048	-0.0012	H
728-287	SF 53	1.72830	28.69	0.0061	-0.0026	S
729-547	TaC8	1.72916	54.67	-0.0046	0.0259	H
729-547	LAL 18	1.72916	54.68	-0.0082	0.0208	O
732-284	FeD D31-28	1.73151	28.41			C
734-265	FDS D34-26	1.73400	26.50			C
734-511	TaC4	1.73400	51.05	-0.0064	0.0209	H
734-514	BCS D34-51	1.73350	51.40			C
734-515	LAL 59	1.73400	51.49	-0.0098	0.0173	O
734-518	LaK 16A	1.73350	51.78	-0.0087	0.0208	S
735-411	LAM 8	1.73520	41.08	-0.0020	-0.0042	O
735-416	LaF N8	1.73520	41.59	-0.0029	0.0072	S
735-498	NbF3	1.73500	49.79	-0.0089	0.0189	H
735-498	LAM 53	1.73500	49.76	-0.0093	0.0241	O
736-322	BaSF 54	1.73627	32.15	0.0059	-0.0040	S
740-281	FeD D40-28	1.74000	28.10			C
740-282	FD3	1.74000	28.24	0.0041	-0.0027	H
740-282	SF 3	1.74000	28.20	0.0056	-0.0032	S
740-283	PBH 3	1.74000	28.29	0.0060	-0.0052	O
740-317	ADF505	1.74000	31.66	-0.0009	0.0096	H
740-317	BPH 50	1.74000	31.71	0.0000	0.0080	O
741-276	SF 13	1.74077	27.60	0.0109	-0.0021	S
741-278	FD13	1.74077	27.76	0.0099	-0.0019	H
741-278	PBH 13	1.74077	27.79	0.0097	-0.0040	O
741-281	SF 54	1.74080	28.09	0.0068	-0.0025	S
741-526	TaC2	1.74100	52.60	-0.0092	0.0171	H
741-527	LAL 61	1.74100	52.65	-0.0092	0.0223	O
743-492	NbF1	1.74330	49.22	-0.0103	0.0142	H
743-493	LAM 60	1.74320	49.31	-0.0094	0.0184	O
744-447	LaF 2	1.74400	44.72	-0.0027	-0.0075	S
744-448	FBS D44-45	1.74400	44.80			C
744-448	LAM 2	1.74400	44.79	-0.0036	-0.0066	O
744-449	LaF20	1.74400	44.87	-0.0052	0.0092	H
744-449	LaF2	1.74400	44.90	-0.0045	-0.0098	H
744-508	LaK 28	1.74429	50.77	-0.0085	0.0189	S
748-277	SF 63	1.74840	27.71	0.0074	-0.0031	S
750-350	LaF7	1.74950	35.04	-0.0043	-0.0065	H
750-350	LAM 65	1.74950	34.96	0.0031	-0.0106	O
750-350	LaF N7	1.74950	34.95	-0.0025	0.0174	S

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
750-353	LAM 7	1.74950	35.27	0.0032	-0.0101	O
751-277	FD41	1.75084	27.69	0.0034	-0.0038	H
751-277	PBH 54	1.75084	27.69	0.0067	-0.0040	O
754-524	LaK 33	1.75398	52.43	-0.0083	0.0182	S
755-272	SF L4	1.75520	27.21	0.0123	0.0043	S
755-275	FD4	1.75520	27.53	0.0042	-0.0017	H
755-275	PBH 4	1.75520	27.51	0.0074	-0.0044	O
755-276	FeD D56-27	1.75520	27.60			C
755-276	SF 4	1.75520	27.58	0.0062	-0.0032	S
755-523	TaC6	1.75500	52.32	-0.0068	0.0195	H
755-523	YGH 51	1.75500	52.33	-0.0088	0.0188	O
756-251	TPH 55	1.75550	25.07	0.0197	0.0044	O
757-317	LaF 11A	1.75693	31.70	0.0002	0.0078	S
757-318	NbFD9	1.75690	31.80	0.0001	0.0040	H
757-363	NbFD6	1.75670	36.34	0.0015	-0.0119	H
757-477	NbF2	1.75700	47.71	-0.0076	0.0120	H
757-478	LAM 54	1.75700	47.82	-0.0076	0.0188	O
757-478	LaF N24	1.75719	47.81	-0.0079	0.0217	S
762-265	TIH 14	1.76182	26.52	0.0151	0.0032	O
762-265	SF 14	1.76182	26.53	0.0134	-0.0037	S
762-266	FD140	1.76182	26.61	0.0118	0.0060	H
762-266	FD14	1.76182	26.55	0.0111	-0.0047	H
762-266	PBH 14	1.76182	26.55	0.0125	-0.0050	O
762-269	FeD D62-27	1.76180	26.90			C
762-269	FDS5	1.76180	26.91	0.0064	-0.0037	H
762-270	SF 55	1.76180	26.95	0.0084	-0.0039	S
762-271	PBH 25	1.76180	27.11	0.0085	-0.0059	O
762-401	LAM 55	1.76200	40.10	-0.0006	-0.0014	O
762-403	NbFD5	1.76200	40.26	-0.0005	-0.0082	H
773-496	TaF1	1.77250	49.62	-0.0086	0.0212	H
773-496	LAH 66	1.77250	49.60	-0.0097	0.0164	O
773-496	LaF N28	1.77314	49.57	-0.0085	0.0126	S
776-378	LaF 13	1.77551	37.84	-0.0026	0.0056	S
782-372	LaF 22A	1.78179	37.20	0.0008	0.0005	S
783-361	NbFD7	1.78300	36.07	0.0015	0.0138	H
783-362	LAM 62	1.78300	36.15	0.0042	-0.0015	O
784-440	LaF N10	1.78443	43.95	-0.0068	0.0194	S
785-257	FD11	1.78472	25.70	0.0098	-0.0053	H
785-257	FD110	1.78472	25.72	0.0138	0.0036	H
785-257	TIH 11	1.78472	25.68	0.0160	0.0018	O

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
785-257	PBH 11	1.78472	25.71	0.0129	-0.0056	O
785-258	FeD D85-25	1.78475	25.80			C
785-258	SF 11	1.78472	25.76	0.0142	-0.0043	S
785-259	FeD D85-26	1.78500	25.90			C
785-261	FDS30	1.78470	26.10	0.0142	-0.0044	H
785-261	FDS3	1.78470	26.06	0.0090	-0.0033	H
785-261	SF 56A	1.78470	26.08	0.0098	-0.0042	S
785-261	SF L56	1.78470	26.08	0.0138	0.0048	S
785-262	PBH 23	1.78470	26.22	0.0092	-0.0066	O
785-263	TIH 23	1.78470	26.30	0.0142	0.0014	O
786-439	NbFD11	1.78590	43.93	-0.0081	0.0120	H
786-442	LAH 51	1.78590	44.19	-0.0069	0.0087	O
787-500	YGH 52	1.78650	50.00	-0.0093	0.0000	O
788-474	FBS D88-47	1.78800	47.40			C
788-474	LAH 64	1.78800	47.38	-0.0086	0.0157	O
788-475	TaF4	1.78800	47.49	-0.0090	0.0157	H
788-475	LaF N21	1.78831	47.47	-0.0079	0.0158	S
789-428	NbFD1	1.78900	42.76	-0.0073	0.0107	H
795-284	LaF 9	1.79504	28.39	0.0059	-0.0015	S
795-453	LAH 67	1.79500	45.29	-0.0083	0.0163	O
795-454	TaF2	1.79450	45.39	-0.0094	0.0114	H
796-432	FBS D96-43	1.79619	43.20			C
797-351	LaSF 36A	1.79712	35.08	0.0011	0.0008	S
797-411	NbFD2	1.79720	41.14	-0.0055	0.0083	H
800-422	LAH 52	1.79952	42.24	-0.0069	0.0192	O
800-423	NbFD12	1.79950	42.34	-0.0072	0.0077	H
800-456	FBS E00-46	1.80000	45.60			C
801-350	LAM 66	1.80100	34.97	0.0011	0.0129	O
802-443	NbFD14	1.80170	44.32	-0.0090	0.0138	H
803-304	LaSF 32	1.80349	30.40	0.0055	0.0094	S
803-464	FBS E03-47	1.80300	46.40			C
803-464	LaSF N30	1.80318	46.38	-0.0084	0.0099	S
803-467	LAH 62	1.80300	46.66	-0.0096	0.0153	O
804-396	LAH 63	1.80440	39.58	-0.0046	0.0116	O
804-465	TaF3	1.80420	46.50	-0.0067	0.0172	H
804-466	LAH 65	1.80400	46.58	-0.0090	0.0139	O
805-254	PBH 6	1.80518	25.43	0.0097	-0.0073	O
805-254	TIH 6	1.80518	25.43	0.0150	0.0017	O
805-254	SF L6	1.80518	25.39	0.0148	0.0032	S
805-254	SF 6	1.80518	25.43	0.0092	-0.0048	S

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
805-255	FeD E05-25	1.80510	25.50			C
805-255	FD6	1.80518	25.46	0.0075	-0.0049	H
805-255	FD60	1.80518	25.46	0.0132	0.0041	H
805-396	NbFD3	1.80450	39.64	-0.0056	0.0083	H
806-333	NbFD15	1.80610	33.27	0.0000	0.0046	H
806-342	LaSF 33	1.80596	34.24	0.0022	-0.0034	S
806-407	NbFD13	1.80610	40.73	-0.0079	0.0101	H
806-410	LAH 53	1.80610	40.95	-0.0055	0.0114	O
807-355	NbFD8	1.80740	35.54	0.0005	0.0038	H
808-406	LaSF 3	1.80801	40.61	-0.0046	0.0184	S
816-444	LAH 54	1.81554	44.36	-0.0084	0.0136	O
816-445	TaFD10	1.81550	44.54	-0.0074	0.0134	H
816-466	TaF5	1.81600	46.57	-0.0083	0.0084	H
816-466	LAH 59	1.81600	46.62	-0.0090	0.0031	O
822-375	LaSF 14A	1.82223	37.45	-0.0023	0.0174	S
829-365	NbFD40	1.82900	36.50	0.0002	-0.0061	H
834-372	LAH 60	1.83400	37.17	-0.0039	0.0106	O
834-373	NbFD10	1.83400	37.34	-0.0021	0.0077	H
835-427	LAH 55	1.83481	42.72	-0.0087	0.0115	O
835-430	TaFD5	1.83500	42.98	-0.0082	0.0122	H
836-302	FBS E37-30	1.83640	30.20			C
847-236	SF L57	1.84666	23.62	0.0177	0.0034	S
847-238	FDS9	1.84666	23.83	0.0096	-0.0067	H
847-238	FDS90	1.84666	23.78	0.0136	0.0029	H
847-238	TIH 53	1.84666	23.78	0.0181	0.0014	O
847-238	SF 57	1.84666	23.83	0.0123	-0.0065	S
847-239	PBH 53	1.84666	23.89	0.0134	-0.0086	O
848-430	TaFD6	1.84750	43.03	-0.0123	0.0029	H
850-322	TaFD9	1.85030	32.18	0.0030	0.0069	H
850-322	LaSF N9	1.85025	32.17	0.0036	-0.0031	S
850-323	LAH 71	1.85026	32.29	0.0027	0.0004	O
855-366	TaFD13	1.85540	36.56	-0.0044	0.0090	H
863-415	LAH 57	1.86300	41.53	-0.0074	0.0049	O
865-401	FBS E65-40	1.86500	40.10			C
874-353	LAH 75	1.87400	35.26	-0.0017	0.0075	O
878-381	LaSF N15	1.87800	38.07	-0.0063	0.0115	S
878-382	TaFD17	1.87800	38.20	-0.0048	0.0053	H
878-385	FBS E78-38	1.87800	38.48			C
881-410	LaSF N31	1.88067	41.01	-0.0084	-0.0046	S
883-408	TaFD30	1.88300	40.80	-0.0093	0.0016	H

Table 2.3. (cont.)

Code	Glass type	Refractive index	Abbe number	Deviation of the partial dispersion from the "normal line"		Producer
		n_d	v_d	$\Delta P_{g,F}$	$\Delta P_{C,t}$	
883-408	LAH 58	1.88300	40.78	-0.0087	0.0012	O
903-298	TaFD20	1.90315	29.84	0.0063	0.0024	H
913-324	LaSF 18A	1.91348	32.36	0.0008	0.0116	S
915-212	PBH 72	1.91536	21.17	0.0220	-0.0123	O
918-215	SF 58	1.91761	21.51	0.0180	-0.0095	S
923-209	FDS1	1.92286	20.88	0.0174	-0.0095	H
923-209	PBH 21	1.92286	20.88	0.0230	-0.0124	O
923-213	PBH 71	1.92286	21.29	0.0199	-0.0111	O
923-360	TaFD31	1.92250	35.95	-0.0056	0.0001	H
933-279	TaFD21	1.93260	27.92	0.0081	0.0077	H
952-204	SF 59	1.95250	20.36	0.0217	-0.0102	S
1003-235	TaFD43	2.00294	23.51	0.0060	-0.0042	H
1022-291	LaSF 35	2.02204	29.06	0.0033	-0.0009	S

2.2 The Chemical Composition of Optical Glasses and Its Influence on the Optical Properties

Marc K. Th. Clement

2.2.1 General Remarks

General Characteristics

The term "glass" is commonly applied to a fusion product of an inorganic material which is cooled to a solid state without crystallizing. It is often associated with fusion products made from silicates, which are used for windows, bottles, lenses, etc. Beside silicates, a lot of other substances can also be obtained in the glassy state. Nevertheless, a lot of non-silicate substances have been made glassy. It is widely understood that the possibility of a material forming glass is not an atomic or molecular property and cannot be measured by chemical or physical properties, but rather is a state of the solid state which can be reached by a sufficiently high cooling rate.

Glasses are normally specified by a certain set of defined properties that are typical of them, but distinguish them from other forms of the solid state. In contrast to crystals, glasses do not have a defined melting point; the viscosity of glasses decreases continuously provided that the temperature increases without discontinuity. Like liquids, glasses are isotropic, which is a very important characteristic for their optical applications. Unlike other non-crystalline materials, glasses are usually made by a fusion process. Other production methods are the chemical vapour deposition and sol-gel processes. By the latter methods it is

possible to produce several materials in the glassy state that cannot be made by a melting process.

Optical glass is defined as a group of glasses with the following characteristics:

- well-defined optical properties,
- optical homogeneous performance,
- absence of bubbles, striae, strain, and inclusions.

These characteristics must be taken into consideration during the production of optical glasses.

Oxides, some of which have been known for more than 5000 years, are the classical glass-forming materials. The oxides B_2O_3 , SiO_2 , GeO_2 , P_2O_5 , As_2O_3 , and Sb_2O_3 form glasses even without the addition of further oxides. They are usually known as glass-formers. Oxides such as TeO_2 , SeO_2 , MoO_3 , WO_3 , Bi_2O_3 , Al_2O_3 , Ga_2O_3 , TiO_2 , and V_2O_3 are known to form glasses if they are melted with a sufficient quantity of at least one other oxide (“conditional glass-formers” [2.33, 34]). Figure 2.11 shows a part of the periodic table whose oxides are glass-formers (hatched area) and conditional glass-formers (crossed area). Reviews on glass formation and the structures and properties of the oxide systems of glass-formers and conditional glass-formers have been given by *Kreidl* [2.35–38].

Chalcogenides [2.39–41] and halides [2.42–47] can also form glasses to be used in optics. Among the fluorides only BeF_2 is a glass-former. There are several conditional glass-formers that produce glasses with interesting optical properties; see Fig. 2.12 [2.47, 48]. Some mixtures of oxides and fluorides also form glasses.

The understanding of the structure of glasses has been strongly influenced by the “network theory” of *Zachariasen* [2.49, 50] and *Warren* [2.51] who explained that the short-term order of the constituents of a glass, i.e. the order of the atoms relative to their neighbours, is nearly the same as the order found in crystals. The long-term order, however, which is typical of crystalline materials, is missing in glasses. Unlike crystals, glasses show a “random network”. The constituents of a glass can be divided into cations that have the function of “network formers”,

1a	2a	3b	4b	5b	6b	7b	8				1b	2b	3a	4a	5a	6a	7a	0
H																		He
Li	Be											B	C	N	O	F	Ne	
Na	Mg											Al	Si	P	S	Cl	Ar	
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	N	Cu	Zn	Ga	Ge	As	Se	Br	Kr	
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe	
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn	

Fig. 2.11. Elements whose oxides act as glass-formers (hatched) and conditional glass-formers (crossed) for oxide glass systems [2.34]

1a	2a	3b	4b	5b	6b	7b	8						1b	2b	3a	4a	5a	6a	7a	0
H																			He	
Li	Be														B	C	N	O	F	Ne
Na	Mg														Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr			
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe			
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn			

Fig. 2.12. Elements whose fluorides act as glass-formers (hatched) and conditional glass-formers (crossed) for fluoride glass systems [2.48]

i.e. they contribute to the basic structure of the glass (e.g. SiO_2 , B_2O_3 , P_2O_5 , ...) and cations that have the function of “network modifiers”, i.e. cations that break up the network (e.g. Na_2O , CaO , PbO). The group of cations that change the network was later extended by the group of “intermediate” cations, which can act as network formers and network modifiers.

The introduction of a large number of network modifiers into a glass leads to an increase in the splitting of the network. An increased mobility of the structural units causes a decrease in the viscosity. Moreover, other physical properties, such as the electrical conductivity, vary greatly. As the network modifiers are irregularly distributed in the whole network, the properties of the glass normally change continuously according to the change in the composition.

Nevertheless, such a continuous change in the properties is not found in all glasses, since the type of network structure is not the same for different compositions. For example, if the content of Na_2O is increased in a SiO_2 glass, a continuous change in the properties of the glass can be found. If, however, the Na_2O content is increased in a B_2O_3 glass, abnormal changes in the properties of the glass can be found. This can be explained by the fact that an increasing alkaline content causes changes in the structural units of the network of the B_2O_3 glass. Normally, a B_2O_3 glass contains units of $[\text{BO}_3]$ which form the network of the glass. If alkaline oxides are introduced into this network, the coordination of the boron changes from $[\text{BO}_3]$ units to $[\text{BO}_4]$ units; this leads to a network with different physical properties. If the alkaline content increases further, the coordination number of boron is reduced again and the network is changed once more [2.52, 53]. Similar structural changes which depend on the glass composition are also reported for other glass constituents [2.54].

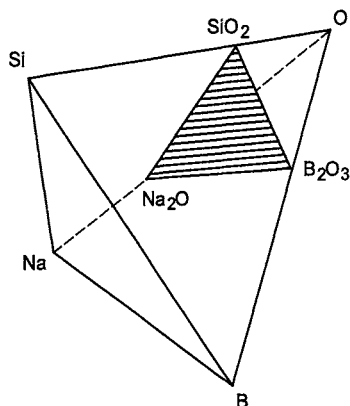


Fig. 2.13. Quaternary system of boron, silicon, sodium and oxygen

Description of the Composition

Most of the optical glasses are based on oxide glass systems which comprise a certain number of cations (e.g. Na^+ , Ca^{2+} , Pb^{2+} , La^{3+}). The composition of a mixture, e.g. of a quaternary system consisting of the elements silicon, sodium, boron, and oxygen, is shown by a regular tetrahedron where any composition is represented by a point inside the tetrahedron. The pure elements are represented by the corners of the tetrahedron. On the sides which combine oxygen with the elements silicon, sodium, and boron, the position of the binary compositions of the cations with oxygen (SiO_2 , B_2O_3 , Na_2O) can be fixed (Fig. 2.13).

In the case of limitation to a subsystem of the quaternary system, where the following stoichiometric condition is fulfilled (n_i is the number of moles of the component i),

$$n_{\text{O}} = 2n_{\text{Si}} + 3n_{\text{B}}/2 + n_{\text{Na}}/2, \quad (2.38)$$

a quasiternary system can be defined whose compositions may be shown by using a triangular cross section. The corners of this triangle denote the three binary compounds SiO_2 , B_2O_3 , and Na_2O . The sides of the triangle represent the binary systems SiO_2 – B_2O_3 , SiO_2 – Na_2O , and B_2O_3 – Na_2O (Fig. 2.14). Because of

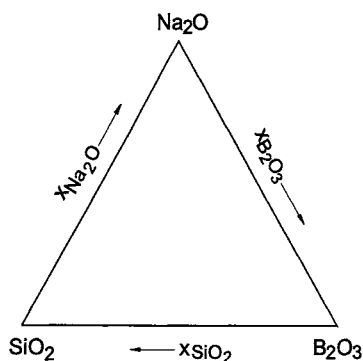


Fig. 2.14. Binary subsystems SiO_2 – B_2O_3 , B_2O_3 – Na_2O and SiO_2 – Na_2O and the quasiternary system SiO_2 – B_2O_3 – Na_2O

condition (2.38) the compositions of this quaternary system can be characterized by the mole fractions x_{SiO_2} , $x_{\text{B}_2\text{O}_3}$, and $x_{\text{Na}_2\text{O}}$. For a sufficient description of the quasiternary system, only two of the three mole fractions are needed, as the third mole fraction is defined by the condition that the sum of all mole fractions has to be unity, i.e.

$$x_{\text{SiO}_2} + x_{\text{B}_2\text{O}_3} + x_{\text{Na}_2\text{O}} = 1. \quad (2.39)$$

So this system can be described by:

$$\text{Si}_a\text{B}_{2b}\text{Na}_{2(1-a-b)}\text{O}_{1+a+2b} \quad \text{with} \quad 0 \leq a, b \leq 1. \quad (2.40)$$

Analogous to the treatment of quasibinary sections as edges of a quasiternary plane, the quasiternary systems can be treated as surfaces of a quaquaternary body which contains five elements.

A very common method for describing the composition of a glass is to quote the content of each oxide by weight (wt%) or mole (mol%). In a system with i oxides, only $i - 1$ concentration values have to be known. The value of the oxide concentration i is fixed by a condition similar to (2.39).

The transformation of the concentration values of a set of oxides i from wt% (w_i) to mol% (m_i) can be realized by

$$m_i = \frac{100 w_i}{M_i \sum_{j=1}^n w_j / M_j}, \quad 1 \leq i \leq n, \quad (2.41)$$

where M_i represents the molar mass of the oxide i (2.42). For the transformation of concentration values from mol% (m_i) to wt% the following equation can be used:

$$w_i = \frac{100 m_i M_i}{\sum_{j=1}^n m_j M_j}, \quad 1 \leq i \leq n. \quad (2.42)$$

Sometimes it is convenient to express the composition of a glass by cations with only the oxygen content being constant.

Thus the system $\text{SiO}_2\text{--B}_2\text{O}_3\text{--Na}_2\text{O}$ can be described by

$$\text{Si}_{3x}\text{B}_{4y}\text{Na}_{12(1-x-y)}\text{O}_6 \quad \text{with} \quad 0 \leq x, y \leq 1. \quad (2.43)$$

If the composition of a quasiternary system is described in this way, the quantity of cations is always related to a fixed amount of oxygen. By this method the molar fractions of a quaternary system can be used to describe the composition of a quasiternary system. The concentration unit of this description is known as equivalent-percent (eq%).

To convert the values a and b of (2.40) into the values x and y of (2.43) the content has to be related to the same amount of oxygen. Thus

$$x = \frac{2a}{1 + a + 2b} \quad (2.44)$$

and

$$y = \frac{3b}{1 + a + 2b} \quad (2.45)$$

and

$$z = (1 - x - y) = \frac{1 - a - b}{1 + a + 2b} . \quad (2.46)$$

These are three methods used to describe the composition of a glass; the first two methods, however, are more common.

Sometimes problems arise with the description of glass systems, especially if other anions besides oxygen, e.g. fluorine, are introduced into a glass. In general, these glasses can be described as discussed above. The only difference, however, is that now a different quaternary system has to be described. Starting again from a regular tetrahedron (Fig. 2.15), which represents the quaternary system, we express the quasiternary system by a square cross section.

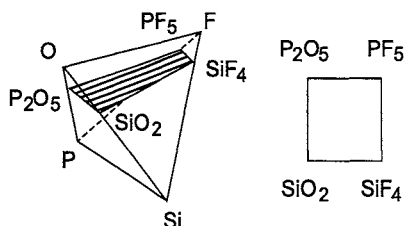
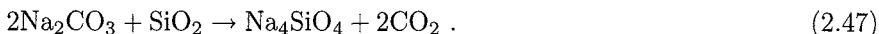


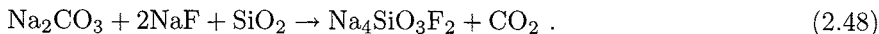
Fig. 2.15. Quaternary system with fluorine

For glasses containing anions other than oxygen, *Geffcken* [2.55] has proposed a special description. An oxide glass consisting of sodium and silica is normally described by the contents of the corresponding oxides (Na_2O , SiO_2). This description can be derived from the following chemical reaction, which starts with the batch materials, e.g.



The product of the reaction Na_4SiO_4 is transformed into $2\text{Na}_2\text{O} \cdot \text{SiO}_2$ and represents the composition of the glass expressed in oxides of the cations.

If fluorine is introduced into the glass and the amounts of sodium and silica are not changed, the batch reaction is as follows:



If the product of the reaction $\text{Na}_4\text{SiO}_3\text{F}_2$ is also expressed in oxides of the cations, as in (2.47), a description like “ $2\text{Na}_2\text{O} \cdot \text{SiO}_2 \cdot \text{F}_2$ ” is not correct, as it refers to the material “ $\text{Na}_4\text{SiO}_4\text{F}_2$ ”, which is not the same as in (2.48).

If, however, the concentration of a glass is to be described by the indication of the oxides of cations, the fictitious component “ (F_2-O) ” (= fluorine minus oxygen) has to be introduced. “ (F_2-O) ” represents the part of oxygen that is replaced by fluorine. With this fictitious component, the product of the reaction (2.48) would be expressed as $2\text{Na}_2\text{O} \cdot \text{SiO}_2 \cdot (\text{F}_2-\text{O})$, which leads to the correct composition of the reaction product $\text{Na}_4\text{SiO}_3\text{F}_2$.

Similarly, a glass consisting, for example, of $\text{CaF}_2\text{-B}_2\text{O}_3$ is described as a ternary system comprising the compounds $\text{CaO-B}_2\text{O}_3$ and the fictitious component (F_2-O) . The fictitious component (F_2-O) can be deducted from (2.49):

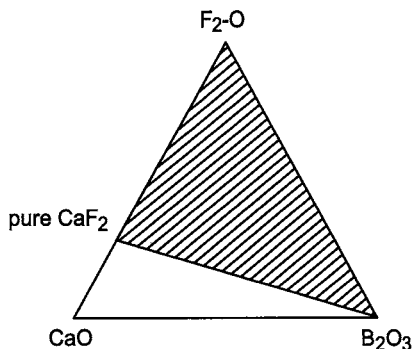


Fig. 2.16. Introduction of the component (F₂-O)

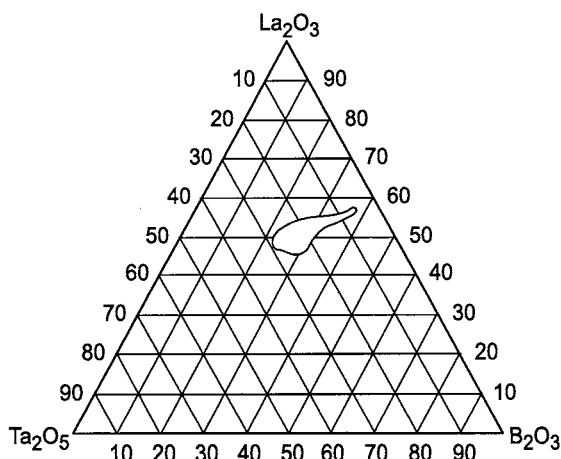


Fig. 2.17. Glass system B₂O₃-La₂O₃-Ta₂O₅ (wt%) [2.57]



In the above-mentioned system the content of fluorine is evidently fixed by the concentrations of CaO and B₂O₃ (Fig. 2.16). If this kind of description is applied, only a part of the concentration triangle can be used to describe the concentration.

The introduction of the component (F₂-O) is useful, since with the help of this component the condition that the sum of the concentrations of all components is unity (or 100%), can be fulfilled in a glass that contains more than one type of anion. Therefore even glasses that contain different concentrations of fluorine can be very easily compared [2.55, 56].

To evaluate the region of a system that forms glassy phases, usually a series of compositions is melted and casted. The casted melts are characterized. Figures 2.17–19 give some examples of glassy regions of various systems [2.55–58]. A review of the optical properties of different glass systems has been given by *Polukhin* [2.59] (Fig. 2.20). This figure includes glasses developed in the laboratory and commercially available glasses.

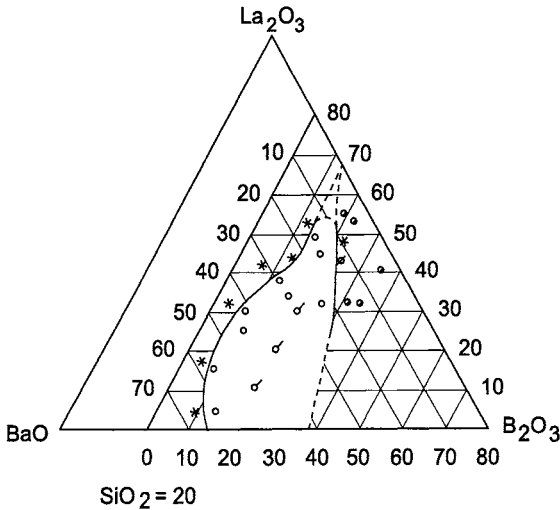


Fig. 2.18. Glass system B_2O_3 - La_2O_3 - BaO -20 SiO_2 (wt%) [2.55] (circle = clear glass, circle with outside dash = surface crystallization, circle with dash = few crystals in the glass, half-filled circle = phase separated glass, star = glass mixed with crystals)

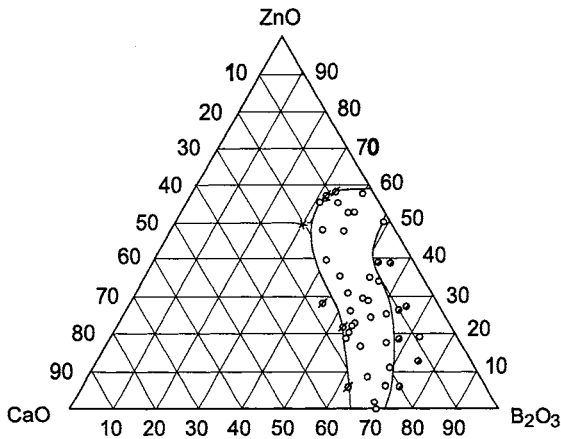


Fig. 2.19. Glass system B_2O_3 - ZnO - CaO (wt%) [2.55] (symbols as in Fig. 2.18)

2.2.2 Glass Types

The historical development of optical glasses can be divided into three steps [2.60]:

- Step 1: Before 1880 optical systems were constructed with simple crown glasses having a low dispersion and flint glasses having a high dispersion.
- Step 2: From 1880 to 1895 *Otto Schott* [2.61] developed new optical glasses by the introduction of B_2O_3 , P_2O_5 , and BaO . Thus many new glasses were obtained.
- Step 3: From 1930 to 1940 optical glasses of the lanthanum-crown and the lanthan-flint group were developed [2.62–64]. A prototype of these glasses was the glass reported by *Brewster* and *Kreidl* [2.63]. Other

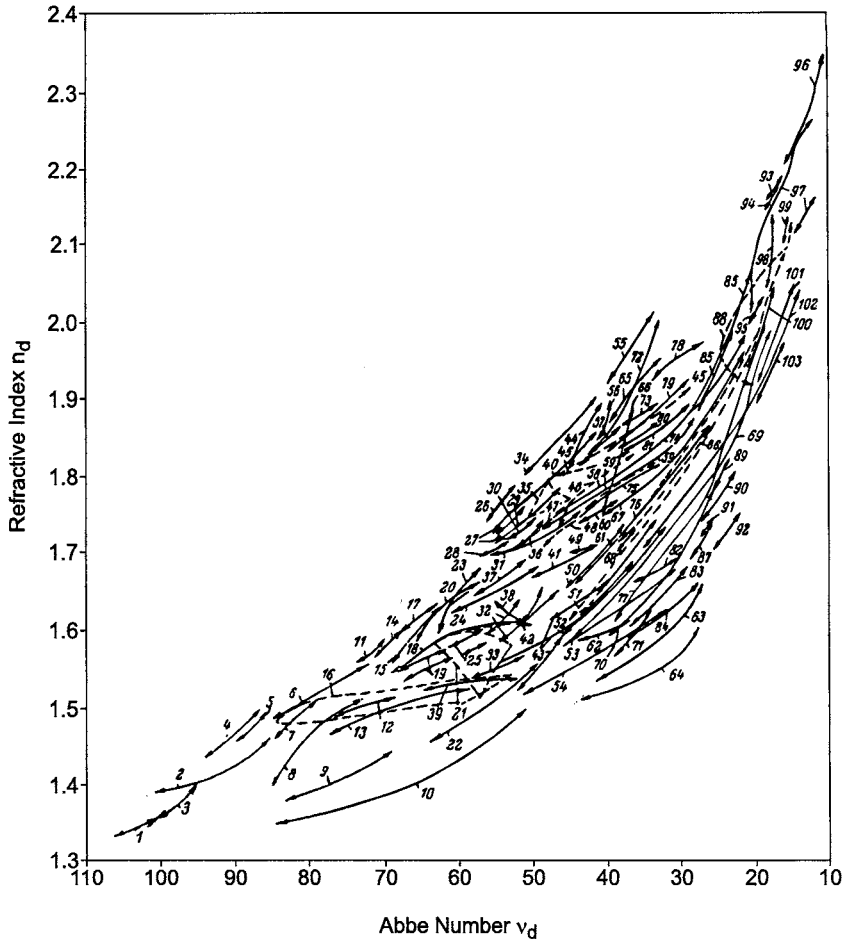


Fig. 2.20. Survey of refractive indices and Abbe numbers of known glass systems [2.59]

glasses based on boron–lanthanum systems [2.55, 59] or fluorine-containing phosphate glasses [2.58] were later developed.

The designation of optical glasses traditionally consists of a combination of the terms “crown” (German: “Kron”) and “flint” (German: “Flint”) with the terms “dense” (German: “Schwer”) denoting a high refractive index and “light” (German: “Leicht”) denoting a low refractive index. The term “crown” originally described glasses belonging to the glass system $\text{SiO}_2\text{--CaO--Na}_2\text{O}$, which were blown as crown-like glasses, whereas the term “flint” was used for glasses of the system $\text{SiO}_2\text{--PbO}$. In the past, only “flint stones” from the British south coast could be used for these flint glasses in order to achieve a transmission that was acceptable for that time.

Subgroups of optical glasses are classified by the chemical components that are characteristic of the optical properties, e.g., “dense barium flint” (BaSF) or “fluorine crown” (FK).

For the fabrication of high-quality optical systems, achromatic and apochromatic systems are required. For an achromatic system the chromatic aberration is minimized for two colours, e.g., red and blue. The other part of the visible spectrum remains uncorrected (“secondary spectrum”). The chromatic aberration in an apochromatic system, however, is corrected through the whole visible region [2.65, 66].

Most of the optical glasses show normal dispersion. These glasses are not suitable for the construction of apochromatic systems. Therefore, for the construction of apochromatic systems, glasses with a short spectrum in the blue range of the visible spectrum have been developed and called KzF and KzFS glasses. The abbreviations denote “short flint” and “short flint special” (German: “Kurz-Flint”, “Kurz-Flint-Sonder”, respectively). Special optical glasses with a long spectrum in the blue range of the visible spectrum are called LgSK, which stands for “long dense crown” (German: “Lang-Schwer-Kron”). KzF and KzFS glasses are glasses with Abbe numbers smaller than 55, whereas the glasses with the characteristics of LgSK glasses have an Abbe number in excess of 55. A combination of glasses with short-flint and long-crown characteristics is suitable for reducing the chromatic aberration.

Today we know of more than 250 different types of optical glasses that can be classified by their chemical composition (Fig. 2.21). The following is an outline of the various glass systems.

1. $\text{SiO}_2\text{--B}_2\text{O}_3\text{--M}_2\text{O}$.

This system is formed by the network formers SiO_2 and B_2O_3 . Alkaline oxides (M_2O) have been introduced as network modifiers. The presence of B_2O_3 and alkaline oxides is essential, as the glass-forming oxide SiO_2 has melting temperatures which are too high for conventional melting techniques. The addition of B_2O_3 and alkaline oxides facilitates the meltability of these glasses. The optical glass BK 7[®] is one of the main representative glasses of this system. It is the optical glass most frequently used and it can be produced with extreme homogeneity.

2. $\text{SiO}_2\text{--B}_2\text{O}_3\text{--BaO}$ and

3. $\text{SiO}_2\text{--BaO--M}_2\text{O}$.

If BaO is used as the main network modifier instead of the alkaline oxide or B_2O_3 , this results in two large glass systems. In contrast to other earth alkaline network modifiers, the introduction of BaO has various advantages. Beside PbO there is no other divalent oxide that increases the refractive index as strongly as BaO. Additionally, BaO neither decreases the Abbe number nor makes it shift the UV-transmission edge significantly to greater wavelengths as PbO does. BaO-

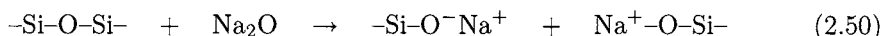
5. $(\text{SiO}_2, \text{B}_2\text{O}_3)\text{--BaO--PbO}$.

This glass system is a mixture of the glass systems $\text{SiO}_2\text{--BaO--M}_2\text{O}$ and $\text{SiO}_2\text{--PbO--M}_2\text{O}$. Thus these glasses represent a transition from one glass system to the other. In the system $\text{SiO}_2\text{--BaO--M}_2\text{O}$, the exchange of PbO for BaO leads to the production of glasses with a lower tendency to crystallize. Compared with similar BaO -containing glasses, PbO -containing glasses are characterized by lower softening points and higher expansion coefficients. In some glasses of this system, BaO is partly replaced by ZnO .

6. $\text{SiO}_2\text{--PbO--M}_2\text{O}$.

Glasses of this system have been known for a very long time. They are widely used for optical glasses and crystal glasses. As a glass component, PbO plays a vital role because on the one hand it increases the refractive index and on the other hand it decreases the Abbe number very strongly. Furthermore, PbO significantly effects the partial dispersion of glasses since a high PbO content leads to a positive deviation from the "normal line". High amounts of PbO lead to a high glass density and a rather small chemical resistance.

PbO can be introduced into glasses to more than 80 wt%, which indicates that it does not only have network modifying properties. Normally, if a network modifier is introduced into a glass, e.g. Na_2O into a SiO_2 glass, bridging oxygen atoms are converted into non-bridging oxygen atoms:



With an increasing content of network modifiers the network progressively weakens and the viscosity of the glass melt is decreased. If the content of the network modifiers is increased further, more and more bridging oxygen atoms are converted into non-bridging atoms. For a ratio of network modifiers to network formers of about 1:1, a three-dimensional network can no longer be formed and glass formation becomes impossible. However, the glass-forming tendency of binary PbO--SiO_2 systems leads to a PbO content of more than 70 mol%. According to *Zachariasen* and *Warren* these glasses would normally not exist if PbO was only a network modifier.

Fajans [2.67] and *Stanworth* [2.68] explained how the introduction of such a high amount of PbO is possible. In small concentrations, PbO has the same function as any other network modifier. High concentrations of PbO , however, cause an increase in the formation of $[\text{PbO}_4]$ tetrahedra. These $[\text{PbO}_4]$ tetrahedra can act as network formers and are incorporated in the three-dimensional network.

Glasses that contain large amounts of PbO are usually more or less yellow in colour. Impurities, such as Fe_2O_3 or Cr_2O_3 , have very strong effects on the colour of PbO -containing glasses, even stronger than in soda-potash-lime glasses [2.69]. *Vogel* and *Heindorf* [2.70] found that, first, beside the different structures of the PbO -containing glasses, an equilibrium between Pb^{2+} and Pb^{4+} exists, and, second, brown PbO_2 is formed if the melt is supersaturated with oxygen. Unlike the yellow colour that is due to impurities, the yellow tint caused by the

equilibrium $\text{Pb}^{2+}/\text{Pb}^{4+}$ can be reversed by a variation of the redox condition of the melt.

For a long time, attempts have been made to increase the chemical stability of optical glasses with high PbO contents. Replacing parts of the PbO content with TiO_2 (in the range of some wt%) has been a common method to increase the chemical durability. A sufficient TiO_2 content not only improves the chemical durability but, unfortunately, also leads to a transmission loss in the UV range. A solution to this problem is reached by replacing parts of the PbO content with both TiO_2 and ZrO_2 [2.71]. The chemical durability is improved and the optical properties of the original glass ($\text{SiO}_2\text{--PbO--M}_2\text{O}$) are obtained again.

7. $\text{P}_2\text{O}_5\text{--MO--B}_2\text{O}_3\text{--Al}_2\text{O}_3$.

Glasses in which P_2O_5 acts as a network former have been known for a long time [2.72]. P_2O_5 forms glasses with rather low dispersions (high Abbe numbers) and high refractive indices, compared with the classic glasses based on B_2O_3 or SiO_2 . Glasses with a high P_2O_5 content have long dispersion spectra in the blue range. These glasses are normally used to correct chromatic aberration. The second and very important property of glasses containing P_2O_5 is their excellent UV transparency.

A disadvantage of phosphate glasses is that their chemical durability is poor compared with the durability of SiO_2 -based glasses.

8. $\text{P}_2\text{O}_5\text{--Al}_2\text{O}_3\text{--MO--(F}_2\text{--O)}$.

In order to obtain glasses with lower dispersions than those of phosphate glasses, one can introduce fluorine. Most of these glasses are based on a $\text{P}_2\text{O}_5\text{--Al}_2\text{O}_3\text{--MO}$ glass system which contains high amounts of fluorine. This glass system shows a relatively high chemical durability and a relatively low devitrification tendency. Fluorine takes the places of oxygen atoms so that one oxygen is replaced by two fluorine atoms. The formation of the fluoro-phosphate glasses of this system is rather complex and can be described with amorphous fluoro-alumino-phosphates [2.73].

As fluorine has a smaller ionic radius and a higher electronegativity than oxygen, the polarization of the cations in these glasses is increased so that the refractive index and the dispersion are decreased. The replacement of oxygen by fluorine leads to glasses that have distinct LgK characteristics.

9. $(\text{B}_2\text{O}_3, \text{SiO}_2)\text{--La}_2\text{O}_3\text{--MO}$.

In connection with B_2O_3 , glasses containing large amounts of La_2O_3 form a very extensive glass system. Glasses belonging to this system normally have a high refractive index with a relatively low dispersion. As their chemical resistance is rather poor, their durability can be increased by replacing SiO_2 with B_2O_3 and by the introduction of divalent oxides. In addition, the introduction of ZrO_2 and TiO_2 is another possibility for achieving better chemical properties. In the past,

ThO₂ and CdO have also been used to achieve this aim, but for environmental protection these oxides have been eliminated from these glasses.

10. B₂O₃–La₂O₃.

These glasses are based on the B₂O₃–La₂O₃ glass system. They all have high refractive indices and high dispersion. In order to obtain high refractive indices, large numbers of oxides that are known to cause a high refractive index, such as Gd₂O₃, Y₂O₃, TiO₂, Nb₂O₅, Ta₂O₅, WO₃, and ZrO₂ are incorporated in the glasses of this system. The large number of network modifiers leads to glasses whose viscosities depend to a high degree on temperature (i.e., they are called short [2.74]) and which show a strong tendency to crystallize. On the other hand, these glasses have a distinct hardness and chemical resistance, which are even increased with higher refractive indices. In this respect, the glasses differ from those of the system SiO₂–PbO–M₂O, which also have a high refractive index.

11. SiO₂–Sb₂O₃–B₂O₃.

The replacement of PbO by Sb₂O₃ leads to an extensive glass system. Sb₂O₃ reduces the dispersion of the blue part of the visible spectrum. Consequently, short-flint glasses are obtained. The introduction of Sb₂O₃ into a glass gives results similar to those we have in the case of PbO, although Sb₂O₃ glasses seem to form long chains of –Sb–O–Sb–O– units [2.75].

12. B₂O₃–PbO–Al₂O₃.

If SiO₂ is replaced by B₂O₃, glasses with primarily very poor chemical properties are obtained. Glasses of this system have an intensive short-flint characteristic, since the PbO content induces absorption in the blue range of the visible spectrum, and B₂O₃ causes strong absorption in the infrared range of the spectrum. Al₂O₃ is added to improve the chemical properties to an acceptable level.

13. SiO₂–M₂O–TiO₂–(F₂–O).

The PbO content in glasses of the system SiO₂–M₂O–PbO can be replaced by TiO₂. As TiO₂ causes an increase in the refractive index, glasses of the system SiO₂–M₂O–TiO₂–(F₂–O) are characterized by high refractive indices. The dispersion of these glasses is increased to a higher degree than that of glasses containing the same amount of PbO. The introduction of TiO₂ into glasses is limited, because when the TiO₂ content is increased, the glasses tend to crystallize strongly. This should be avoided by the introduction of alkaline oxides into the glass. In spite of the large content of alkaline oxides, these glasses are highly resistant to the influences of water and acids. Glasses with high TiO₂ contents show a strong absorption in the UV range. This absorption is stronger than that of glasses containing PbO or Sb₂O₃. Both glass types are characterized by a positive deviation of the partial dispersion from the normal line.

2.2.3 Reduction of the Density

The PbO content of glasses of the $\text{SiO}_2\text{-PbO-M}_2\text{O}$ system can also be replaced by the oxides WO_3 , TiO_2 , ZrO_2 , and Nb_2O_5 . The glasses obtained normally show excellent chemical durability, low density, and good mechanical properties, which allow good grinding and polishing. The production of these glasses, however, is much more difficult than that of the traditional glasses with large PbO content. Because of the low density, the glasses of the SF group and the LaK group are called “lightweight glasses” in the Schott programme, and they are indicated by the letter “L” preceding the glass number.

Additionally, in glasses which originally do not contain any PbO, a decreased density may be achieved by optimizing the glass compositions. The compounds responsible for a high density are replaced by compounds which contribute little to the glass density (e.g., by replacing BaO with CaO and MgO). Table 2.4 gives a summary of the lightweight glasses of the Schott programme. Other glass manufacturers have also developed lightweight glasses.

Table 2.4. Lightweight glasses of the Schott programme compared with the original “heavy” glasses (ρ , glass density; wr, weight reduction; SR, acid resistance (ISO 8424); AR, alkali resistance (ISO in draft); T_g , transformation temperature; H_K , Knoop hardness)

Type	n_d	ν_d	ρ g/cm ³	Weight reduct.	SR	AR	T_g	H_K
F N11	1.62096	36.18	2.66	-26%	1	1.0	577	610
F2	1.62004	36.37	3.61		1	2.3	438	420
SF L4	1.75520	27.21	3.17	-34%	1.0	1.3	569	580
SF 4	1.75520	27.58	4.79		4.3	2.3	424	390
SF L6	1.80518	25.39	3.37	-35%	2	1	605	570
SF 6	1.80518	25.43	5.18		51.3	2.3	426	370
SF L56	1.78470	26.08	3.28	-33%	1	1.3	592	570
SF 56A	1.78470	26.08	4.92		3.2	2.2	433	380
SF L57	1.84666	23.62	3.55	-36%	1	1	598	580
SF 57	1.84666	23.83	5.51		52	2.3	402	350
SF 64A	1.70585	30.30	3.00	-26%	1	1.2	580	590
SF 15	1.69895	30.07	4.06		1	1.2	455	420
BaSF 64A	1.70400	39.38	3.20	-19%	3.2	1.2	582	650
BaSF 13	1.69761	38.57	3.97		52.3	1.2	584	530
LaSF 32	1.80349	30.40	3.52	-28%	1	1.0	552	660
LaSF 8	1.80741	31.61	4.87		52	1.2	476	420
LaSF 36A	1.79712	35.08	3.60	-20%	3.3	1.0	651	670
LaSF 33	1.80596	34.24	4.48		51	1.2	543	540
LaK L12	1.67790	54.92	3.32	-19%	58.3	2.2	651	700
LaK N12	1.67790	55.20	4.10		53.3	3.3	614	560
LaK L21	1.64048	59.75	2.97	-21%	53.3	3.3	641	710
LaK 21	1.64049	60.10	3.74		53.2	4.3	639	600

2.2.4 Development of Optical Glasses

Calculation of the Optical Properties

In modern glass development, properties have be computed from the chemical composition with the help of models relating the composition to the properties. The experiments necessary for the evaluation of these models are based on statistical methods which allow the experiments to be kept to a minimum.

For the development of a glass it is necessary to know the concentration range of the components that will be introduced into the glass. The model is evaluated only for this concentration range, but this makes it possible to forecast the values of the properties within this concentration range with high accuracy.

Thus, as some data are known, we are able to forecast the optical properties. These data are based on the assumption that it is possible to calculate the optical properties with constant or nearly constant influences of the glass components within a certain concentration range.

Refractive Index and Main Dispersion

One method for calculating the refractive indices of an optical glass is based on the assumption that the refractive index can be calculated if the nature and the concentration of the various glass components are known. In this case the term "component" is used rather arbitrarily and may refer to oxides or to structural units.

A simple but nevertheless very efficient way to calculate the refractive index n_d has been introduced by *Appen* [2.76–85]:

$$n_d = \frac{\sum_{i=1}^n n_{d,i} c_i}{100}, \quad (2.51)$$

where $n_{d,i}$ is a factor which refers to a glass oxide and c_i is the oxide concentration in mol%. Some factors are shown in Table 2.5. In addition, Appen also introduced factors for the main dispersion,

$$D = n_F - n_C, \quad (2.52)$$

which is calculated similarly to (2.51):

$$D = \frac{\sum_{i=1}^n D_i c_i}{100}. \quad (2.53)$$

Some factors of this calculation are also given in Table 2.5.

The Abbe number ν_d can be calculated from equations (2.51) and (2.53) by

$$\nu_d = \frac{n_d - 1}{D}. \quad (2.54)$$

Another method for the calculation of the refractive index n_d by means of the composition has been proposed by *Demkina* [2.86]:

$$n_d = \frac{\sum_{i=1}^n c_i n_{d,i} / t_i}{\sum_{i=1}^n c_i / t_i}. \quad (2.55)$$

Table 2.5. Factors necessary to calculate the refractive index n_d and the dispersion D of glasses by the composition [2.79–85]

Component	$n_{d,i}$	D_i
SiO ₂	1.4585–1.475	0.00675
B ₂ O ₃	1.470–1.710	0.0066–0.0090
Al ₂ O ₃	1.520	0.0085
Li ₂ O	1.695 (1.6555)	0.0138 (0.0130)
Na ₂ O	1.590 (1.575)	0.0142 (0.0140)
K ₂ O	1.575 (1.595)	0.0130 (0.0132)
MgO	1.610 (1.570)	0.0111
CaO	1.730	0.0148
SrO	1.770	0.0163
BaO	1.880	0.0189
ZnO	1.710	0.0165
TiO ₂	2.08–2.23	0.050–0.062
PbO	2.5–2.35	0.0528–0.0744
n_{R_2O} , D_{R_2O} :	values in parentheses are valid for glasses in the M ₂ O–SiO ₂ system	
n_{K_2O} , D_{K_2O} :	values in parentheses are valid for those glasses that contain more than 1 mol% Na ₂ O	
n_{MgO} :	the value in parentheses is valid for parts of the glasses of the systems Na ₂ O–MgO–SiO ₂ and K ₂ O–MgO–SiO ₂	
n_{SiO_2} :	$n_{SiO_2} = 1.5085 - 0.0005 c_{SiO_2} 0.$ for $100 \geq c_{SiO_2} \geq 67$ $n_{SiO_2} = 1.475$ for $c_{SiO_2} \leq 67$	
n_{TiO_2} , D_{TiO_2} :	$n_{TiO_2} = 2.480 - 0.005 c_{SiO_2}$ $D_{TiO_2} = 0.082 - 0.0004 c_{SiO_2}$	
n_{PbO} , D_{PbO} :	with $\alpha = c_{SiO_2} + c_{B_2O_3} + c_{Al_2O_3}$ is: $n_{PbO} = 2.685 - 0.0067 \alpha$ $D_{PbO} = 0.1104 - 0.00072 \alpha$ $n_{PbO} = 2.350$ $D_{PbO} = 0.0528$	
$n_{B_2O_3}$, $D_{B_2O_3}$:	with $\beta = (\sum c_{R_2O} + \sum c_{RO} + \sum c_{Al_2O_3})/c_{B_2O_3}$ is: – for $64 \geq c_{SiO_2} \geq 44$: $n_{B_2O_3} = 1.710$ $D_{B_2O_3} = 0.0090$ $n_{B_2O_3} = 1.518 + 0.048 \beta$ $D_{B_2O_3} = 0.0064 + 0.00065 \beta$ $n_{B_2O_3} = 1.616 + 0.048 \beta$ $D_{B_2O_3} = 0.0064 + 0.00065 \beta$ $n_{B_2O_3} = 1.470$ $D_{B_2O_3} = 0.0066$ – for $80 \geq c_{SiO_2} \geq 71$: $n_{B_2O_3} = 1.710$ $D_{B_2O_3} = 0.0090$ $n_{B_2O_3} = 1.518 + 0.12 \beta$ $D_{B_2O_3} = 0.0090$ $n_{B_2O_3} = 1.760 - 0.12/\beta$ $D_{B_2O_3} = 0.0064 + 0.00065 \beta$ $n_{B_2O_3} = 1.614 - 0.48/\beta$ $D_{B_2O_3} = 0.0064 + 0.00065 \beta$ $n_{B_2O_3} = 1.470$ $D_{B_2O_3} = 0.0066$	

Table 2.6. Data for the calculation of optical properties (after [2.86])

Oxide	t_i	$n_{d,i}$	D_i	ν_i
SiO ₂ (I)	60	1.475	0.00695	68
SiO ₂ (II)	60	1.458	0.00678	68
B ₂ O ₃ (I)	43	1.61	0.00750	81
B ₂ O ₃ (II)	70	1.464	0.00670	69
Al ₂ O ₃	59	1.49	0.00850	58
Na ₂ O	62	1.59	0.01400	42
K ₂ O	94	1.58	0.01200	48
MgO	140	1.64	0.01300	49
CaO	86	1.83	0.01750	47
BaO	213	2.03	0.02280	45
ZnO	223	1.96	0.02850	34
PbO (I)	343	2.46	0.07700	19
PbO (II)	223	2.46	0.07700	19
PbO (III)	223	2.50	0.11600	13

In (2.55) c_i is the concentration of the oxide i in wt% in the glass, t_i is a special conversion coefficient used to transform weight units into so-called “volume fractions” so that it will be possible to apply (2.55) to numerous glass compositions. $n_{d,i}$ is a coefficient determined by experimental measurements and characterizes the contribution of the oxide i to the refractive index. A similar equation can also be used for the main dispersion ($D = n_F - n_C$) and the Abbe number (ν_d). Some coefficients are shown in Table 2.6.

The presence of two values for one glass component indicates that the contribution of this component to the optical properties depends on special concentration ranges. In order to select the correct values from Table 2.6, Demkina drew up a number of rules which have to be obeyed.

Huggins and *Sun* [2.87, 88] have developed a method for calculating the optical properties of a glass. This method is related to the concept of the specific volume of glass. The refractive index n_d is calculated from

$$n_d = 1 + \frac{\rho}{\rho_0} \sum_{i=1}^n n_{d,i} c_i, \quad (2.56)$$

where ρ is the density of the glass, ρ_0 the unit density 1 g cm^{-3} , $n_{d,i}$ the contribution of the oxide i to the refractive index and c_i the concentration of the oxide i in weight fracture. The main dispersion is calculated by

$$D = \frac{\rho}{\rho_0} \sum_{i=1}^n D_i c_i \quad (2.57)$$

The density of the glass can be calculated according to

$$\frac{1}{\rho} = \sum_{i=1}^n a_i c_i, \quad (2.58)$$

where a_i is the influence of the oxide i on the density and c_i the concentration of the oxide in weight fracture. The values of a_i depend on the number of silicon atoms, N_{Si} , per oxygen atom in the glass. For the determination of this number (2.59) is used:

$$N_{\text{Si}} = \frac{c_{\text{SiO}_2}}{60.06 \sum_{i=1}^n R_i c_i}, \quad (2.59)$$

where c_i is the weight fraction of the oxide i , R_i is the ratio of the number of oxygen atoms in one formula unit of the oxide to the molecular weight of the oxide, c_{SiO_2} is the concentration of SiO_2 in the glass in weight fracture and 60.06 is the molecular mass of SiO_2 . Table 2.7 shows some values for R_i and a_i (dependent on the value of N_{Si}). Table 2.8 shows some values for $n_{\text{d},i}$ and D_i .

Gilard and Dubrud [2.89] have proposed a non-linear model for the calculation of the refractive index:

$$n_{\text{d}} = \sum_{i=1}^n \{a_{1,i} c_i + a_{2,i} (c_i - a_{3,i})^2\}, \quad (2.60)$$

where the concentration c_i is used in wt%. The coefficients for several oxides are shown in Table 2.9.

Partial Dispersion

As mentioned above, the refractive index is a function of the wavelength. The partial dispersion is used to express the variation of the the refractive index in different parts of the optical spectrum. Normally a nearly linear relationship between the partial dispersion and the Abbe number is found for most optical glasses. Glasses that show large deviations from this linear relationship are called glasses with an abnormal partial dispersion.

The oscillation frequency in the IR range of the main network formers influences the dispersion of the glass significantly. *Reitmayer* [2.90] and *Izumitani* [2.91] explained the negative values of the deviation from the “normal line” of the glasses KzFS 2, KzFS 4 and LaK 9 by a shift of the oscillation wavelength to lower values due to the very high B_2O_3 contents of these glasses. *Brewster* [2.92] found that a negative deviation occurs if SiO_2 is replaced by B_2O_3 . The replacement of SiO_2 by P_2O_5 leads to a positive deviation from the “normal line”. Measurements on silicate glasses with different water concentrations show an increase in the deviation of the partial dispersion from the “normal line” with increasing water content [2.93].

The reasons for the abnormal partial dispersion may be explained by Lorentz's theoretical equation. With this equation the refraction index n can be expressed as a function of the wavelength λ in a region where no absorption occurs:

$$n^2 - 1 = \frac{\sum_{i=1}^n K N_i f_i}{1/\lambda_i^2 - 1/\lambda^2}. \quad (2.61)$$

Here λ_i are the wavelengths of the inherent absorption, f_i is the oscillator strength of the inherent absorption, N_i is the number of atoms per volume and

Table 2.7. Coefficients for the calculation of the optical properties (after [2.87, 88])

Oxide	$R_i \times 10^2$	a_i for various N_{Si}			
		0.270–0.345	0.345–0.400	0.400–0.435	0.435–0.500
SiO ₂	3.3300	0.4063	0.4281	0.4409	0.4542
B ₂ O ₃ (BO ₄)	4.3079	0.5900	0.5260	0.4600	0.3450
B ₂ O ₃ (BO ₃)	4.3079	0.7910	0.7270	0.6610	0.5460
Al ₂ O ₃	2.9429	0.4620	0.4180	0.3730	0.2940
Li ₂ O	3.3470	0.4520	0.4020	0.3500	0.2610
Na ₂ O	1.6131	0.3730	0.3490	0.3240	0.2810
K ₂ O	1.0617	0.3900	0.3740	0.3570	0.3290
MgO	2.4800	0.3970	0.3600	0.3220	0.2560
CaO	1.7832	0.2850	0.2590	0.2310	0.1840
SrO	0.9650	0.2000	0.1850	0.1710	0.1450
BaO	0.6521	0.1420	0.1320	0.1220	0.1040
TiO ₂	2.5032	0.3190	0.2820	0.2430	0.1760
ZrO ₂	1.6231	0.2220	0.1980	0.1730	0.1300
PbO	0.4480	0.1060	0.0955	0.0926	0.0807

Table 2.8. Coefficients for $n_{d,i}$ and D_i using the method given in [2.87, 88]

Oxide	$n_{d,i}$	$D_i \times 10^2$
SiO ₂	0.2083	0.3095
B ₂ O ₃ (BO ₄)	0.2150	0.2580
B ₂ O ₃ (BO ₃)	0.2530	0.4310
Al ₂ O ₃	0.2038	0.3630
Li ₂ O	0.3080	0.7028
Na ₂ O	0.1941	0.5013
K ₂ O	0.2025	0.4320
MgO	0.2100	0.4540
CaO	^a	^a
SrO	0.1592	0.3140
BaO	^a	^a
TiO ₂	0.3130	1.6800
ZrO ₂	0.2090	0.6900
PbO	^a	^a

^a $n_{d,i}$ and D_i of CaO, BaO and PbO are calculated as follows:

– CaO:

$$n_{d, \text{CaO}} = 0.2257 + 0.4770 \times 10^{-4} c_{\text{CaO}} / \sum_{i=1}^n R_i c_i$$

$$D_{\text{CaO}} = 0.4636 + 0.0013 c_{\text{CaO}} / \sum_{i=1}^n R_i c_i$$

– BaO:

$$n_{d, \text{BaO}} = 0.1290 + 2.1300 \times 10^{-4} c_{\text{BaO}} / \sum_{i=1}^n R_i c_i$$

$$D_{\text{BaO}} = 0.2220 + 0.0030 c_{\text{BaO}} / \sum_{i=1}^n R_i c_i$$

– PbO:

$$n_{d, \text{PbO}} = 0.1272 + 2.0440 \times 10^{-6} c_{\text{PbO}} / (\sum_{i=1}^n R_i c_i)^2$$

$$D_{\text{PbO}} = 0.582 + 0.5044 \times 10^{-4} c_{\text{PbO}} / (\sum_{i=1}^n R_i c_i)^2$$

Table 2.9. Data for the calculation of the refractive index achieved by the method given in [2.89]

Oxide	$a_{1,i}$	$a_{2,i}$	$a_{3,i}$	Conc. range of oxide ^a
SiO ₂	0.01458	0	0	0–100
B ₂ O ₃	0.01725	-50×10^{-6}	0	0–40
Al ₂ O ₃	0.01510	0	0	0–20
Li ₂ O (I)	0.01820	0	0	0–25
Li ₂ O (II)	0.01980	-64×10^{-6}	0	25–45
Na ₂ O (I)	0.01618	0	0	0–20
Na ₂ O (II)	0.01646	-14×10^{-6}	0	20–55
K ₂ O (I)	0.01575	0	0	0–20
K ₂ O (II)	0.01575	$+130 \times 10^{-6}$	-20	20–60
MgO	0.01700	0	0	0–15
CaO	0.01785	0	0	0–40
BaO	0.01690	0	0	0–50
ZnO	0.01675	0	0	0–18
TiO ₂	0.0200	0	0	0–10
PbO (I)	0.01760	0	0	0–40
PbO (II)	0.01760	$+140 \times 10^{-6}$	-40	40–80

^a in wt%

K is a constant. For optical glasses a model with 4 inherent absorptions [2.94] can be used, namely, for the absorption by bridging oxygen atoms (index 1), the absorption by non-bridging oxygen atoms (index 2), the absorption in the infrared region (index 3) and the absorption in the ultraviolet region (index 4), provided that the glass contains PbO:

$$n^2 - 1 = \frac{\sum_{i=1}^4 K N_i f_i}{1/\lambda_i^2 - 1/\lambda^2} . \quad (2.62)$$

Using (2.62) Izumitani explained that the shift of the absorption peaks resulting from different structural groups in a glass causes a change in the run of the refractive index depending on the wavelength. For a calculation of the partial dispersion of a glass system, the constants λ_i , K and f_i must be determined.

If the constants are determined, the dispersion of the refractive index can be calculated by the number of atoms N_i that cause absorption bands.

Linear models represent another way to calculate the partial dispersion of glasses. In principle there are two ways to determine the partial dispersion, e.g. $P_{g,F}$:

$$P_{g,F} = \frac{n_g - n_F}{n_F - n_C} . \quad (2.63)$$

One possibility is to use linear models to determine first the refractive indices n_j as in Appen's model:

$$n_j = \sum_{i=1}^n n_{j,i} c_i, \quad j \in \{g, F, C\} . \quad (2.64)$$

From these refractive indices n_j the partial dispersion is determined with (2.63).

The other possibility is to calculate the partial dispersion by using a linear model for the partial dispersion itself:

$$P_{x,y} = \sum_{i=1}^n P_{x,y,i} c_i, \quad (2.65)$$

where $P_{x,y,i}$ is the contribution of the oxide i to the partial dispersion $P_{x,y}$ and c_i is the concentration of the oxide. The index “ x,y ” indicates the wavelength combination of the partial dispersion to be calculated.

Glass systems with high PbO contents might show a negative deviation value from the “normal line”. This is shown in Fig. 2.22, where the ratio of B_2O_3 to PbO, starting with KzFS N4, is changed continuously and the partial dispersion is calculated by a model similar to (2.65). This “simple” variation of the B_2O_3 :PbO ratio leads to values of the partial dispersion which can also be obtained by the other KzFS glasses of the Schott programme [2.95].

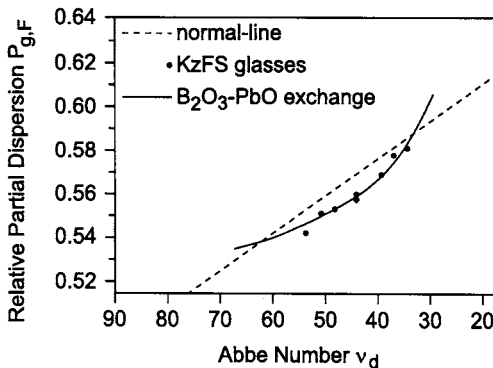


Fig. 2.22. Change in the values of $P_{g,F}$ and ν_d upon exchanging B_2O_3 and PbO

Although the partial dispersion can be calculated by a linear model, in a plot of $P_{g,F}$ versus ν_d the curve does not follow a straight line, but occurs as the typical curve indicated in Fig. 2.22.

Optimization of the Glass Composition

During recent years a lot of data on the properties of optical glasses have been collected. They can be used for the regression analysis and the determination of regression coefficients. These coefficients, e.g. for linear models, represent a mathematical relationship between the glass properties and the glass composition, and they provide information about the influence of the different glass compositions on the glass properties. Moreover, it is possible to calculate the optimum glass composition even if more than one property of the glass is desired.

In this case one needs to know the regression coefficients, a procedure for optimization and an error function. For linear models the glass properties can be calculated with

$$Q_{j,\text{cal}} = \sum_{i=1}^n Q_{j,i} c_i , \quad (2.66)$$

where $Q_{j,\text{cal}}$ is the calculated glass property j , c_i the concentration of the component i and $Q_{j,i}$ the regression coefficient of the component i for the property j .

In practice, however, (2.66) is unsuitable, as normally the development of a glass starts from a given glass with defined properties. For this reason, it is more convenient to use the deviation of the desired properties from the properties of the known glass for the optimization [2.96]:

$$Q_{j,\text{cal}} = Q_{j,\text{d}} + \sum_{i=1}^n Q_{j,i} \Delta c_i , \quad (2.67)$$

where $Q_{j,\text{d}}$ is the property j of the given glass and Δc_i is the deviation of the concentration of the component i of the desired glass from the concentration of the component i of the given glass. With this transformation, errors arising from the use of simple statistical models may be reduced.

Before starting the optimization, the order of magnitude of the properties $Q_{j,\text{d}}$ which are calculated must be considered. Normally, for the optimization procedure only properties with a large magnitude and not those with a small magnitude are considered. For this reason the properties have to be normalized to unity:

$$F_j = 1/Q_{j,\text{d}} , \quad (2.68)$$

$$Q'_{j,\text{cal}} = F_j Q_{j,\text{cal}} . \quad (2.69)$$

In order to develop a glass, the scientist tries to achieve the desired properties by changing the glass composition. Usually the test melts, which are essential to glass development, will not immediately lead to the goal aimed at. For the mathematical treatment of this process an error function, which has to be minimized, is defined [2.97]:

$$E = \sum_{j=1}^n (Q'_{j,\text{cal}} - Q'_{j,\text{a}})^2 w_j , \quad (2.70)$$

where $Q'_{j,\text{a}}$ is the aspired final value of the glass property j and w_j a weighting factor for the property j . The weighting factor is used to influence the error function with respect to the special glass properties. For a new glass, the demanded property profile is normally not expressed by an exact value, but by upper and lower limits of the properties. Consequently, (2.70) has to be changed to

$$E_j = \begin{cases} (Q'_{j,\text{cal}} - Q'_{j,\text{a,u}})^2 w_j, & \text{for } Q'_{j,\text{cal}} > Q'_{j,\text{a,u}} , \\ 0, & \text{for } Q'_{j,\text{a,l}} < Q'_{j,\text{cal}} < Q'_{j,\text{a,u}} , \\ (Q'_{j,\text{cal}} - Q'_{j,\text{a,l}})^2 w_j, & \text{for } Q'_{j,\text{cal}} < Q'_{j,\text{a,l}} , \end{cases} \quad (2.71)$$

and

$$E = \sum_{j=1}^n E_j, \quad (2.72)$$

where $Q'_{j,a,l}$ is the normalized value of the lower limit of the property j and $Q'_{j,a,u}$ the normalized value of the upper limit of the property j .

The error function of (2.72) has to be minimized to meet the required properties of the glass. For the minimization of the error function various mathematical tools can be used.

One method, which can be applied to various tasks and with which a personal computer can easily be programmed, is the Simplex method [2.98, 99]. For the optimization of a glass with n components, $n + 1$ different glass compositions are set up. During the calculation an error function, see (2.72), is calculated for every starting composition. The starting composition with the highest error function is now replaced by a new starting composition by reflection (Fig. 2.23). Thus, a new set of $n + 1$ different glass compositions is obtained for which the calculation of the error function and a new replacement can be performed. After several replacements (iteration steps) have been made, one of the calculated starting compositions reaches an optimum and the error function is minimized.

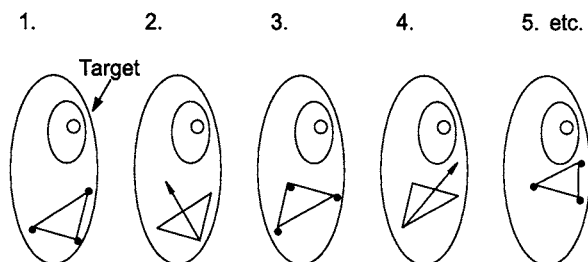


Fig. 2.23. Operation of the simplex method for a two-component system

The disadvantage of the simplex method is the large number of iteration steps necessary to reach the optimum. However, since a common iteration procedure is done within a few minutes it presents an acceptable method for the identification of melt experiments.

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2.3 Transmission and Reflection

Norbert Neuroth

2.3.1 General Relations

Let us assume that a light beam with the intensity I_0 falls on a glass plate having a thickness d . At the entrance surface a part of the beam is reflected (Fig. 2.24).

After that the intensity of the light beam is $I_e = I_0(1 - r)$ with r being the reflectivity. Inside the glass the beam is attenuated according to the exponential function. At the exit surface the beam intensity is

$$I_d = I_e \exp(-kd) , \quad (2.73)$$

where k is the absorption constant. At the exit surface another reflection occurs. The transmitted beam has the intensity

$$I_t = I_d(1 - r) . \quad (2.74)$$

These formulae give the following relation:

$$I_t = I_e(1 - r) \exp(-kd) = I_0(1 - r)^2 \exp(-kd) . \quad (2.75)$$

The beam reflected at the exit surface returns to the entrance surface and is divided into a transmitted and a reflected part (Fig. 2.25). This process is repeated infinitely. With the multiple reflections taken into account, the transmission of the glass plate is

$$\tau = \frac{I_t}{I_0} = P\tau_i , \quad (2.76)$$

with

$$\tau_i = \frac{I_d}{I_e} = \text{internal transmission} \quad (2.77)$$

and

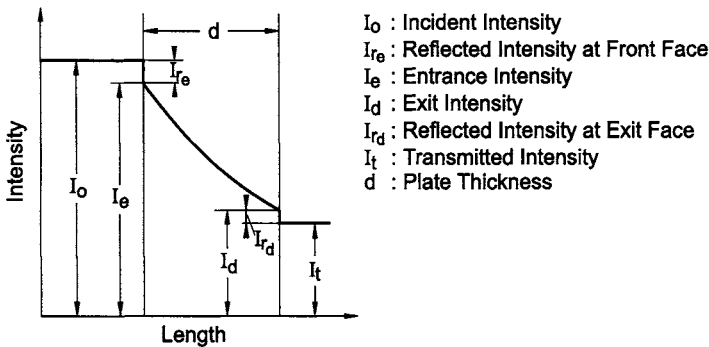


Fig. 2.24. Change in the intensity of a light beam passing through a glass plate

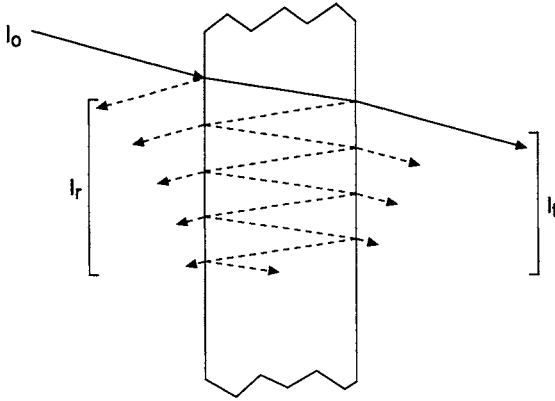


Fig. 2.25. Multiple reflection of a light beam at the surfaces of a glass plate

$$P \approx \frac{(1-r)^2}{1-r^2} = \frac{2n}{n^2+1}. \quad (2.78)$$

The internal transmission is that part of the beam attenuation that is caused by the glass itself, whereas the “reflection factor” P is caused by the reflections. The last formula has been derived from Fresnel’s formula which describes the relation between the reflectivity r and the refractive index n . The reflected intensity of the beam incident perpendicularly upon a plate consisting of dielectric material with the refractive index n is

$$r = \left(\frac{n-1}{n+1} \right)^2. \quad (2.79)$$

This relation is used in formula (2.78). The approximation of the first part of formula (2.78) causes deviations from the exact value of the order of 10^{-3} to 10^{-2} (depending on the r and τ_1 value) [2.100]. In Fig. 2.26 the reflection factor P is given as a function of the refractive index n .

Usually, the transmission τ of the plate is measured by a spectrophotometer [2.101–105]. In order to calculate the internal transmission τ_1 and the absorption constant k it is essential to know the refractive index, which is used to calculate the reflection factor P according to formula (2.78). If τ_1 is known, the absorption constant

$$k = \frac{1}{d} \ln \frac{1}{\tau_1} \quad (2.80)$$

can be computed.

When d is given in cm, k has the dimension cm^{-1} . If the refractive index is unknown, it is necessary to measure the transmission of another plate that is of the same material but has a smaller thickness d_2 . The ratio of the transmission τ_1 of the plate with the thickness d_1 to the transmission τ_2 of the plate with the thickness d_2 is the internal transmission of a plate with the thickness $d_1 - d_2$:

$$\tau_{1,1-2} = \frac{\tau_1}{\tau_2}. \quad (2.81)$$

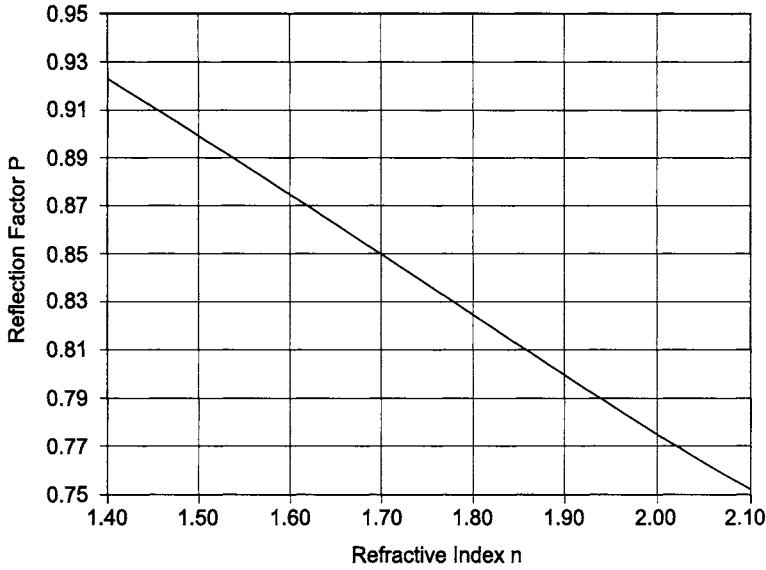


Fig. 2.26. Reflection factor P of a glass plate as a function of its refractive index

Then the absorption constant is

$$k = \frac{1}{d_1 - d_2} \ln \frac{1}{\tau_{11-2}} = \frac{1}{d_1 - d_2} \ln \frac{\tau_2}{\tau_1}, \quad d_2 < d_1. \quad (2.82)$$

k is the natural absorption constant. It is also common to use 10 as the base of the exponential function:

$$I_d = I_e \times 10^{-Ad}. \quad (2.83)$$

The absorbance A is calculated from

$$A = \frac{1}{d} \log \frac{1}{\tau_1}. \quad (2.84)$$

Note that k is 2.3 times larger than A .

Sometimes the attenuation of a glass plate is expressed as

$${}^{10}\log \frac{I_0}{I_t} = \text{optical density}. \quad (2.85)$$

This expression includes losses by absorption and reflection.

If the internal transmission τ_{11} of a plate with the thickness d_1 is known, it is possible to calculate the internal transmission τ_{12} of a plate with a thickness d_2 :

$$\log \tau_{12} = \frac{d_2}{d_1} \log \tau_{11}. \quad (2.86)$$

The transmission τ_2 is obtained by multiplying τ_{12} with the reflection factor P .

2.3.2 Sources of Optical Loss in Glass

Intrinsic Absorption

Glasses are dielectric materials. Normally (in the case of oxide glasses) they are transparent in the near-ultraviolet, the visible and the near-infrared. Unlike metals, they are transparent because the electrons of the atomic shells are bound to the individual atom. With metals there are free electrons which can oscillate with the frequency of an incident electromagnetic wave. Therefore the light is strongly reflected. Although the electrons of the glass are bound they can be excited to higher energy levels by ultraviolet radiation. In Fig. 2.27 the ultraviolet reflection spectra of glasses of the system $x\text{Na}_2\text{O}-(1-x)\text{SiO}_2$ are given [2.106]. Pure SiO_2 glass has relatively strong and narrow reflection peaks at 122 and 108 nm and shorter wavelengths. If sodium oxide is added, these bands become broader. According to Sigel [2.107, 108] this is due to a change in the oxygen surrounding. In a pure SiO_2 glass SiO_4 tetrahedra are the basis of the molecular structure. They are forming a special network. Each oxygen atom has two neighbouring Si atoms (bridging oxygens). The presence of sodium in glass produces oxygen atoms that have at their sides a Si atom and a Na atom, respectively. They are called non-bridging oxygen atoms. The excitation energy levels of their electrons are varied. Consequently, with an increasing sodium content, the UV-reflection bands are broadened and an additional band appears at about 170 nm. Figure 2.28 shows the change in the UV transmission of $\text{Na}_2\text{-SiO}_2$ glasses. Even a very small Na_2O content reduces the UV transmission considerably. If the glass

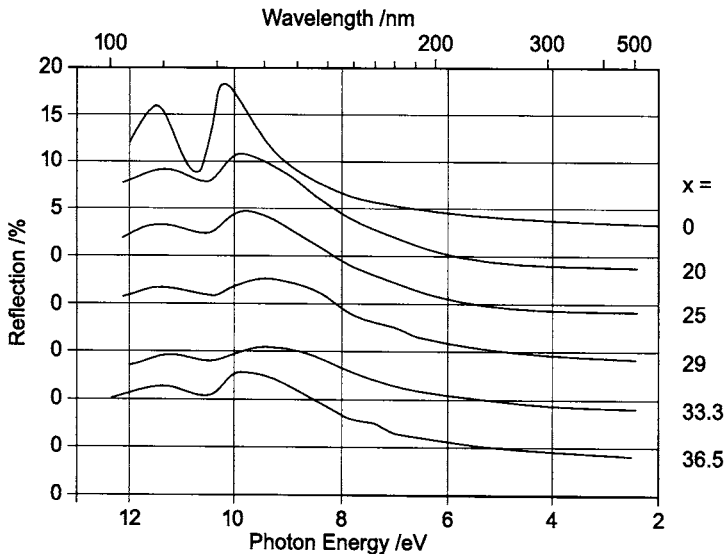


Fig. 2.27. Ultraviolet reflection spectra of glasses of the system $x\text{Na}_2\text{O}-(1-x)\text{SiO}_2$ [2.106]

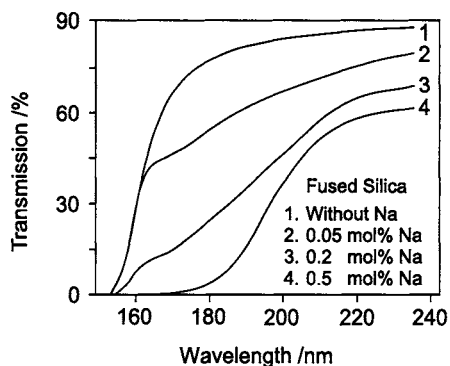


Fig. 2.28. Ultraviolet transmission of silica glass with small amounts of Na_2O [2.107]; sample thickness about 1 mm

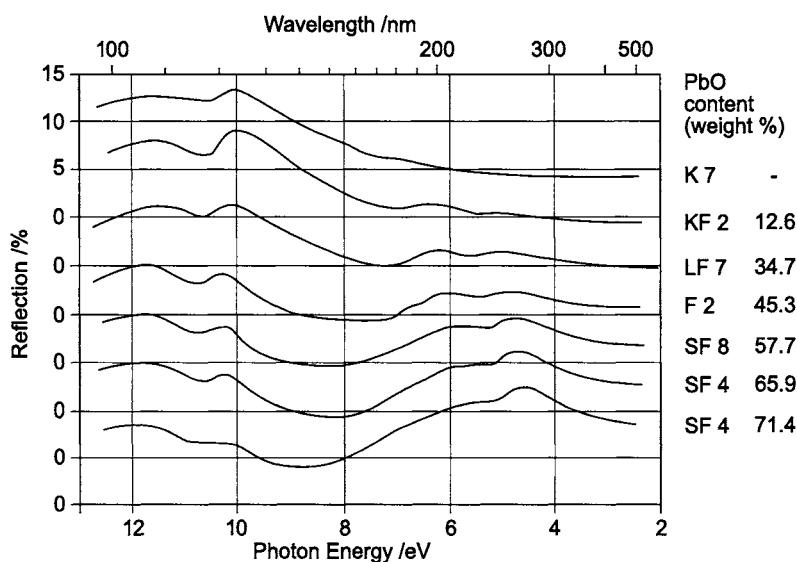


Fig. 2.29. Ultraviolet reflection of silicate glasses [2.106]

contains heavier elements, e.g. lead oxides, two other reflection bands appear between 180 and 280 nm; see Fig. 2.29 [2.106].

The other source of intrinsic loss is molecular vibrations. Figure 2.30 presents two vibration modes of the SiO_4 tetrahedra: the Si-O stretching and bending vibrations of the SiO_4 tetrahedra [2.109]. In the right hand side of Fig. 2.30 the infrared reflection spectra of the SiO_2 glass and of the $x\text{Na}_2\text{O}-(1-x)\text{SiO}_2$ glass system are shown [2.110]. The stretching vibration $\text{Si}\rightarrow\text{O}$ causes the $8.9\mu\text{m}$ reflection peak. If Na_2O is added, the peak becomes weaker as well as broader and is shifted to longer wavelengths. Another peak occurs in the range of 10.4 to $10.8\mu\text{m}$; it is due to vibrations of non-bridging oxygens. In this range the absorption of glass is very strong: the absorption constant is between 10^4 and 10^5 cm^{-1} . Such a strong absorption of the basic oscillation also causes attenu-

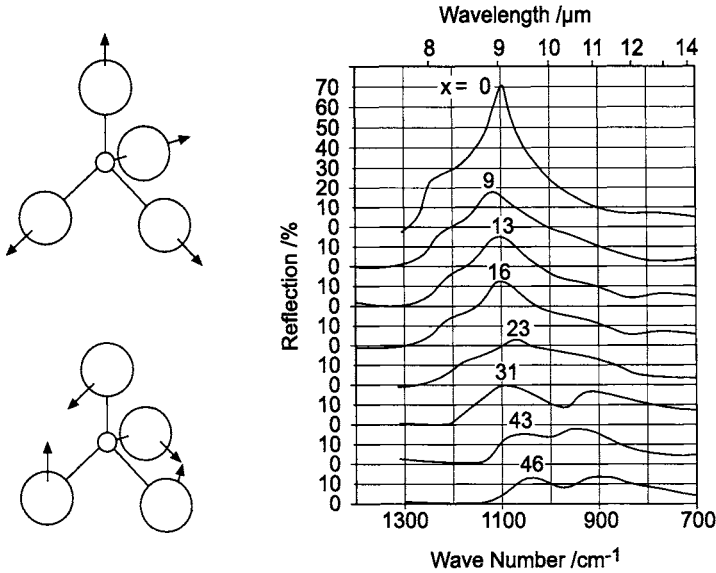


Fig. 2.30. (left) Two vibration modes of a SiO_4 tetrahedron [2.109]. (right) Infrared reflection spectra of glasses of the system $x\text{Na}_2\text{O}-(1-x)\text{SiO}_2$ [2.110]

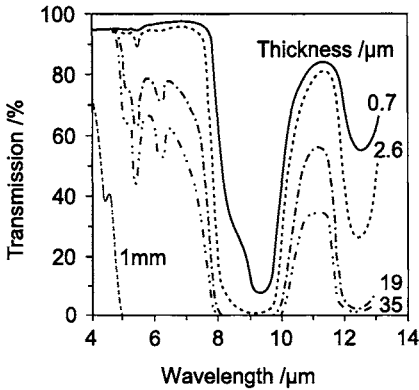


Fig. 2.31. Infrared fundamental oscillations and first overtone oscillations of fused silica [2.111, 112]

ations at the overtone frequencies. In Fig. 2.31 the transmission of fused silica with thickness values of 0.7 μm to 1 mm is shown [2.111, 112]. The fundamental oscillations induce absorption in the range of 8 to 13 μm (maxima at 8.2 (shoulder), 9.4 and 12.4 μm). Even the first overtones at 5.0, 5.3 and 6.2 μm lead to strong absorption (up to 200 cm^{-1}). Therefore a glass sample that is 1 mm thick becomes transparent at wavelengths smaller than 4.5 μm , i.e., at half of the wavelength of the fundamental oscillation.

The strong UV- and IR-absorption phenomena that have been described determine the limitations of the spectral transmission. They are caused by the material itself (intrinsic loss).

Impurities

Some special oxides show absorption in the transparent spectral region of a glass. These are the oxides of the transition elements Ti, V, Cr, Mn, Fe, Co, Ni, Cu and many of the rare earths [2.113–117]. Figure 2.32, for instance, shows the spectral internal transmission of a BK 7 glass doped with 1 ppm of the oxides of the different transition elements [2.118]. The absorption spectra of several transition elements depend on the valency of the specific element. In optical glasses, the presence of colouring elements has to be avoided although they may be useful for the production of a coloured glass. Usually, the content of iron oxide in optical glasses is in the range of 1 to 10 ppm, whereas the concentrations of the other transition elements are less than 1 ppm. As mentioned above, these impurity levels are still too high to be acceptable for glass fibres used in telecommunication, since for this usage impurity levels within the range of ppb are required. The traditional production techniques are unsuitable. By means of the chemical vapour deposition technique, however, it is possible to produce optical fibres with a loss that is 10^{-4} times lower than that arising in the production of standard optical glasses.

Scattering

The phenomenon of light scattering in glass may be caused by undissolved particles (such as stones, metals, crystals), or by bubbles, phase separation and/or density fluctuations. The first three reasons for scattering given here are as a result of the melting technique. The density fluctuations are provoked by the Brownian movement of the molecules in the melt [2.119]. Heavy components (e.g., PbO) and light components (e.g., SiO₂, Na₂O) may separate as a result of the thermal movements of the molecules in the melt. These small separations of lighter and heavier components cause fluctuations of the density and of the refractive index. If the melt is annealed, these fluctuations of the refractive index are frozen in.

The size of the fluctuations is much smaller than that of the wavelengths. According to Rayleigh, the scattering in this case depends on the wavelength λ corresponding to λ^{-4} . The intensity of the scattered light (wavelength 546 nm) at an angle of 90° to the direction of the incident light is within the range of 2 to 70×10^{-6} of the incident intensity, depending on the type of the glass. The chemical composition and the melting technique influence the exponent m in the relationship

$$I_{\text{scat}} \approx \frac{1}{\lambda^m}, \quad (2.87)$$

where m varies between 3.5 and 4.8 [2.120]. Crown glasses have the lower m values, flint glasses the higher ones. Figure 2.33 shows the light scattering of some optical glasses as a function of the wavelength. If the size of the particles is comparable with the wavelength, the curves are less steep (m is lower than 4, Mie scattering).

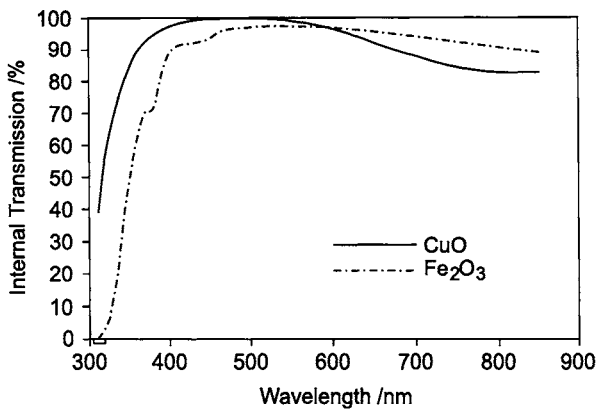
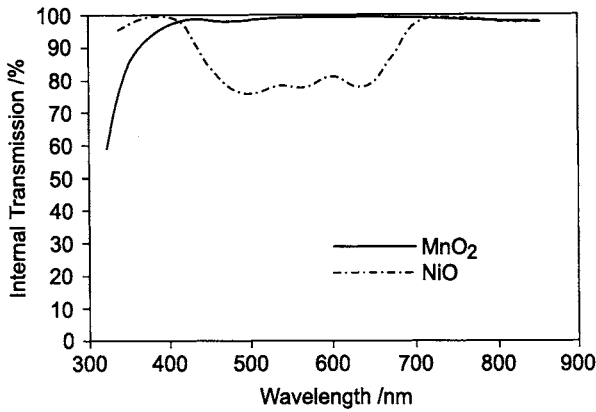
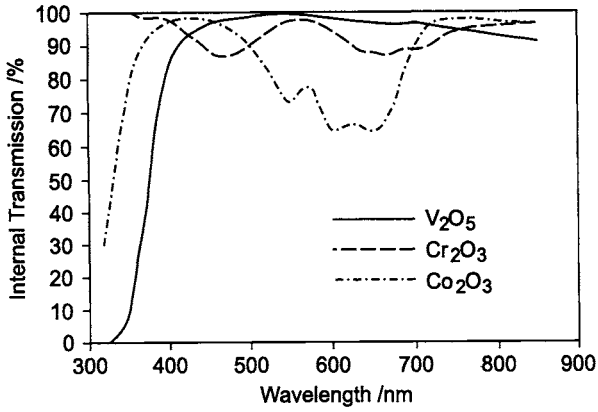


Fig. 2.32. Spectral internal transmission of a coloured BK 7 glass by different oxides; sample thickness 100 mm [2.118]

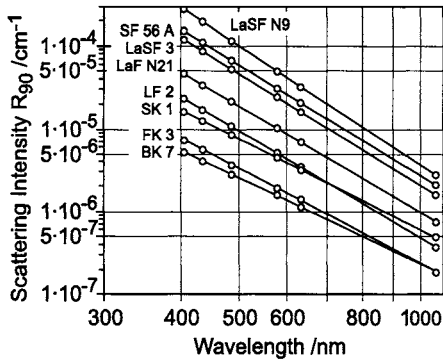


Fig. 2.33. Light scattering R_{90} of optical glasses as a function of the wavelength [2.120]

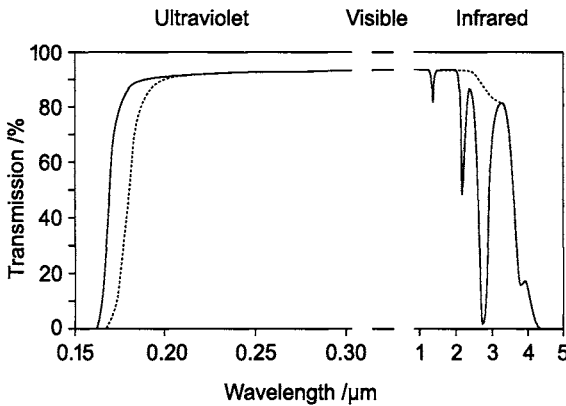


Fig. 2.34. Transmission of fused silica; thickness 10 mm (— OH-containing, --- with a low OH content) [2.112]

2.3.3 Examples of Transmission Spectra

Figure 2.34 illustrates the transmission of 10 mm thick plates consisting of fused silica with different water contents. They represent typical glasses with a good UV transmission. Fused silica that has been produced synthetically is free of impurities apart from the OH content, which causes absorption bands in the infrared region. By using a special production method, the OH content, can also be reduced significantly.

In Fig. 2.35 the spectral transmission of three types of optical glass with different refractive indices is shown. The glass with the lower refractive index achieves the better UV transmission. High-index glasses containing heavier elements have UV absorption at longer wavelengths. The UV-absorption edge of optical glasses that are 5 mm thick is to be found between 280 and 390 nm. Lenses for cameras are up to 20 mm thick. The high-index glasses can show a weak absorption in the blue region. Pieces that are 20 mm thick may have a yellowish colour. As described above, there are two reasons for this loss: the tail

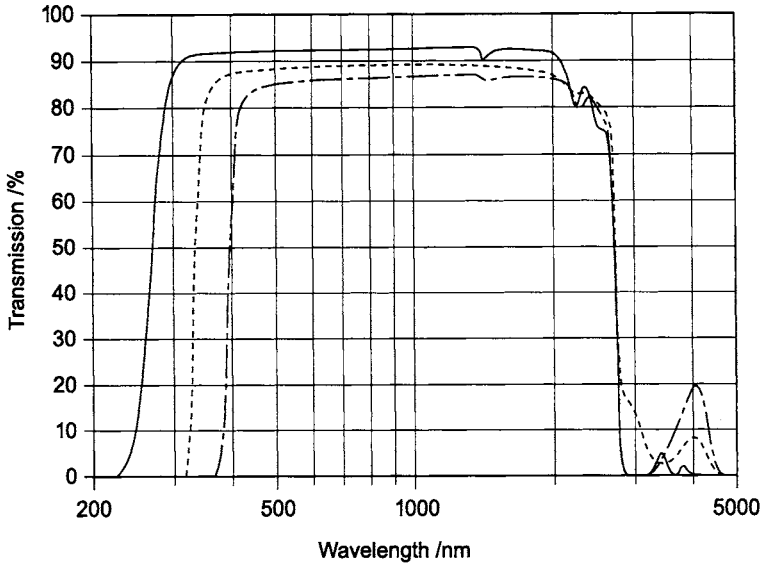


Fig. 2.35. Transmission of three types of optical glass; thickness 5 mm (— FK 5, --- SF 2, - · - · SF 11) [2.118]

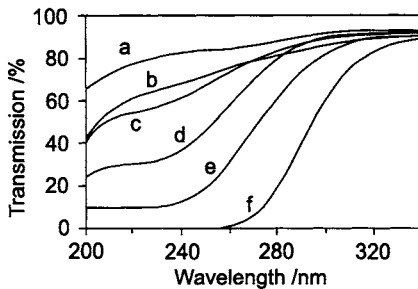


Fig. 2.36. Transmission of optical glasses with improved ultraviolet transmission; thickness 5 mm. (a) Ultran® 10, (b) Ultran® 30, (c) Ultran® 20 (preliminary values, see Sect. 8.5), (d) Ultran® 15, (e) Ultran® 25, (f) UBK-7 [2.121]

of the short-wavelength absorption, caused by the electronic transitions of heavy elements in the glass and/or impurities such as iron (Fe^{3+}), platinum, titanium or cerium.

If the glass is to have a good UV transparency, it must contain neither heavy elements nor any of the impurities mentioned. In Fig. 2.36 the transmission of optical glasses with an improved ultraviolet transmission is given [2.121]. The UV transmission of the types a, b and c nearly reaches that of fused silica or CaF_2 crystal, which are standard UV materials. These glass types also have an extremely anomalous dispersion (see Fig. 2.37), a low temperature coefficient of the optical pathlength and a low stress-optical constant (see also Sect. 8.3).

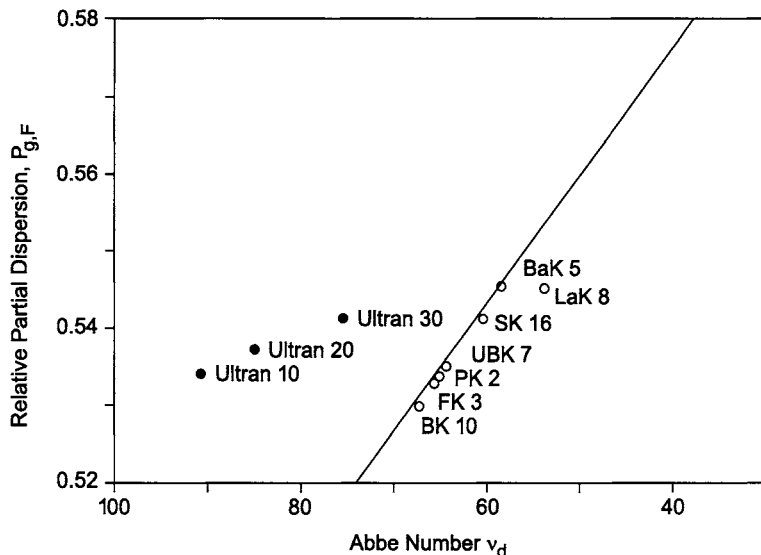


Fig. 2.37. Relative partial dispersion of UV-transparent and other optical glasses [2.121]. The line indicates the position of glasses with “normal partial dispersion”

The impurity most crucial for the infrared transmission of glasses is the OH content. In Fig. 2.35 typical OH-absorption bands are to be seen. The OH absorption is sensitive to the atomic surroundings [2.122]. If the distance between the proton and another oxygen is great ($\geq 3 \text{ \AA}$), the OH-stretching vibration causes an absorption band at $2.9 \mu\text{m}$ (Fig. 2.35, glass SF 11). If the distance is smaller (about $2.5 \text{ \AA}$), the force constant of the OH-stretching vibrations is weaker and the absorption bands occur at about $3.6 \mu\text{m}$ (sometimes also at $4.2 \mu\text{m}$). Often both types of OH bands (2.9 and $3.6 \mu\text{m}$) with varying ratios of intensities are present at the same time in a glass (see Fig. 2.35, glass SF 2). If the water content is high and/or the sample is thick, the overtones of OH vibrations are also to be seen at 2.2 and $1.4 \mu\text{m}$ (see Fig. 2.34).

Other sources of infrared absorption are the overtone of the B–O stretching vibration at $3.7 \mu\text{m}$ and the electron transition band of Fe^{2+} at $1.1 \mu\text{m}$ (very broad).

One possibility to improve the infrared transmission of glasses is to reduce the water content and minimize the Fe content. A further step is to select glass formers with lower molecular vibration frequencies, (e.g., the oxides of Ge, As, Te, Bi, Pb, Ga, the fluorides of Zr, Y, Hf, the chalcogenides of As, Ge, Sb, Ga). In Fig. 2.38, a selection of typical infrared glasses is given (for further information see Sect. 8.4). It is possible to produce glasses with a transmission up to $35 \mu\text{m}$ (not yet commercially available).

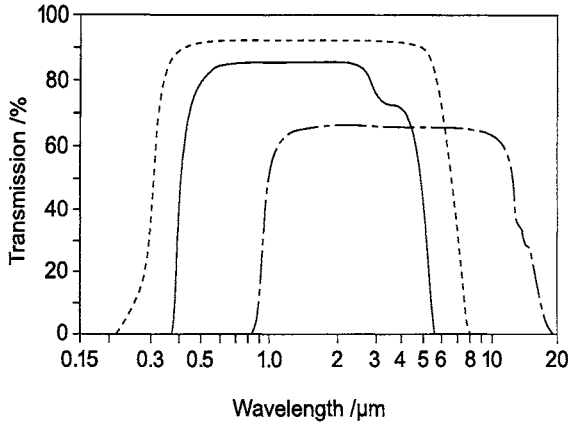


Fig. 2.38. Three types of infrared glasses; thickness 5 mm (— germanium oxide, --- fluoride, - · - · chalcogenide)

2.3.4 Influence of Temperature

If the temperature of a glass is increased, the ultraviolet absorption edge is shifted to longer wavelengths (Fig. 2.39) and the infrared-absorption edge to smaller wavelengths (Fig. 2.40). This results from the broadening of the strong ultraviolet- and infrared-absorption bands. Also the OH-vibration bands are influenced by temperature. At higher temperatures the atomic distances are en-

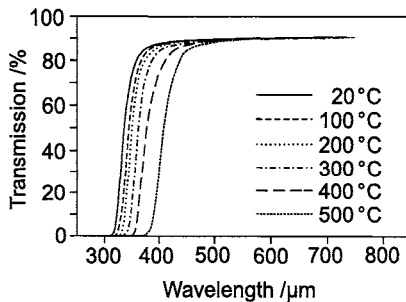


Fig. 2.39. Influence of temperature on the ultraviolet transmission of flint glass F2; thickness 10 mm [2.118]

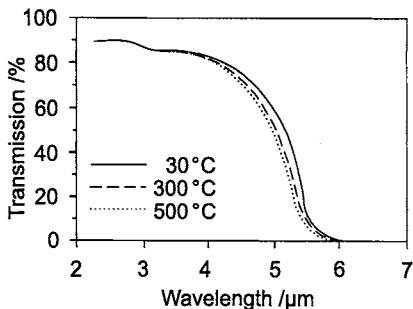


Fig. 2.40. Influence of temperature on the infrared transmission of the glass IRG2; thickness 5 mm [2.118]

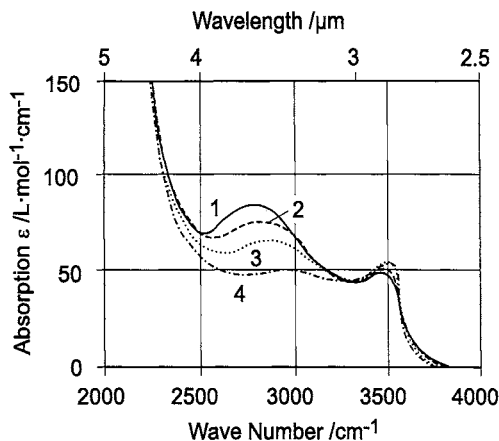


Fig. 2.41. Influence of temperature on the OH-absorption band in a $\text{Na}_2\text{O}-\text{CaO}-\text{SiO}_2$ glass. 1: 30 °C. 2: 276 °C. 3: 378 °C. 4: 576 °C [2.122] part IV

larged. Thus the number of OH groups whose protons are influenced by other oxygen atoms is diminished. That means that the 3.6 and 4.2 μm absorption bands become less intense at higher temperatures and the number of undisturbed OH groups with absorption at 2.9 μm increases (Fig. 2.41 [2.122]).

2.3.5 Influence of Radiation on the Transmission

Electromagnetic radiation may influence the transmission of optical glasses. Highly intensive visible light radiation can reduce the transmission especially in the blue region. The irradiation with electromagnetic radiation with shorter wavelengths generally has a greater influence: ultraviolet radiation, X-rays, gamma rays and particle beams (electrons, protons or other ions and neutrons). The irradiation can generate defect centres of different nature: ionization, trapped electrons, trapped holes, ruptured Si–O bonds, non-bridging oxygens [2.123]. Impurities in the glass can also be influenced by the radiation and therefore induce absorption (for instance by changing the valency). In Fig. 2.42 (left), the transmission of SK 5 glass is shown before and after irradiation with ultraviolet light (200–400 nm) of a xenon lamp, whose power density is 1.2 W/cm² [2.124]. Polyvalent ions in glass, especially cerium, are able to diminish or to prevent the discolouration by irradiation. Figure 2.42 (right) shows the transmission of SK 5 glass doped with CeO_2 before and after irradiation with UV light in the same manner. The doping causes the absorption edge to shift from approximately 290 nm (undoped glass) to 390 nm. After that the influence of the UV radiation is virtually zero.

In Fig. 2.43 the influence of gamma radiation (^{60}Co source, 10⁴ Gy) on the glass BK 7 is illustrated [2.125]. The darkening is very severe (curve b). The doping with CeO_2 protects the glass against darkening. Curve c shows the transmission of Ce doped BK 7-G 25 that has not been irradiated. The doping makes the absorption edge shift from 310 to 370 nm. The influence of gamma radiation (curve d) is now very small. A CeO_2 -doped BK 7 glass shows similarly

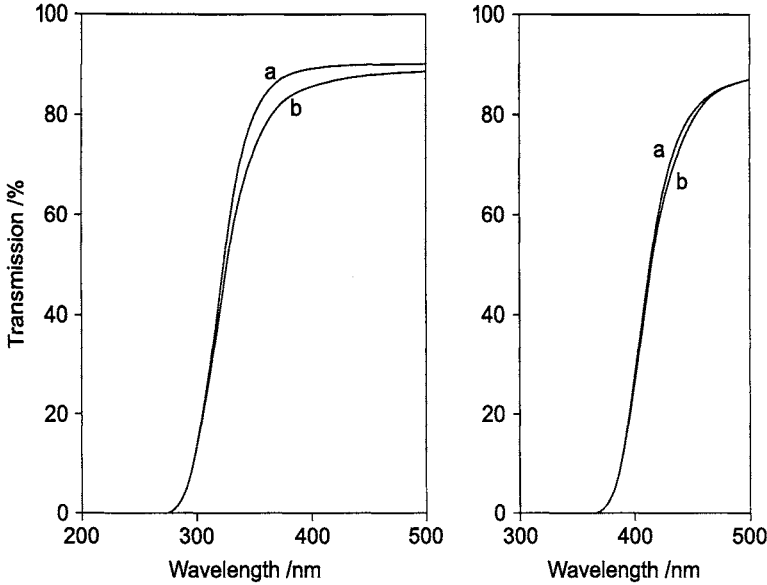


Fig. 2.42. Influence of UV radiation of a xenon lamp (200–400 nm; 1.2 W/cm^2) on the transmission of (left) SK 5 glass, (right) SK 5-G 6 glass, protected by CeO_2 : (a) unsolarized, (b) solarized [2.124]; sample thickness 10 mm

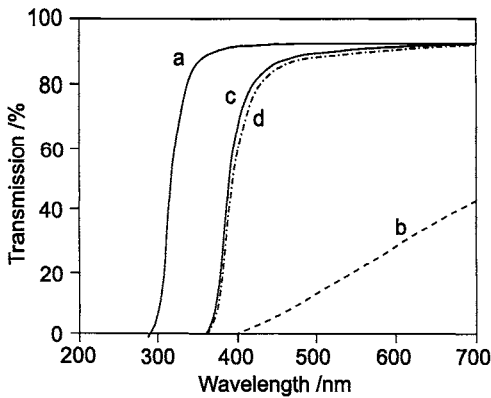


Fig. 2.43. Influence of gamma radiation (source ^{60}Co , 10^4 Gy) on the spectral transmission of BK 7 glass, sample thickness 10 mm, without Ce doping: (a) not irradiated; (b) irradiated. With Ce doping (BK 7-G 25): (c) not irradiated; (d) irradiated [2.125]

small changes in transmission after irradiation with proton or electron beams. A stronger darkening is caused by neutron irradiation [2.125]. More details are given in the literature [2.124–133].

Acknowledgement

I thank R. Haspel, H.J. Becker, T. Korb and H. Schwartz for the measurement and calculations, respectively, and representation of the Figs. 2.26, 2.32, 2.35, 2.39, 2.40 and I am grateful to W. Jochs and G. Westenberger for Fig. 2.33.

2.4 Differential Changes of the Refractive Index

Hans-Jürgen Hoffmann

2.4.1 General Remarks

The presence of matter is prerequisite to a refractive index n different from 1. On the other hand, the use of matter is correlated with absorption of electromagnetic waves. This necessarily causes dispersion of n , i.e., the refractive index changes with the wavelength λ . Dispersion is the most prominent and most important variation of the refractive index because of its order of magnitude, which can be as large as 0.1 over the visible spectral region. It is the main reason for the rich variety of optical glasses used to achieve a good quality of achromatic imaging, which has been discussed in Sect. 2.1. In the following, other causes for minor – differential – changes of the refractive index will be considered.

Since for applications of glasses in the atmosphere the refractive index is given relative to air by

$$n_{\text{rel}}(\lambda) = \frac{n_{\text{abs}}(\lambda)}{n_{\text{air}}(\lambda)}, \quad (2.88)$$

differential changes in $n_{\text{rel}}(\lambda)$ can be caused by changes in both the (absolute) index of the material, n_{abs} , and the index of the ambient medium, n_{air} . The influence of possible changes in n_{air} is considered in the next subsection. The other differential changes to be discussed here are due to the optical material itself. Changes of the refractive index caused by the thermal treatment are specific to glasses (we do not consider changes due to small variations of the chemical compositions, to different raw materials or accidental impurities or to diverse melting parameters). Knowledge of the modifications induced by thermal treatment of glasses is important to achieve reproducible properties of optical elements. In order to obtain homogeneous materials, all parts of a glass sample have to undergo the same temperature–time programme.

Another homogeneous change of n can be caused by a shift in the ambient temperature. If the optical elements are to be used in an environment with changing temperature, the refractive index shifts because of both a shift in the intrinsic material properties and in the refractive index of the ambient medium. To predict this shift, the temperature coefficient of n or thermo-optical coefficient as a function of λ and of the temperature T has to be known.

Some external intensive parameters (“forces”) cause anisotropies in a glass. Since optical glasses are (statistically) isotropic materials, the anisotropic behaviour can be characterized by a minimum number of independent tensor parameters. Stress-optical effects are well known in this respect. Stress changes n for the different planes of vibration of linearly polarized electromagnetic waves. Thus, in order to keep the good homogeneity and isotropy of optical glasses, it is mandatory to avoid mechanical stress in optical elements.

Electric fields, too, can cause anisotropy in glasses for linearly polarized electromagnetic waves. In order to effect appreciable changes, however, large electric field strengths are required. Under normal circumstances this hardly occurs. Therefore, this effect is of minor importance.

Anisotropies for circularly polarized electromagnetic waves are caused by magnetic fields. The difference between the changes for right- and left-handed circularly polarized electromagnetic waves can be detected by investigations of the Faraday rotation. This effect can be used for the modulation of light beams and for non-reciprocal optical elements.

The contributions of the phenomena mentioned so far will be considered in the following.

2.4.2 The Refractive Index of Air and Its Variations

The absolute refractive index of air has been the subject of many studies in the past. Its value and its dispersion are relevant for imaging optics in the atmospheric environment (see, [2.134–139] and references cited therein). Since the refractive index of air, n_{air} , can be determined very precisely by interferometric methods with an accuracy of the order of 10^{-7} or better, whereas the accuracy for the refractive index of glass is usually limited to the range of 10^{-5} to 10^{-6} , many dispersion formulae for n_{air} are useful. In the following we refer to the relationship

$$n_{\text{air}}(15^\circ\text{C}, p_0) = 1 + 10^{-8} \left(6432.8 + \frac{2\,949\,810\,\mu\text{m}^{-2}\lambda^2}{(146\,\mu\text{m}^{-2}\lambda^2 - 1)} + \frac{25\,540\,\mu\text{m}^{-2}\lambda^2}{(41\,\mu\text{m}^{-2}\lambda^2 - 1)} \right), \quad (2.89)$$

which was accepted by the Joint Commission for Spectroscopy in Rome in September 1952 as cited in [2.134]. Relation (2.89) applies in the range of wavelengths between $0.2\,\mu\text{m}$ and $1.35\,\mu\text{m}$ for dry air containing 0.03% CO_2 by volume at a temperature of 15°C and a pressure $p_0 = 0.101325 \times 10^6\,\text{Pa}$ (corresponding to $760\,\text{Torr}$). Curve a in Fig. 2.44 shows n_{air} as a function of the wavelength for these parameters.

Its value is roughly 1.0003 in the visible part of the spectrum. Consequently, the contribution of n_{air} to n_{rel} of any glass cannot be neglected if high precision is required. The refractive index of air changes with the temperature, T_c (in degrees centigrade), the atmospheric pressure, p , and the partial pressure of additional gaseous components, of which, for example, the partial pressure w of

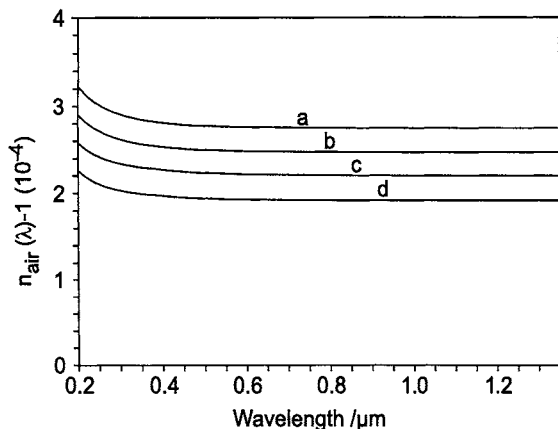


Fig. 2.44. $n(\lambda) - 1$ (n : refractive index) of dry air containing 0.03% CO_2 as a function of the wavelength according to (2.89) and (2.90) at 20°C . (a) $p_0 = 0.101325 \times 10^6 \text{ Pa}$, (b) $0.9 p_0$, (c) $0.8 p_0$, (d) $0.7 p_0$

water vapour is important. The changes due to T_c , p and w can be described according to [2.134] by

$$n_{\text{air}} = 1 + \frac{n_{\text{air}}(15^\circ\text{C}, p_0) - 1}{1 + \frac{\alpha}{1+15^\circ\text{C}\alpha}(T_c - 15^\circ\text{C})} \frac{p}{p_0} - \frac{413 \times 10^{-12} \text{ Pa}^{-1} w}{1 + \alpha T_c}, \quad (2.90)$$

where $\alpha = 3.67 \times 10^{-3}/^\circ\text{C}$. The last term in (2.90) applies only to the visible spectral region; see [2.134] p. 408.

If there are other constituents i in the ambient atmosphere, their contribution to the refractive index of the surrounding medium can be taken into account by the relation

$$\Delta n_i(T, p) = n_i(T, p) - 1 = \frac{n_i(T_0, p_0) - 1}{\rho_i(T_0, p_0)} \rho_i(T, p), \quad (2.91)$$

where $\rho_i(T_0, p_0)$ and $\rho_i(T, p)$ are the mass densities of additional constituents at the reference partial pressure p_0 and the temperature T_0 , and at the partial pressure p and temperature T , respectively. Equation (2.91) is based on the dispersion formula of *Gladstone-Dale* [2.140],

$$n - 1 = \text{const.} \times \rho, \quad (2.92)$$

which was developed to describe the refractive index of liquids. It applies also to less-dense materials, such as gases. However, there are no general a priori rules for the constant of proportionality or its dispersion. Therefore, the data have to be determined experimentally or taken from the literature.

Any variation of the composition of the ambient atmosphere, its temperature or the atmospheric pressure changes not only the relative refractive index of the optical glass but also its dispersion to a small extent. Fluctuations of the atmospheric pressure by about 2% correspond to changes of n_{rel} by $(5-6) \times 10^{-6} n_{\text{rel}}$.

Other variations may be due to changes of the height, h , above sea level. The atmospheric pressure of each component of the air decreases – in the isothermal case and considered as an ideal gas – according to the barometric pressure equation

$$p_i(h) = p_i(0) \exp\left(-\frac{M_i g}{k_B T} h\right) \quad (2.93)$$

(for a derivation of this formula, see e.g. [2.141]; M_i , mass of a molecule of kind i ; g , acceleration of gravity; k_B Boltzmann's constant; T , absolute temperature). In addition, the temperature and the relative composition of the atmosphere change. With these changes neglected, a moderate shift of 300 m in the position above sea level causes a variation of about 10^{-5} of n_{air} and – as a consequence – a shift in the relative refractive index of optical glasses by $10^{-5} n_{\text{rel}}$.

2.4.3 Thermal Treatment of Glasses and the Refractive Index

The stress relief as a function of time, t , has been described by a linear and a quadratic relaxation model. In a linear model one assumes that the rate of stress relief is proportional to the stress, σ , that is left, namely

$$\frac{d\sigma}{dt} = -\frac{\sigma(t)}{\tau(T)}, \quad (2.94)$$

with time variable t and the relaxation time constant, τ , which depends on the temperature, T . If τ does not depend on the stress, the solution of (2.94) is

$$\sigma(t) = \sigma_0 \exp\left(-\frac{t}{\tau}\right), \quad (2.95)$$

with the initial value of the stress, σ_0 .

In a quadratic model one assumes for the rate of stress relief [2.142]

$$\frac{d\sigma}{dt} = -a \sigma^2, \quad (2.96)$$

with the constant of proportionality a . The solution of (2.96) is

$$\sigma(t) = \frac{\sigma_0}{1 + \sigma_0 a t} \quad (2.97)$$

if $\sigma = \sigma_0$ for $t = 0$.

Both equations (2.95) and (2.97) are approximations to the relaxation of the stresses as pointed out by *Berger* [2.143].

Stresses in glass relax within about 15 minutes if the viscosity is 10^{12} dPas [2.144]. The viscosity reaches this value in many cases at about 20 to 80 °C above the transformation temperature, T_g . Technical glasses that are 1 cm or less thick are annealed at those temperatures with constant rates of the order of 1 °C/min. This is sufficient for the stress relief in technical glasses. On the other hand, even small stresses due to thermal gradients in the annealing range cause permanent strains and inhomogeneities of the refractive index. This can

be detrimental to optical glasses of high quality. Thus, optical glasses have to be annealed very carefully in order to provide the good optical homogeneity required for transmissive optical elements. The effects of precision annealing on the homogeneity of optical glasses have been investigated by *Reitmayer* and *Schuster* [2.145]. They succeeded in reducing the inhomogeneity of the refractive index to values as low as $\pm 3 \times 10^{-7}$ in a BK 7 disk (diameter 400 mm, thickness 47 mm).

Even if there are no inhomogeneities due to stresses, there still exist driving forces for the glass to relax. The reason for this is that the glass reaches a non-equilibrium state in the process of solidification from the melt, since the minimum of Gibbs' free energy for solids is obviously in the ordered (i.e. crystalline) state. Glasses are solids without long-range order. Therefore, there are driving forces on an atomic scale to reach a state of advanced order. As a consequence, glasses tend to relax to a state with less Gibbs' free energy. This relaxation is impeded by kinetic parameters which depend on the composition of the glass. Since the relaxation requires structural rearrangements of the constituents, such effects are predominantly induced within the temperature range near T_g (but they can also occur at much lower temperatures, as will be shown later). The fact that the specific volume of a glass depends on the annealing rate near T_g (see textbooks on glass or ceramics such as [2.146, 147]) is very well known. Corresponding to the variation of the specific volume, many other parameters of a glass depend more or less on the state of relaxation. Among these parameters are the mass density, the coefficient of linear thermal expansion, the refractive index, the hardness, the transformation temperature, thermodynamic data such as the specific heats, etc. *Berger* [2.143] gives a survey of the early work on the dependence of physical parameters of glass as a function of the thermal history.

There are numerous experimental investigations of the changes of parameters by thermal treatment [schlag] thermal stresses in the transformation range. The following is an arbitrary selection of publications.

Brandt has determined the refractive index of a borosilicate crown glass as a function of the annealing time at a given temperature in order to derive annealing schedules for predetermined index homogeneity [2.148].

Lillie and *Ritland* have studied the influence of different thermal treatments on a borosilicate glass [2.149]. In particular they applied the methods of "constant-temperature soaking" and "constant cooling rate". They recommended the method of constant cooling rate, because this procedure insures a uniform thermal history of the glass and it requires less total time. Each part of a glass sample experiences to a very good approximation the same temperature-time programme if one starts the annealing at a constant (slow) rate sufficiently above T_g . The temperature differences have to be as small as possible in order to avoid thermal stresses during and after the annealing process. For an infinite plate of thickness d which is annealed at a constant rate $v = dT/dt$, *Lillie* and *Ritland* published the relation [2.149]

$$\Delta T = \frac{v}{8C} d^2 \quad (2.98)$$

for the maximum difference in temperature between the centre and the surface of the plate with the thermal diffusivity $C = k/\rho c_p$, where k is the coefficient of thermal conductivity, ρ is the mass density and c_p is the specific heat capacity at constant pressure. Inserting typical values for optical glasses yields, with $d = 1$ cm and $v = 1^\circ\text{C/h}$, temperature differences ΔT of the order of 0.01°C and 1°C if v is 100 times as fast.

Stresses σ caused by temperature differences ΔT can be estimated using [2.144]

$$\sigma = \frac{\alpha E}{1 - \mu} \Delta T, \quad (2.99)$$

where the coefficient of linear thermal expansion is α , Young's modulus is E and Poisson's ratio is μ . Inserting $\alpha = 10^{-5}/^\circ\text{C}$, $E = 7.5 \times 10^4 \text{ N/mm}^2$, $\mu = 0.25$ and $\Delta T = 0.01^\circ\text{C}$ yields stresses of about 10^{-2} N/mm^2 . (In this respect it is worth mentioning that for large pieces, the weight of the glass itself can cause even larger stresses.) Stresses of that order of magnitude, however, do not cause the large differences in the refractive index observed when the annealing rates are varied.

From a practical point of view it is important to predict the changes of the refractive index, e.g. with the parameters of the thermal treatment in the transformation range. For this purpose many correlations between experimental quantities have been investigated.

Tool, *Tilton* and *Saunders* [2.150] and *Spinner* and *Waxler* [2.151] have determined the change of the mass density, ρ , and of the refractive index, n , of various optical glasses induced by annealing and by hydrostatic pressure. According to their results, the increase of n was larger for a given increase of ρ upon annealing than for the same increase of ρ upon compression. Thus, the change of n upon annealing is not caused by a variation of the density alone. Instead one must consider that some restructuring also takes place during annealing.

Prod'homme has determined the temperature coefficient of the refractive index, n , the mass density, ρ , and the coefficient of linear thermal expansion, α , of various optical glasses and their modifications caused by thermal treatment near T_g [2.152]. In his comprehensive work he reported on measurements of n and ρ as a function of time at different temperatures and discussed several possibilities to interpret the relaxation. In particular, the relaxation was correlated with the change of the viscosity as a function of time. The increase of the viscosity with time had been reported earlier by *Lillie* [2.153, 154].

To describe the relaxation behaviour of glasses near T_g , *Tool* has introduced the concept of the *fictive temperature* [2.155]. Some limitations of this concept have been discussed by *Ritland* [2.156]. More-recent models also take advantage of the concept of the fictive temperature (see [2.157–159]). As these models unfortunately require too many parameters they are in most cases not useful for predictions.

Since the refractive index, n , depends on the parameters of the thermal treatment, such as temperature, duration and annealing rate, one would like to profit

from this effect to shift and correct n by small amounts, as can be necessary to meet the requirements in production. One thus needs accurate prescriptive data on how much one has to change the annealing parameters in order to reproduce a given value of n . For the special case of a constant annealing rate v this can be achieved. For a special glass, *Lillie* and *Ritland* published the relation [2.149]

$$\Delta n_d \left(\frac{v}{v_1} \right) = m_d \log \left(\frac{v}{v_1} \right), \quad (2.100)$$

with the reference annealing rate v_1 and the constant of proportionality m_d for the d line. Relation (2.100) has been applied to other glasses by *Rötger* and *Besen* [2.160, 161] and *Danyushevskii* [2.162] (as cited in [2.160, 161, 163]).

Figure 2.45 shows for an arbitrary selection of optical glasses the variation of Δn_d with the (uniform) annealing rate v .

For a standard annealing rate v_2 different from v_1 the corresponding curves in Fig. 2.45 are just shifted by $-m_d \log(v_2/v_1)$, since one obtains from (2.100)

$$\begin{aligned} \Delta n_d \left(\frac{v}{v_2} \right) &= \Delta n \left(\frac{v}{v_1} \frac{v_1}{v_2} \right) = m_d \log \left(\frac{v}{v_1} \frac{v_1}{v_2} \right) \\ &= m_d \log \left(\frac{v}{v_1} \right) - m_d \log \left(\frac{v_2}{v_1} \right). \end{aligned} \quad (2.101)$$

In addition to n_d one can also adjust the Abbe number ν_d of the optical glass by an appropriate choice of the annealing rate v [2.161, 164] by applying the relation

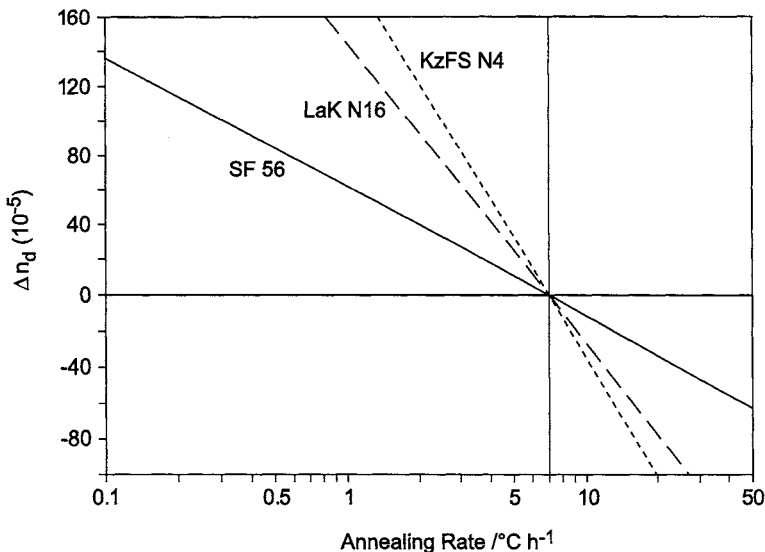


Fig. 2.45. Change of the refractive index n_d ($\lambda_d = 587.56$ nm) as a function of the annealing rate v . Reference annealing rate: $v_1 = 7^\circ\text{C/h}$; data from [2.164]

$$\Delta\nu_d \left(\frac{v}{v_1} \right) = m_{\nu_d} \log \left(\frac{v}{v_1} \right), \quad (2.102)$$

which is similar to (2.100) (see Fig. 2.46). However, it is not possible to adjust n_d and ν_d independently by variation of the annealing rate v .

Equations (2.100) and (2.102) cannot be applied to the limits $v \rightarrow 0$ and $v \rightarrow \infty$, since both relations diverge in both cases. In principle one can remove these divergences by suitable factors. These limits, however, are not relevant in practice; for $v \rightarrow 0$ one needs impractically slow annealing rates and for $v \rightarrow \infty$ the temperature gradients in the glass samples are too large.

It has to be emphasized that n_d and ν_d can be changed with the annealing rate according to (2.100) and (2.102) if the annealing starts above the transformation temperature T_g . However, glasses can also relax at temperatures far below T_g . Therefore, variations of the refractive indices must be taken into account if the glasses are subjected to high temperatures with subsequent fast annealing during the manufacturing process of optical elements. Such components may be subjected, for example during coating processes, to temperatures well above 300°C , which is within 100°C of the T_g of some types of glasses. Very often the components are rapidly annealed from the maximum process temperature. In this case small changes of the refractive index may occur. Table 2.10, taken from [2.165], illustrates the changes of the refractive index Δn_d that occur if the optical component is rapidly annealed from 100°C below T_g . In all cases, n_d decreases as a result of fast annealing. The main cationic components are also included in that table. It is worth noting that glasses containing large amounts of TiO_2 show a rather large decrease of n_d upon fast annealing. The

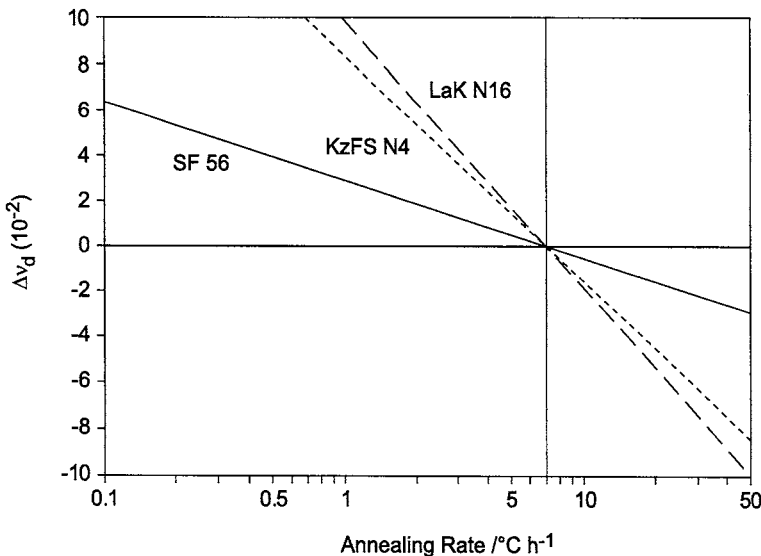


Fig. 2.46. Change of the Abbe number $\Delta\nu_d$ ($\lambda_d = 587.56 \text{ nm}$) as a function of the annealing rate v . Reference annealing rate: $v_1 = 7^\circ\text{C/h}$; data from [2.164]

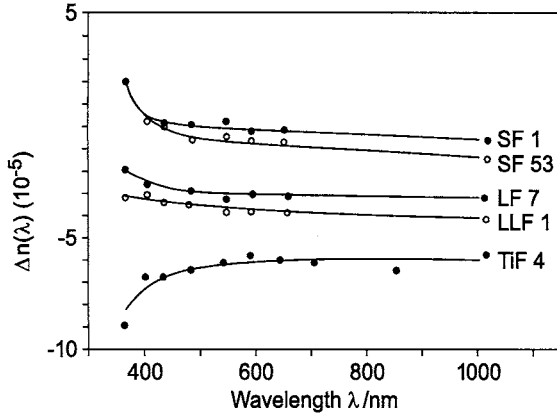


Fig. 2.47. Changes of the refractive indices Δn as a function of wavelength after thermal cycling of different optical glasses (heating to 250 °C for 30 minutes and subsequent fast annealing) (data from [2.165])

fast annealing process may also change the dispersion, as can be seen from Fig. 2.47 [2.165].

In order to avoid detrimental changes of the refractive index and its dispersion, it is mandatory to reduce the maximum process temperature and the annealing rates.

2.4.4 The Thermo-Optical Coefficient

The refractive indices of optical glasses and their dispersion, as well as the parameters of the dispersion equations in the catalogues, refer to a fixed temperature (usually 20 °C, see [2.164]). If the optical elements are used at various ambient temperatures, it is necessary to know how much the refractive index and its dispersion changes with the temperature. Such variations are described very often by the average thermo-optic coefficients

$$\frac{n(\lambda_k, T_{j+1}) - n(\lambda_k, T_j)}{T_{j+1} - T_j} = \frac{\Delta n(\lambda_k, T_{j,j+1})}{\Delta T_{j,j+1}}, \quad (2.103)$$

which are determined from the refractive indices measured at different temperatures T_j for a given set of wavelengths λ_k or directly from the incremental or decremental differences $\Delta n(\lambda_k, T_{j,j+1})$ according to the method of *Pulfrich* [2.166]. Data for many optical glasses are compiled in the catalogues for the spectral lines h = 404.66 nm, g = 435.83 nm, F' = 479.99 nm, e = 546.07 nm, d = 587.56 nm, C' = 643.85 nm, s = 852.11 nm, t = 1013.98 nm and 1060 nm and for steps of typically 20 °C in temperature ranges from -40 °C to +80 °C and in some cases within even larger intervals [2.167–173].

Instead of data determined for a small number of discrete wavelengths and averaged over a given temperature interval, a dispersion formula of the thermo-optical coefficient for practical applications is preferred. In this respect it is

Table 2.10. Decrease of the refractive index n at 20 °C and for $\lambda_d = 587.56$ nm due to heat treatment at 100 K below the glass transition temperature T_g and fast annealing. Only the cations of the oxides are given. R stands for alkali ions and E for alkaline earth ions (data from [2.165])

Type of glass	Main components of the chemical composition				T_g °C	$-\Delta n_d$ (10^{-6})
FK 1, 5	Si	B	R	F	390, 464	51, 56
FK 51, 52	P	E	Al	F	405, 434	16, 18
PK 2	Si	B	R		568	41
PK 50	P	Al	R		496	24
PSK 3	Si	B	Ba		602	51
BK 6, 7	Si	B	R		537, 559	41, 48
K 5, 7, 10, 50	Si	R			459–553	43–71
ZK 1, N7	Si	R	Zn		528–562	40, 18
BaK 4	Si	Ba	Zn		552	52
SK 16, 52	Si	B	E		638, 624	20, 4
KF 1, 9	Si	Pb	R		478, 445	58, 39
BaLF 4	Si	Ba	Zn		569	48
SSK 52	Si	Ba	B		639	18
LLF 1	Si	Pb	R		448	61
BaF N10, 53	Si	Ba			630	14, 22
LF 2	Si	Pb	R		440	43
F 2, 9	Si	Pb			432, 468	36
F N11	Si	Ti			572	64
BaSF 1, 50	Si	Pb	E		493, 500	1–9
LaSF N9	Si	Ba	La	Ti	698	54
SF 6, 13, 15, 58, 59	Si	Pb			362–472	6–39
SF L6, L56, N64	Si	Ti	Ba		578–585	120–134
TiF 2, 3, N5	Si	Ti	R(F)		419–472	135–201
TiF 6	P	Ti	R		410	96
TiSF 1	Si	Ti	K		495	230
TiK 1	Si	B	Al	R	340	55
KzFS 1, N4 N7	B	Pb	Al		472–512	13–16
LgSK 2	As	B	Al	F	512	10

reasonable to assume that the parameters of the dispersion formulae given in Sect. 2.1 depend on the temperature [2.174]. Among these dispersion formulae, the three-term Sellmeier equation provides a very good fit of n from the UV to NIR with an accuracy better than 10^{-5} :

$$n^2(\lambda) - 1 = a \sum_{i=1}^3 \frac{N_i}{V} f_i \frac{\lambda^2 \lambda_{0i}^2}{\lambda^2 - \lambda_{0i}^2}, \quad (2.104)$$

where N_i/V , f_i and λ_{0i} are the density, strength and effective resonance wavelength of the oscillators of kind i ; λ is the vacuum wavelength of the electromagnetic wave and “ a ” is a constant. In general, two resonances in the UV and one in the IR describe the dispersion in the visible and NIR spectral regions sufficiently well.

With the abbreviation $A_i = a(N_i/V)f_i$ ($i = 1, 2, 3$) as a parameter, one has to determine a total of six parameters (A_i and λ_{0i}) as a function of the temperature. The expansion of these unknown functions to second order requires a total of 18 parameters which have to be determined by a fitting routine. This large number of fitting parameters can be reduced if one applies a Sellmeier equation with only one term to describe the variations of n with respect to the temperature. This is justified since n varies only slightly with the temperature. Under this approximation one obtains [2.168]

$$\frac{dn(\lambda, T)}{dT} = \frac{n^2(\lambda, T_0) - 1}{2n(\lambda, T_0)} \times \left(D_0 + 2D_1(T - T_0) + 3D_2(T - T_0)^2 + \frac{E_0 + 2E_1(T - T_0)}{\lambda^2 - \lambda_0^2} \right). \quad (2.105)$$

This equation can be integrated in order to give the increments or decrements Δn of the refractive index as a function of both the temperature and the wavelength, with respect to a standard dispersion equation for the temperature T_0 :

$$\Delta n(\lambda, T - T_0) = \frac{n^2(\lambda, T_0) - 1}{2n(\lambda, T_0)} \left(D_0(T - T_0) + D_1(T - T_0)^2 + D_2(T - T_0)^3 + \frac{E_0(T - T_0) + E_1(T - T_0)^2}{\lambda^2 - \lambda_0^2} \right). \quad (2.106)$$

Both equations (2.105) and (2.106) can be simplified further, since the dependence of the prefactor on λ can be taken into account – at least to a good approximation – by slightly different fitting parameters yielding

$$\frac{dn(\lambda, T)}{dT} = D_{s0} + 2D_{s1}(T - T_0) + 3D_{s2}(T - T_0)^2 + \frac{E_{s0} + 2E_{s1}(T - T_0)}{\lambda^2 - \lambda_{s0}^2} \quad (2.107)$$

and

$$\Delta n(\lambda, T) = D_{s0}(T - T_0) + D_{s1}(T - T_0)^2 + D_{s2}(T - T_0)^3 + \frac{E_{s0}(T - T_0) + E_{s1}(T - T_0)^2}{\lambda^2 - \lambda_{s0}^2}. \quad (2.108)$$

Equations (2.105) to (2.108) have been derived to describe the thermo-optic coefficient and the increments or decrements of the refractive index Δn_{abs} with respect to vacuum. These quantities can be recalculated without difficulty for other ambient media of the glasses by use of the relationships

$$n_{\text{abs}} = n_{\text{rel}} n_{\text{med}} \quad (2.109)$$

and

$$\frac{d}{dT} n_{\text{abs}} = n_{\text{med}} \frac{d}{dT} n_{\text{rel}} + n_{\text{rel}} \frac{d}{dT} n_{\text{med}}, \quad (2.110)$$

Table 2.11. Coefficients to be inserted into the dispersion formulae (2.105) and (2.106) (data from [2.167])

Type of glass	$D_0 \times 10^6$	$D_1 \times 10^8$	$D_2 \times 10^{11}$	$E_0 \times 10^7$	$E_1 \times 10^{10}$	$\lambda_0/\mu\text{m}$
FK 52	-20.9	-0.855	3.41	3.11	4.02	0.204
PK 3	1.08	1.29	-2.65	4.37	7.07	0.138
PK 51A	-19.8	-0.606	1.60	4.16	5.01	0.134
BK 7	1.81	1.20	-2.30	4.95	7.71	0.149
K 5	-0.413	1.03	-3.40	4.73	5.19	0.213
K 7	-1.67	0.880	-2.86	5.42	7.81	0.172
K 10	4.86	1.72	-3.02	3.82	4.53	0.260
BaK 4	2.90	1.33	-3.10	5.14	7.37	0.213
SK 15	0.669	1.24	-2.62	3.76	5.03	0.233
SK 16	-0.0951	1.13	-1.19	3.86	5.92	0.192
BaF 51	-0.130	0.891	-0.290	7.94	11.1	0.196
F 2	1.51	1.56	-2.78	9.34	10.4	0.250
BaSF 10	-0.433	1.24	-2.20	9.40	9.51	0.230
BaSF 64	1.60	1.02	-2.68	7.87	9.65	0.229
LaF 2	-4.09	0.904	-1.06	6.67	7.07	0.213
LaF 3	-2.35	1.07	-0.938	5.72	6.01	0.220
SF 3	3.72	1.74	-3.21	14.9	14.1	0.260
SF 6	7.04	1.93	-4.03	17.8	18.8	0.272
SF 14	8.85	1.81	-4.12	13.9	15.3	0.279
SF L56	-4.30	0.840	-2.28	10.0	15.2	0.287
SF 58	4.98	1.68	-4.44	23.7	23.9	0.280
SF N64	-2.05	1.03	-3.15	9.26	12.3	0.269

where n_{med} and n_{rel} are the refractive indices of the ambient medium with respect to vacuum and of the glass with respect to the medium. For air as the ambient medium one can use (2.89) and (2.90) cited above.

Representative mean values of the coefficients to be inserted into the dispersion formulae (2.105) and (2.106) are given for a selection of optical glasses in Table 2.11. The fitting parameters of further types of optical glasses can be found in [2.167]. Figures 2.48–50 show the thermo-optic coefficients dn_{rel}/dT and dn_{abs}/dT and the increments Δn_{rel} for a special sample of the optical glass type LaSF N9. Comparing Figs. 2.48 and 2.49 one can see how much the thermo-optic coefficient can be changed when air is the ambient medium. In addition to quantitative modifications, even qualitative changes of the dispersion of the thermo-optic coefficient can be observed.

2.4.5 Photoelastic Properties of Glasses

In order to analyse stresses in optical glasses one has to know the stress-optical coefficient. Figure 2.51 shows a rectangular glass sample under external uniaxial stress σ (tensile stresses are assumed to be positive). The stress renders the glass anisotropic. Thus, the refractive index depends on the orientation of the plane of vibration if for example a linearly polarized electromagnetic wave penetrates the glass sample perpendicularly to the direction of the applied stress. For the

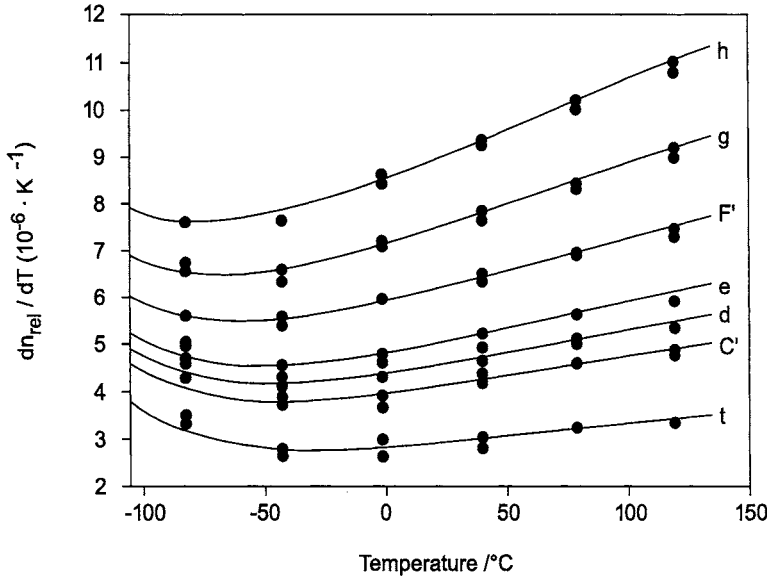


Fig. 2.48. Thermo-optic coefficient of the optical glass type LaSF N9 with respect to dry air as a function of temperature with the wavelength as a parameter. The full curves represent the results of calculations using (2.105), (2.109), (2.110), (2.89) and (2.90). Wavelengths: h = 404.66 nm, g = 435.83 nm, F' = 479.99 nm, e = 546.07 nm, d = 587.56 nm, C' = 643.85 nm, t = 1013.98 nm. $D_0 = -9.44 \times 10^{-7}/^\circ\text{C}$, $D_1 = 1.14 \times 10^{-8}/(^\circ\text{C})^2$, $D_2 = -1.87 \times 10^{-11}/(^\circ\text{C})^3$, $E_0 = 9.22 \times 10^{-7}/^\circ\text{C}$, $E_1 = 1.22 \times 10^{-9}/(^\circ\text{C})^2$, $\lambda_0 = 255 \text{ nm}$; $n(\lambda, 20^\circ\text{C}) = 3.2994326 - 1.1680436 \times 10^{-2} \mu\text{m}^{-2} \times \lambda^2 + 4.0133103 \times 10^{-2} \mu\text{m}^2 \times \lambda^{-2} + 1.3263988 \times 10^{-3} \mu\text{m}^4 \times \lambda^{-4} + 4.7438783 \times 10^{-4} \mu\text{m}^6 \times \lambda^{-6} + 7.8507188 \times 10^{-5} \mu\text{m}^8 \times \lambda^{-8}$

plane of vibration parallel (subscript $\parallel$) or perpendicular (subscript $\perp$) to σ the refractive index, n , changes to

$$n_{\parallel} = n + \frac{dn_{\parallel}}{d\sigma} \sigma \quad (2.111)$$

and

$$n_{\perp} = n + \frac{dn_{\perp}}{d\sigma} \sigma \quad (2.112)$$

as long as the stress is sufficiently small and Hooke's law can be applied. The derivatives in (2.111) and (2.112) are sometimes denoted in the literature as photoelastic coefficients [2.175]

$$K_{\parallel} = C_1 = \frac{dn_{\parallel}}{d\sigma} \quad (2.113)$$

and

$$K_{\perp} = C_2 = \frac{dn_{\perp}}{d\sigma} . \quad (2.114)$$

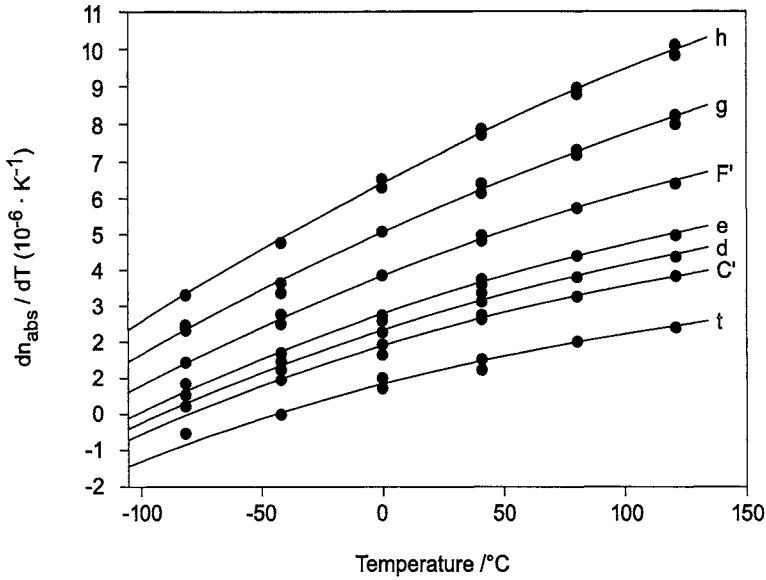


Fig. 2.49. Thermo-optic coefficient of the optical glass type LaSF N9 with respect to vacuum. The full curves represent the results of calculations using (2.105) with the same fitting parameters as given in Fig. 2.48

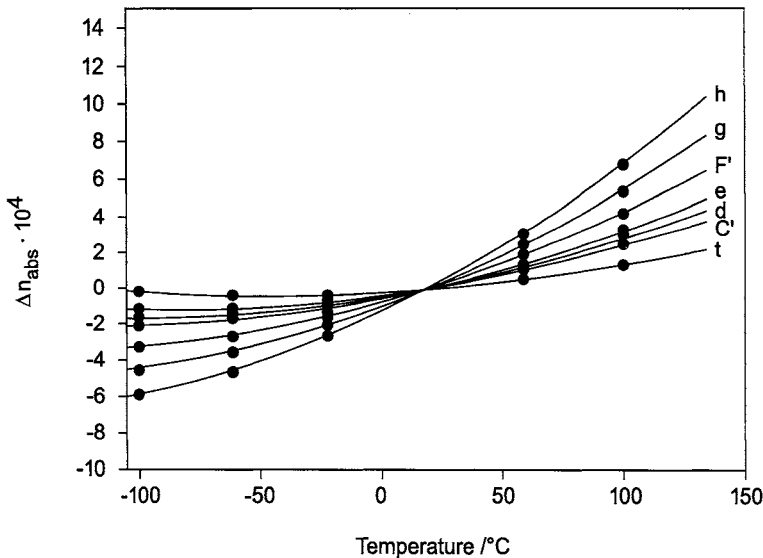


Fig. 2.50. Increments and decrements Δn_{abs} of the absolute refractive index of the optical glass type LaSF N9 with respect to $n(\lambda)$ at 20 °C as a function of temperature with the wavelength as a parameter. The full curves represent the results of calculations using (2.106) with the same fitting parameters as given in Fig. 2.48

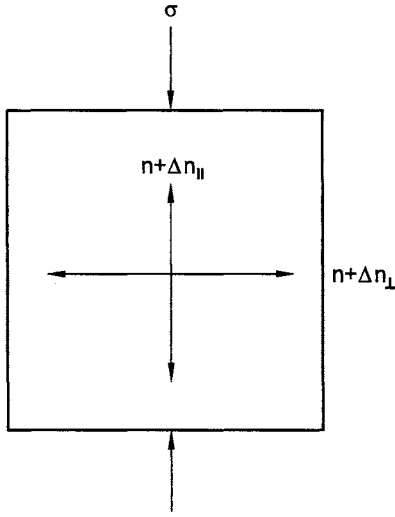


Fig. 2.51. Orientation of the stress σ and the limiting cases of the refractive index

Other relationships (see [2.175–182]) are also known for these derivatives, such as

$$\begin{aligned} \frac{dn_{\parallel}}{d\sigma} &= -\frac{n^3}{2}q_{11} = \frac{n^2}{Ec}(q - 2\mu p) \\ &= -\frac{n^3}{2E}(p_{11} - 2\mu p_{12}) \end{aligned} \quad (2.115)$$

and

$$\begin{aligned} \frac{dn_{\perp}}{d\sigma} &= -\frac{n^3}{2}q_{12} = \frac{n^2}{Ec}(-\mu q + (1 - \mu)p) \\ &= -\frac{n^3}{2E}(-\mu p_{11} + (1 - \mu)p_{12}), \end{aligned} \quad (2.116)$$

where q_{11} and q_{12} are the piezo-optic coefficients, q and p are Neumann's coefficients and p_{11} and p_{12} are the elasto-optic (strain-optical) coefficients. The other symbols E , μ , c and n denote Young's modulus, Poisson's ratio, the velocity of light in vacuum and the refractive index without stress.

These coefficients can be determined experimentally by several methods, e.g. the four-point bending method. The experimental methods are described in [2.178–185]. Experimental data have been compiled by *Lindig* in [2.175]. A more-recent collection of data for many optical glasses is given in [2.186]. In most cases, the data of the photoelastic coefficients have been determined for a single wavelength in the visible, very often for the sodium D lines at $\lambda = 589.3 \text{ nm}$. According to *Pockels* [2.178], who has determined these coefficients as a function of wavelength for different glass compositions, their dispersion can in most cases be neglected. For the heavy flint glasses, however, the dispersion of these coefficients is considerable.

Mostly the refractive index increases with growing compressive stress for both directions of the plane of vibration, i.e. $K_{\parallel} = C_1 = dn_{\parallel}/d\sigma$ and $K_{\perp} = C_2 = dn_{\perp}/d\sigma$ are negative with $|K_{\parallel}| < |K_{\perp}|$ if a compressive (tensile) stress is considered to be negative (positive). In [2.186], very large values up to $9 \times 10^{-12} \text{ Pa}^{-1}$ are reported for the heavy flints. Furthermore, one observes that $|K_{\parallel}|$ increases with the PbO content stronger than $|K_{\perp}|$ does. Consequently, it is possible to melt a glass for which $K_{\parallel} = K_{\perp}$ at least for a given wavelength (because of the dispersion of $K_{\parallel}$ and $K_{\perp}$). Such a glass is nearly free of birefringence since the changes of the refractive index for both directions of the plane of vibration are the same. A possible composition of a glass with this property has already been given by *Pockels* [2.178]. Therefore, this type of glass is called Pockels' glass. A commercial glass, which is nearly free of birefringence under stress, is available under the designation SF 57 [2.187] (see right hand side of Fig. 2.52).

In the case of hydrostatic pressure, one has to consider that the refractive index of both the glass and the ambient medium can change. The derivative of n with respect to the hydrostatic pressure, P , is given simply by

$$\frac{dn}{dP} = K_{\parallel} + 2 K_{\perp} . \quad (2.117)$$

The right-hand side can be transformed into a sum of other coefficients by using (2.113) to (2.116).

By taking advantage of the changes of the refractive index under stress, glasses can be used in acousto-optical modulators [2.188, 189]. To characterize the efficiency for acousto-optic deflection, which depends on the design of the devices, one defines the following figures of merit (Sect. 2.5):

$$\begin{aligned} M_1 &= \frac{n^7 p_{ij}^2}{\rho v} , \\ M_2 &= \frac{n^6 p_{ij}^2}{\rho v^3} , \\ M_3 &= \frac{n^7 p_{ij}^2}{\rho v^2} , \end{aligned} \quad (2.118)$$

where v is the velocity of the acoustic wave, ρ is the mass density of the material and p_{ij} is the appropriate elasto-optic (strain-optical) coefficient depending on the orientation of the acoustic wave and the plane of vibration of the optical wave.

Data for M_1 and M_2 are collected in [2.186]. The glasses SF 58 and SF 59, which have the largest PbO contents in comparison with the other glasses, show the largest values of M_1 and M_2 among the data of [2.186].

For practical stress analysis of glasses it is necessary to know the stress-optical coefficient, K . In the simple case of Fig. 2.51, an electromagnetic wave penetrates the rectangular parallelepiped under normal incidence. The plane of vibration is oriented at an angle of 45° towards the direction of the uniaxial stress σ . The wave can be considered as being superposed by two coherent partial waves with

planes of vibration parallel and perpendicular to σ . Since the refractive indices for both partial waves are different, a difference Δs in the optical pathlength occurs. It is given by

$$\Delta s = \left(n + \frac{dn_{\parallel}}{d\sigma} \sigma - n - \frac{dn_{\perp}}{d\sigma} \sigma \right) \ell = (K_{\parallel} - K_{\perp}) \sigma \ell, \quad (2.119)$$

where ℓ is the geometrical path or the thickness of the sample. Δs transforms into a phase shift

$$\Delta \varphi = \frac{2\pi}{\lambda} (K_{\parallel} - K_{\perp}) \sigma \ell = \frac{2\pi}{\lambda} K \sigma \ell \quad (2.120)$$

with the stress-optical coefficient

$$K = K_{\parallel} - K_{\perp}. \quad (2.121)$$

Thus K can be determined from the photoelastic coefficients. On the other hand, it can be determined directly from the relative phase shift of electromagnetic partial waves with the plane of vibration parallel and perpendicular to the uniaxial stress [2.178, 180, 183, 190–192].

There are numerous collections of data for the stress-optical coefficient of glasses in the literature. An arbitrary selection of references is [2.175, 178–181, 190–199]. There is also an abundance of work dealing with compositional dependence of this coefficient [2.175, 178, 190, 191, 194–199]. It is especially worth noting that K decreases with increasing PbO content until it reverses its sign for a very large lead content. Heavy flint glasses with about 70%–75% PbO by weight do not show appreciable birefringence in the visible (Pockels' glasses, as mentioned above).

The temperature dependence of K has been investigated in [2.192] for soda-lime glasses and silicate glasses [2.193]. As long as the temperature is sufficiently below T_g , K varies by only some percent, whereas near T_g stronger variations are observed. This is to be expected, since in this case the viscoelastic properties of glasses become more important [2.143, 152].

The stress-optical as well as the photo-elastic coefficients depend on the wavelength [2.175, 178]. For most compositions of glasses, however, the dispersion is only of the order of several percent in the visible spectral region [2.178, 199, 200]. In a recent investigation of K as a function of wavelength at 20 °C, this has been confirmed for a large number of optical glasses with different compositions [2.187]. The experimental data points for an arbitrary selection of optical glasses are shown in Fig. 2.52.

Dispersion plays a major role for the high-lead-content glasses, such as SF 56 and SF 57. In order to fit the data points, the following dispersion formula has recently been derived [2.187, 200]:

$$K(\lambda) = \frac{n^2(\lambda) - 1}{2 n(\lambda)} \left(A + \frac{B}{\lambda^2 - \lambda_{\text{soc}}^2} \right), \quad (2.122)$$

with the fitting parameters A , B and λ_{soc} and the refractive index $n(\lambda)$. If the dispersion is small, only the parameter A needs to be considered, because the

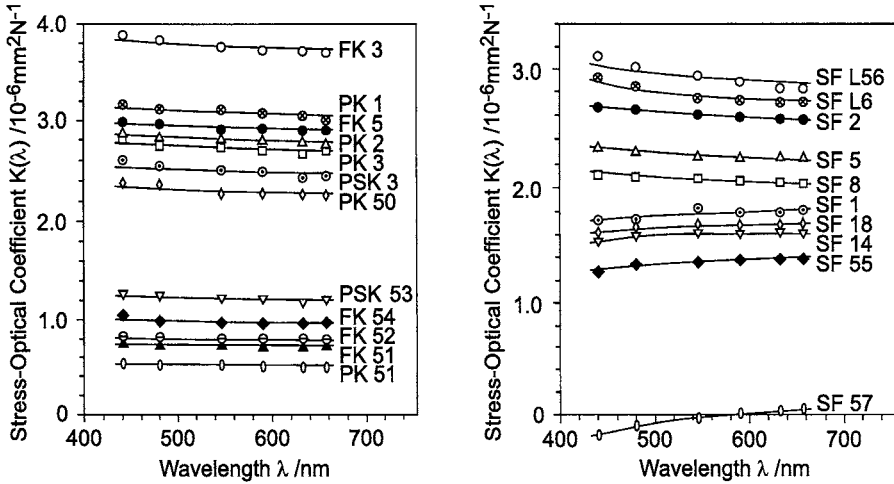


Fig. 2.52. Stress-optical coefficient as a function of wavelength λ at 20°C for optical glasses; calculated with one fitting parameter A ; except for SF 1, SF 14, SF 18, SF 55 and SF 57, where all three fitting parameters of (2.122) have been used, see text and Table 2.12 (data from [2.187])

parameter B can be assumed to be zero. Most of the full curves in Fig. 2.52 have been calculated under this assumption with only one fitting parameter A . A small dispersive effect is due to the prefactor $(n^2(\lambda) - 1)/2n$. For a fit of the data for the heavy flints, however, all of the three fitting parameters A , B and λ_{soc} have been determined. The fitting parameters are given in Table 2.12. In all of these cases, the experimental data can in fact be fitted reasonably well by the new dispersion formula (2.122), as can be seen by the full curves in the right hand side of Fig. 2.52.

Table 2.12. Fitting parameters for the stress-optical coefficient of heavy flints using (2.122) (data from [2.187])

Glass type	$A/10^{-6} \text{ mm}^2/\text{N}$	$B/10^{-14} \text{ mm}^4/\text{N}$	$\lambda_{\text{soc}}/\mu\text{m}$
SF 1	3.4893	-9.9215	0.1447
SF 14	2.9517	-8.0661	0.1528
SF 18	3.2276	-8.2146	0.1452
SF 55	2.6038	-9.1004	0.1516
SF 57	0.3249	-9.6815	0.1617

Neglecting the wavelength dependence of the prefactor $(n^2(\lambda) - 1)/2n$ in the visible spectral range, one can simplify (2.122) further to

$$K(\lambda) = A_s + \frac{B_s}{\lambda^2 - \lambda_{\text{soc},s}^2} \quad (2.123)$$

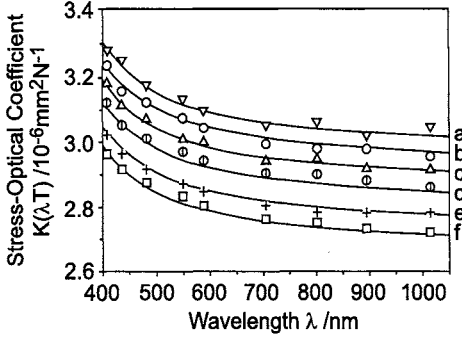


Fig. 2.53. The stress-optical coefficient of BK7 as a function of wavelength for different temperatures. The full curves represent a fit of the experimental data points by the new dispersion formula (2.124) using the fitting parameters $A_{s0} = 2.67 \times 10^{-6} \text{ mm}^2/\text{N}$, $A_{s1} = 8.19 \times 10^{-10} \text{ mm}^2/\text{N}^\circ\text{C}$, $A_{s2} = -4 \times 10^{-13} \text{ mm}^2/\text{N}^\circ\text{C}^2$, $B_s = 3.08 \times 10^{-14} \text{ mm}^4/\text{N}$, $\lambda_{\text{soc},s} = 237.2 \text{ nm}$. Temperatures: (a) 21 °C, (b) 103 °C, (c) 207 °C, (d) 310 °C, (e) 424 °C, (f) 537 °C (according to [2.201])

The dispersion of the stress-optical coefficient of several optical glasses has been investigated recently as a function of the temperature [2.201]. An example is given in Fig.2.53 for the optical glass BK7. The data points show the experimental results for the temperature range from 21 °C to 537 °C. In order to fit the data points, (2.123) had to be extended. All three fitting parameters A_s , B_s and $\lambda_{\text{soc},s}$ can vary with the temperature. However, it was observed that the dominant temperature dependence in (2.123) is due to the parameter A_s . By using an expansion to second order for A_s , the dispersion of the stress-optical coefficient and its temperature dependence can be described by the dispersion formula

$$K(\lambda, T_c) = A_{s0} + A_{s1}T_c + A_{s2}T_c^2 + \frac{B_s}{\lambda^2 - \lambda_{\text{soc},s}^2}, \quad (2.124)$$

where T_c is the temperature difference with respect to some reference temperature (°C in Fig. 2.53). The data points in Fig. 2.53 have been fitted by (2.124). The result is shown by the full curves. Indeed, the fit of the data points is very satisfactory over the full temperature range between 21 °C and 537 °C, and the wavelength range between 406.2 and 1021.6 nm.

2.4.6 The Electro-Optic Effect in Glasses (Kerr Effect)

Applying a static electric field instead of mechanical stress to a glass in a similar configuration as in Fig. 2.51 causes a uniaxial anisotropy of the refractive index, which is known as the Kerr effect [2.202, 203]. The refractive index depends on the orientation of the plane of vibration of the electromagnetic wave towards the electric field. This anisotropy of the refractive index increases with the square of the electric field strength E . (It must not be confused with the linear electro-optic effect occurring in crystals with unit cells that do not possess

inversion symmetry. These effects are much larger than those dealt with in the present subsection.) The difference of the refractive index $n_{\parallel} - n_{\perp}$ or the field-induced birefringence can be determined experimentally in a similar way as the stress-optical coefficient. For this purpose a linearly polarized electromagnetic wave penetrates a rectangular parallelepiped perpendicular to the electric field $\mathbf{E}$ (instead of stress σ in Fig. 2.51). The plane of vibration is oriented at an angle of 45° towards $\mathbf{E}$. The electromagnetic wave can be decomposed into two coherent partial waves with planes of vibrations parallel and perpendicular to $\mathbf{E}$. Due to the anisotropy $n_{\parallel} - n_{\perp} \neq 0$, a difference Δs occurs in the optical pathlength for both partial waves. It is given by

$$\Delta s = (n_{\parallel} - n_{\perp})\ell, \quad (2.125)$$

where ℓ is the geometrical pathlength. This difference (2.125) of the optical pathlength transforms into a phase shift

$$\Delta\varphi = 2\pi \frac{\Delta s}{\lambda} = \frac{2\pi}{\lambda} (n_{\parallel} - n_{\perp})\ell, \quad (2.126)$$

which can be measured by the application of polariscopic methods.

Since $n_{\parallel}$ and $n_{\perp}$ vary with the square of the electric field strength $\mathbf{E}$, $n_{\parallel} - n_{\perp}$ is written as

$$n_{\parallel} - n_{\perp} = \lambda B_K E^2 \quad (2.127)$$

with the Kerr constant B_K .

For oxide glass, B_K is of the order of 0.1 to $3 \times 10^{-12} \text{ cm/V}^2$, whereas for the chalcogenide glass As_2S_3 it is much larger, namely $8.7 \times 10^{-12} \text{ cm/V}^2$ [2.202]. Depending on the type of glass, both signs of B_K have been observed [2.203, 204].

With $B_K \approx 10^{-12} \text{ cm/V}^2$, one estimates a field strength larger than 10^5 V/cm to induce a birefringence of the order of 10^{-6} in the visible spectral region. Thus, very large electric field strengths are necessary for glasses to render this effect useful for the modulation of polarized electromagnetic waves in the visible spectrum.

For a given field strength one can assume that both the induced birefringence, $n_{\parallel} - n_{\perp}$, and the changes of $n_{\parallel}$ or $n_{\perp}$ separately, are of the same order. Thus, the influence of an electric field on the refractive index can be neglected except for very large field strengths.

The Kerr constant is known to show dispersion [2.203, 204]. Its value decreases considerably with increasing wavelength in the visible spectral range. With respect to the Kerr constant as a function of specific chemical constituents, it is well-known that its value increases with increasing PbO content in oxide glasses [2.203, 204].

2.4.7 Magneto-Optic Effects in Glasses

The plane of polarization of an electromagnetic wave that propagates in an optical medium can be rotated by a magnetic field. In the Faraday configuration

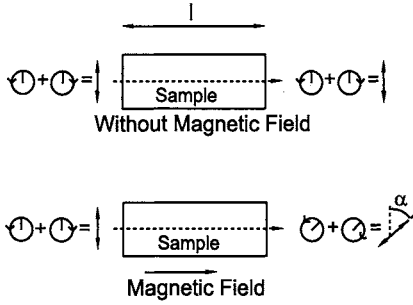


Fig. 2.54. Scheme of the Faraday rotation

(see Fig. 2.54), a linearly polarized electromagnetic wave travels in a transparent medium parallel to the magnetic field vector. The angle of rotation α is given by

$$\alpha = V \ell B, \quad (2.128)$$

with the absolute value B of the magnetic flux density vector $\mathbf{B}$, the distance ℓ which the wave travels in the sample and the Verdet constant V , which is the factor of proportionality. The sign of V – and thus that of the rotation angle – is positive (negative) if the rotation is clockwise (counterclockwise) looking parallel to the vector of the magnetic flux density for both directions of the propagation of the electromagnetic wave, parallel as well as antiparallel to $\mathbf{B}$. Positive Verdet constants correspond to diamagnetic materials, whereas V is negative for paramagnetic samples. As the rotation is independent of the direction of the wave propagation with respect to $\mathbf{B}$, the Faraday rotation is used for non-reciprocal optical elements, i.e., optical isolators.

In conjunction with linear polarizers the Faraday rotation can be exploited to modulate the intensity of linearly polarized light and to construct optical isolators, which transmit electromagnetic waves in one direction only, whereas they block in the reverse direction. This effect is useful to suppress the interaction between the reflected laser beam and the laser itself or other components.

Since a linearly polarized electromagnetic wave can be represented by a superposition of a left-handed (subscript $-$) and a right-handed (subscript $+$) circularly polarized wave (see Fig. 2.54), the rotation angle α can be expressed for the standard Faraday configuration by

$$\alpha = \frac{\pi}{\lambda} \{n_+(B) - n_-(B)\} \ell. \quad (2.129)$$

Since $n_+(B) \neq n_-(B)$, the optical material is anisotropic with respect to circularly polarized electromagnetic waves, whereas stresses and electric fields cause anisotropies for linearly polarized electromagnetic waves. For reviews of magneto-optic effects see [2.205–207].

In general, one has to take into account that $n_+(B)$ and $n_-(B)$ are nonlinear functions of the magnetic flux density, B . Nonlinearities can be observed for paramagnetic materials especially at low temperatures and at large flux densities [2.208], which can result in a rotation angle saturating with B . In this case, the Verdet constant depends both on the temperature and on B . On the other

hand, the Verdet constant of diamagnetic materials is expected to be nearly independent of the temperature and of B [2.206, 209].

Recently, the following dispersion formula of the Verdet constant has been developed on the basis of a single oscillator model [2.210, 211]:

$$V = \frac{\pi}{\lambda} \left\{ \frac{n^2 - 1}{2n} \right\} \left\{ k_1 + \frac{k_2}{\lambda^2 - \lambda_0^2} \right\}, \quad (2.130)$$

with

$$k_1 = \frac{d(\ln f_+ - \ln f_-)}{dB} + 2 \frac{d(\ln \lambda_{0+} - \ln \lambda_{0-})}{dB}$$

and

$$k_2 = 2 \lambda_0^2 \frac{d(\ln \lambda_{0+} - \ln \lambda_{0-})}{dB}.$$

Here f and λ_0 are suitable average values of the strength and of the resonance wavelength of the oscillators.

Since the variation of the prefactor $(n^2 - 1)/2n$ with λ is small compared to the variation of $V(\lambda)$, (2.130) can be simplified further to

$$V = \frac{\pi}{\lambda} \left\{ a + \frac{b}{\lambda^2 - \lambda_0^2} \right\}, \quad (2.131)$$

with the fitting parameters

$$a = k_1 \frac{n^2 - 1}{2n} \quad \text{and} \quad b = k_2 \frac{n^2 - 1}{2n}.$$

The data points in Fig. 2.55 represent experimental results of the Verdet constant of optical glasses in the visible spectral region. The curves result from a fit by (2.131). In fact, the dispersion of the Verdet constant can be described adequately by (2.131). The fitting parameters of a selection of optical glasses are given in Table 2.13.

The Verdet constant decreases monotonically with increasing wavelength λ in nearly all cases. For glasses of the SF type the Verdet constant increases monotonically with the PbO content. Very large values of V have been measured for the heavy flint glass type SF 59. This glass has an extremely large PbO content, namely about 80% by weight. Because of its large Verdet constant, this glass type is excellently suited as a Faraday rotator material. It is to be mentioned that almost all glasses of Table 2.13 show diamagnetic behaviour. On the other hand, special glasses with a large content of rare-earth ions and large magnetic moments such as Tb^{3+} have been developed, providing larger Verdet constants [2.206–208, 213, 214]. Since these glasses are paramagnetic, the Verdet constant and consequently the Faraday rotation angle depend on temperature.

Taking advantage of the rotation in opposite directions, the diamagnetic and paramagnetic behaviour may compensate for appropriate compositions in yielding a Verdet constant close to zero. This has been almost perfectly achieved for the optical glasses SF L6 and SF L56 at room temperature. Thus, in these

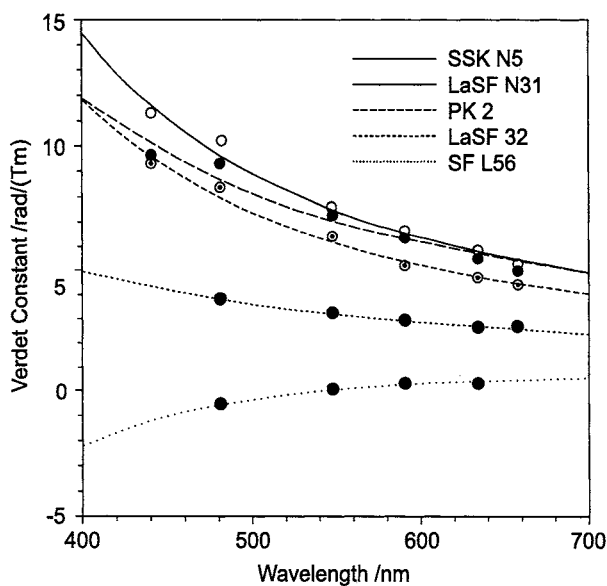
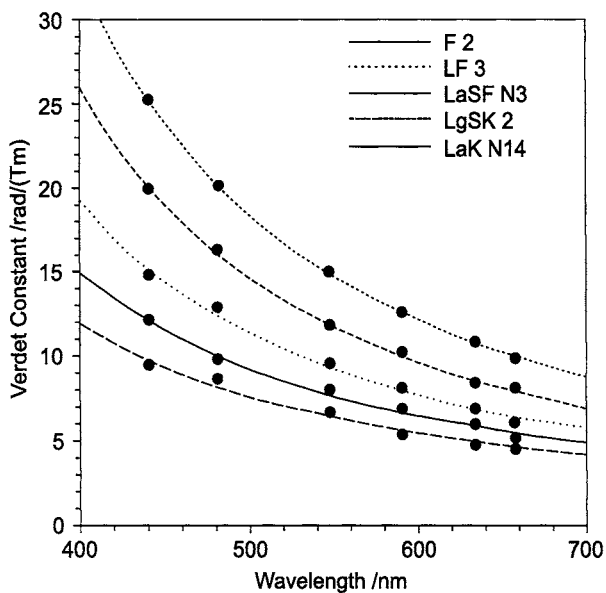


Fig. 2.55. The Verdet constant V of different optical glasses as a function of wavelength in the visible spectral range. Experimental results are shown by the data points. The curves represent the results of fitting calculations according to (2.131)

Table 2.13. Parameters a , b and λ_0 for fitting the experimental data of the Verdet constant by the dispersion formula (2.131) (data from [2.212])

Type of glass	$a/10^{-9}/\text{T}$	$b/10^{-20} \text{ m}^2/\text{T}$	$\lambda_0/10^{-9} \text{ m}$
FK 3	584.86	10.1922	95.3
FK 5	590.95	10.7136	92.3
FK 51	449.47	10.3782	84.7
FK 52	324.24	14.0808	86.2
PK 2	620.50	13.2578	96.4
BK 3	558.73	13.1224	96.1
BK 7	445.56	19.0396	97.0
BaLK N3	499.68	20.0342	100.0
K 3	489.92	21.7512	101.0
BaK 50	630.79	19.3622	102.6
SK 16	519.45	22.6678	101.2
SSK N5	753.71	15.8852	110.6
LaK N12	714.57	20.8496	106.5
LaK N14	701.16	12.2336	106.5
LF 3	807.24	35.9836	120.4
F 2	1007.83	44.5356	129.7
F N11	515.34	0.7574	130.1
F 13	967.22	46.9302	130.4
LaSF N3	791.71	23.8658	125.1
LaSF N31	938.42	8.3116	125.4
LaSF 32	480.47	2.0898	143.9
SF 1	1297.31	68.6416	144.7
SF 2	575.35	64.9114	134.6
SF 5	1142.58	57.7910	138.2
SF 6	1682.15	88.7082	156.4
SF 8	1379.42	56.7350	140.5
SF 14	1261.61	65.9212	152.8
SF 18	1078.09	73.7398	145.2
SF 53	1041.75	74.4138	146.7
SF L56	273.20	-7.4974	154.3
SF 57	1867.83	98.1516	161.7
SF 58	2164.80	121.3394	170.5
SF 59	2843.20	112.5192	175.3
SF N64	367.16	-3.7126	142.8
TiK 1	697.09	12.6900	100.8
TiF 3	534.72	1.0448	119.9
TiF 6	371.12	2.4132	140.6
KzFS N4	822.45	30.2408	117.8

glasses, the plane of polarization of an electromagnetic wave is nearly unaffected by magnetic fields.

From a practical point of view, the dispersion formulae (2.130) or (2.131) can also be applied to paramagnetic materials as long as the Verdet constant is independent of B . A possible dependence of the temperature can be taken into account by a suitable variation of the fitting parameters a , b and λ_0 with temperature.

If the electromagnetic wave propagates in a direction perpendicular – instead of a parallel – to the magnetic flux density vector, $\mathbf{B}$, we have the configuration of the Cotton–Mouton effect [2.215]. With respect to the orientation of the biasing field and to the direction of the propagating electromagnetic wave, this effect is analogous to the Kerr effect. The material becomes anisotropic for linearly polarized electromagnetic waves. Corresponding to (2.127) the anisotropy can be formulated as

$$n_{\parallel} - n_{\perp} = \lambda K_{\text{KM}} B^2 . \quad (2.132)$$

Data for the Cotton–Mouton constant K_{KM} , however, are not yet available for glasses, but they are for gases and liquids [2.216].

2.4.8 Further Possibilities

In this contribution various effects have been described which can change the refractive index. Some of the effects may be detrimental for applications, such as the variation of the refractive index due to the ambient temperature, to an annealing rate that is tfast, or to a changing environmental atmosphere. Such differential modifications of the refractive index may limit the performance of imaging optical systems. On the other hand, some of these effects, such as the Faraday effect, have advantageously been exploited to build measuring, sensing or protecting devices.

An important positive aspect is the possibility that glasses can be melted in large volumes and can be tailored to meet the requirements of specific applications. The following examples illustrate some of the achievements.

1. The temperature coefficient with respect to standard air can be made approximately zero [2.164]. In this case, the deflection by a prism is insensitive to variations of temperature.

2. The temperature coefficient of the refractive index, dn_{abs}/dT , the refractive index, n , and the thermal expansion coefficient, α , can be matched in order to make the temperature coefficient of the optical pathlength within a glass sample nearly zero, i.e.,

$$\frac{d}{dT}n_{\text{abs}} + (n_{\text{abs}} - 1)\alpha \simeq 0 . \quad (2.133)$$

Such glasses are called “athermal glasses” [2.217]. They are useful, for example, for athermal Fabry–Perot etalons.

3. The parameters of laser glasses as active media can be adjusted by the chemical composition so that the thermal lensing of laser rods is a minimum [2.218]. The refractive power of a laser rod is given in the steady state by [2.219]

$$\begin{aligned} \frac{1}{f_r} &= \frac{P}{kA} \left[\frac{1}{2} \frac{dn}{dT} + \alpha C_r n^3 + \frac{\alpha R(n-1)}{L} \right] , \\ \frac{1}{f_{\phi}} &= \frac{P}{kA} \left[\frac{1}{2} \frac{dn}{dT} + \alpha C_{\phi} n^3 + \frac{\alpha R(n-1)}{L} \right] , \end{aligned} \quad (2.134)$$

where f_r , f_ϕ , are the focal lengths of the radial and azimuthal components of an electromagnetic wave in the rod; P is the total heating power in the rod; k is the coefficient of heat conduction; A is the cross section of the rod; dn/dT is the temperature coefficient of the absolute refractive index; n is the refractive index; α is the coefficient of linear thermal expansion; R and L are the radius and length of the rod; C_r and C_ϕ , are the radial and azimuthal photoelastic coefficients.

For a rod with $R/L = 1/30$ made of phosphate laser glass LG 750 the values of the sums in the bracket of (2.134) are very small in order to minimize the thermal lensing.

Besides the effects discussed above, there exist further possibilities for changing the refractive index. These possibilities, however, would deserve more detailed consideration with respect to experimental techniques and practical applications than can be given in this survey. An important technical procedure is the variation of the refractive index due to ion exchange. This is the basis for integrated optics. Special optical glasses have been developed to meet the corresponding requirements [2.220]. Ion implantation is another possibility to vary the refractive index in thin layers, typically of the order of some tenths of a micrometer [2.221].

By leaching a glass the refractive index can be reduced in a surface layer. This has been used to produce antireflective surfaces [2.222, 223].

Furthermore, the refractive index of glass can be modified by irradiation with UV and γ photons [2.224], α and β particles or neutrons [2.225]. This creates defects and disturbs the nearest-neighbouring order. On the other hand, neutrons can be captured by nuclei which subsequently transmute into nuclei with their atomic number increased by unity, thus changing the isotopic or chemical composition of a glass.

Doping with suitable electron donors and acceptors enhances the possibility to change the refractive index locally. Photons may initiate the transfer of charge carriers from one site to another and change the refractive index. In glasses this effect plays only a minor role. Large effects are observed in electro-optic and photorefractive crystals (for a survey on photorefractive see [2.226]). The refractive index can also be changed by photodecomposition, which has been observed in PbI_2 films [2.227]. This effect may also play a role for surfaces of glasses that can be decomposed by the interaction of radiation.

In amorphous WO_3 films, the refractive index can be changed by injection of charge carriers [2.228]. Changes of the refractive index of the order of 0.01 have been observed in small semiconductor devices where free charge carriers have been injected by electric fields [2.229].

Furthermore, the refractive index can be changed by optical transfer of electrons into excited states, as has been discussed by *Baldwin* and *Riedel* [2.230]. For Eu-doped glasses, a permanent change of the refractive index has been observed under focused laser excitation [2.231].

As in the Kerr effect, where the refractive index depends on the static or quasi-static electric field strength, it is possible that the electric field of the electromagnetic wave itself can modify n . Since the intensity I of an electromagnetic wave is proportional to the time average of the square of the electric field strength E ,

$$I = \left[\frac{C}{4\pi} \right] n_0 |E|^2, \quad (2.135)$$

the refractive index increases proportionally to the I of the electromagnetic waves [2.219]:

$$\begin{aligned} n(I) &= n_0(\lambda) + \gamma I \\ &= n_0(\lambda) + 4.19 \times 10^{-3} \frac{n_2}{n_0} \left(\frac{\text{cm}^2}{\text{W}} \right) I, \end{aligned} \quad (2.136)$$

with the nonlinear refractive index n_2 . Typical values of γ are of the order of $4 \times 10^{-16} \text{ cm}^2/\text{W}$. Thus, intensities I of several 10^9 W/cm^2 are necessary to modify n in the sixth decimal place.

By comparing (2.127) with (2.136) and (2.135), the intensity-dependent refractive index can be considered as a degenerate Kerr effect since the frequencies of the applied electric field and of the electromagnetic wave are the same.

The intensity-dependent refractive index belongs to a whole series of possible nonlinear effects. A survey of nonlinear effects occurring in glasses is given in Sect. 2.6 [2.232]. In glasses, mainly nonlinear absorptive effects can be exploited. Since glasses are statistically isotropic materials, phase matching of waves with different frequencies is not possible. Therefore, frequency mixing or multiplication cannot be used efficiently.

If the intensity, I , in (2.136) decreases monotonically as a function of the radius from the centre of an intense laser beam, we have a gradient index lens and the beam may be focused as a result of the self-induced inhomogeneity of the refractive index (self-focusing). With sufficiently large intensities this may cause the destruction of the material [2.233, 234].

Narrow focusing of intense laser beams can induce the destruction of any optical material. An overview of the damage threshold of optical glasses has been given by *Hack* and *Neuroth* [2.233]. These authors found correlations between (a) the damage threshold for internal destruction of glasses and the nonlinear refractive index n_2 if very short pulses of the order of nanoseconds were used, and (b) the absorption constant at the laser wavelength (typically 1060 nm) if longer pulses of the order of milliseconds were applied. The interpretation of these correlations is the following. The larger n_2 is, the stronger the focusing effect and, consequently, the intensity in the focus are, which causes dielectric breakdown in the glass. If, on the other hand, the absorption constant is large, the heating power in the glass is increased, which may induce thermal breakdown (melting process).

In addition to the internal destruction, damage can also occur on the surface of the sample. The threshold for this type of damage, however, is correlated with the surface quality and the polishing method.

Acknowledgements

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2.5 Acousto-Optical Properties of Glasses

Hans-Jürgen Hoffmann

2.5.1 Acousto-Optical Effects and Applications

Acoustic waves propagate in gases and liquids normally as compressive waves, i.e., the acoustic wave causes periodic density variations in the direction of propagation. The length of the period is given by the wavelength

$$\Lambda_{ac} = \frac{v_{ac}}{f_{ac}} , \quad (2.137)$$

where v_{ac} and f_{ac} are the velocity and the frequency of the acoustic wave. Since the density and the refractive index of the gas or the liquid are modulated concomitantly, a phase grating is created. In a standard configuration, shown in Fig. 2.56, a transducer excites plane acoustic waves in a tank. The planes of constant refractive index are indicated by dashed lines. Electromagnetic waves that pass the tank perpendicularly to the propagation vector of the acoustic waves become partially deflected or diffracted, depending on the wavelength of the acoustic wave (or the period of the phase grating) and the maximum differences between the optical pathlengths in the layers of the phase grating [2.235–243]. As long as there are no additional anisotropies (e.g. dynamic anisotropies in liquids) the diffraction of the electromagnetic waves is independent of whether their plane of polarization is parallel or perpendicular to the vector of propagation of the acoustic waves. The diffraction angles Θ_m are given by the relation

$$\sin \Theta_m = m \frac{\lambda}{n \Lambda_{ac}} , \quad (2.138)$$

where m is an integer, λ is the wavelength of the electromagnetic wave in vacuum and n is the refractive index of the medium. To achieve a deflection of at least 1° in the first order ($m = 1$) for an electromagnetic wave in the visible part of the spectrum ($\lambda = 550 \text{ nm}$, refractive index $n = 1.5$), the wavelength

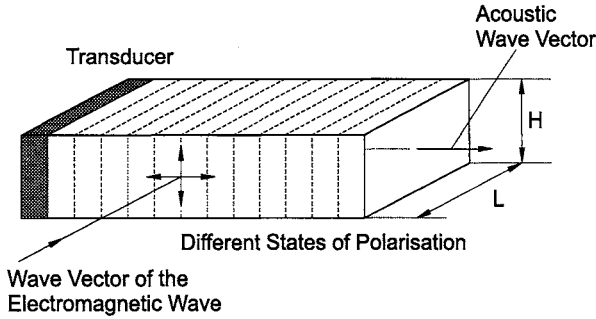


Fig. 2.56. Schematic configuration for an electromagnetic wave interacting with an acoustic wave. The different states of polarization of the electromagnetic wave are indicated by the vertical and horizontal arrows. The schematic device may extend further to the right for an acoustic absorber to avoid reflection of the acoustic wave. L : interaction length of the electromagnetic wave with the acoustic wave. H : lateral dimension of the acoustic field

λ_{ac} of the acoustic wave must be about $20\text{ }\mu\text{m}$, corresponding to a frequency of about 200 MHz if the velocity of the acoustic wave is 4000 m/s . Unfortunately, the modulation depth of the refractive index is low in gases and the damping of the acoustic wave is rather large in liquids, which makes it difficult to achieve effective and homogeneous diffraction of light beams using these materials. Since the damping of acoustic waves is at least one or two orders of magnitude smaller in crystalline and amorphous solids than in gases and liquids, one prefers these solid materials for applications in acousto-optic devices. In this case, however, both longitudinal compressive waves and elastic shear waves can be excited by a transducer. For simplicity, we consider the volume in Fig. 2.56 to be filled with an isotropic solid, e.g. homogeneous, statistically isotropic glass. The longitudinal compressive waves are excited by the transducer in the medium. Even in this simple case, one has to take into account that the refractive index n is modulated by the acoustic wave by different amounts depending on the direction of polarization of the electromagnetic wave. In this respect, consider Fig. 2.51 in Sect. 2.4, where the changes of the refractive index are indicated to be different depending on whether the plane of vibration of the electromagnetic wave is parallel or perpendicular to the propagation vector of the (compressive) stress. This obviously also occurs in an acoustic wave.

An acousto-optic device operates in the Raman–Nath regime if $2\pi\lambda/n\lambda_{ac} \gg \lambda_{ac}/L$ (which corresponds to the Klein–Cook parameter $Q = 2\pi L\lambda/n\lambda_{ac}^2 \gg 1$ [2.240–244]). Since the interaction length L is relatively short in this regime, the diffraction efficiency is rather poor. In addition, multiple orders of diffracted beams are observed ($m = \pm 1, \pm 2, \pm 3, \dots$).

For practical use one prefers the Bragg regime, for which $2\pi\lambda/n\lambda_{ac} \ll \lambda_{ac}/L$ (corresponding to the Klein–Cook parameter $Q = 2\pi L\lambda/n\lambda_{ac}^2 \ll 1$ [2.240–244]). In this case, the device is usually tilted by the Bragg angle, so that the elec-

tromagnetic wave is incident obliquely with respect to the wave vector of the acoustic wave.

In the Bragg regime, the interaction length for the acoustic and the electromagnetic waves is long and the beam intensity of the electromagnetic wave deflected into the first order can be very large compared to that of the higher orders and the zero order, provided that the refractive index is sufficiently strongly modulated by the acoustic wave.

Both limiting cases, the Raman-Nath regime and the Bragg regime, are discussed in textbooks, of which [2.240–243] are just an arbitrary selection. The transition range from $Q \ll 1$ to $Q \gg 1$ has been treated by complete numerical calculations in [2.244].

Acoustic waves can be excited (and consequently propagate) not only in the bulk but also on the surface of solids. Thus, the propagating acoustic strain can be confined to a layer near the surface. This is the basis of surface acoustic wave devices. It is obvious that application of the interaction between acoustic and electromagnetic waves in surface layers is an interesting field of integrated optics. Surface acoustic wave interactions are treated for instance in [2.242, 243].

For applications in acousto-optical devices one varies the frequency and the intensity of the acoustic wave. In this connection, however, one has to point out that the frequency of the diffracted electromagnetic wave is modulated or shifted with respect to the incident wave by the frequency of the acoustic wave.

Though essentially only two parameters of the acoustic wave can be varied, there are numerous applications of acousto-optic devices. Just a few elements are to be mentioned, starting from simple to more elaborate devices: switching elements, mode lockers and Q switches in laser cavities, modulators, deflectors, scanners, wavelength-tunable filters, spectrum analyzers, convolvers and correlators, optical computing matrix processors and Fourier transform processors [2.242, 243].

2.5.2 Characterization of Glasses

In the Bragg regime, the ratio of the beam intensities in the first and zero order is given approximately by [2.240]

$$\frac{I_1}{I_0} = \sin^2 \left(\frac{\pi}{2} \frac{L}{H} \left(\frac{n^6 p_{1i}^2}{\rho v_{ac}^3} \right) \frac{P_{ac}}{\lambda^2} \right)^{\frac{1}{2}} = \sin^2 \left(\frac{\pi}{2} \frac{L}{H} M_{2i} \frac{P_{ac}}{\lambda^2} \right)^{\frac{1}{2}} \quad (2.139)$$

or, for $\left(\frac{\pi}{2} \frac{L}{H} M_{2i} \frac{P_{ac}}{\lambda^2} \right)^{\frac{1}{2}} \ll 1$,

$$\frac{I_1}{I_0} \simeq \left(\frac{\pi}{2} \frac{L}{H} M_{2i} \frac{P_{ac}}{\lambda^2} \right). \quad (2.140)$$

In (2.139) and (2.140) $i = 1$ ($i = 2$) stands for the polarization of the electromagnetic wave parallel (perpendicular) to the propagation vector of the acoustic wave, L is the interaction length of the electromagnetic wave with the acoustic wave, H the lateral dimension of the acoustic field, λ the wavelength of the

electromagnetic wave in vacuum, n the refractive index, p_{1i} the photo-elastic or strain-optical coefficients, ρ the mass density, P_{ac} the total acoustic power and v_{ac} the velocity of the acoustic wave.

The figure of merit [2.245–247]

$$M_{2i} = \frac{n^6 p_{1i}^2}{\rho v_{ac}^3} \quad (2.141)$$

in (2.139) and (2.140) characterizes the diffraction efficiency, which depends only on material parameters. The larger M_{2i} is, the lower the power of the acoustic wave needed to achieve a large intensity of the deflected beam.

For acousto-optic scanning and switching elements, one has to take into account, in addition to the diffraction efficiency, the time constant corresponding to the bandwidth of the acousto-optic element. Therefore, one introduces in this case the figure of merit [2.246, 247]

$$M_{1i} = \frac{n^7 p_{1i}^2}{\rho v_{ac}} \quad (2.142)$$

which characterizes the product of the bandwidth and diffraction efficiency.

Considering the lateral dimension or the width H of the acoustic field, there is still a further figure of merit based on the following idea [2.246]. Since the required acoustic power decreases with the width H of the transducer (Fig. 2.56), H should be as small as possible. The lower limit of H corresponds to the waist of a (typically laser) beam focused in the acousto-optic interaction region. Then the modulation bandwidth Δf is about the reciprocal travel time of the acoustic wave across the beam waist $\Delta f = v_{ac}/H$. This yields [2.246]

$$M_{3i} = \frac{n^7 p_{1i}^2}{\rho v_{ac}^2} \quad (2.143)$$

It characterizes an acousto-optical material with respect to both the diffraction efficiency and the bandwidth for minimum lateral dimensions under focusing.

The figures of merit M_{1i} , M_{2i} and M_{3i} have been calculated for a series of optical multicomponent glasses at 20 °C [2.248]. A selection of data is given in Table 2.14. For the velocity of the acoustic wave v_{ac} in (2.141–143)

$$v_{\text{long}} = \sqrt{\frac{E(1-\mu)}{\rho(1+\mu)(1-2\mu)}} \quad (2.144)$$

has been inserted, which is the velocity of longitudinal acoustic waves in isotropic solids extending to infinity. These values are larger than

$$v_d = \sqrt{\frac{E}{\rho}} \quad (2.145)$$

which is valid for longitudinal acoustic waves propagating along the axis of long, slender rods.

From the formulae (2.141–143) one concludes that the figures of merit are large if the refractive index n and the strain-optical coefficients p_{1i} ($i = 1, 2$)

are large whereas the mass density ρ and the velocity of the acoustic waves v_{ac} are small. However, these quantities are not all independent of each other. An analysis of these parameters has been made for crystalline materials in [2.249]. Inserting the relation (2.144) into (2.141–143), we obtain immediately the most important guideline for a large figure of merit. Since the dependence of the mass density is rather small in the denominators of (2.141–143), large figures of merit are expected if the refractive index n is large and Young's modulus E is small (in this case the strain-optical coefficients are also very often small). Therefore it is not surprising that TeO_2 , PbMoO_4 , $\alpha\text{-HfO}_3$, As_2S_3 , GaP , GaAs and Tl_3AsS_4 are among the materials with large figures of merit. The figures of merit M_{2i} of these materials are typically of the order of 20 to $500 \times 10^{-18} \text{ s}^3/\text{g}$ [2.246, 249, 250]. As compared to these large values most data of optical glasses are rather small. This can be understood by the fact that optical glasses have not been developed for a large acousto-optical figure of merit but for special requirements of the refractive index, dispersion and anomalous partial dispersion. Since, compared to crystals, the figures of merit of many different types of optical glasses are smaller by at least two orders of magnitude, one may generally conclude that glasses are not suitable for acousto-optical applications. Nevertheless, optical glasses that possess a large figure of merit do exist, namely the heavy flint glasses SF 6, SF 56, SF 57, SF 58, SF 59 and SF 61, for which the figures of merit M_{2i} are between 5 and $17 \times 10^{-18} \text{ s}^3/\text{g}$. In this respect, the reader is referred to [2.251, 252]. Glasses have the advantage over crystalline materials in that they can be produced in large volume and very homogeneously. Furthermore, for the Pockels' glass (SF 57, see Sect. 2.4) the piezo-optical coefficients q_{11} and q_{12} are approximately the same in the visible spectral range, since the stress-optical coefficient K is approximately zero [2.253]. The figures of merit and the diffraction of an acousto-optical device using this type of glass are consequently independent of the polarization of the incident electromagnetic wave with its wavelength in the visible spectral range. By variation of the content of lead oxide one can shift that range towards the infrared spectral region. The possibility of adjusting the figure of merit for both planes of vibration by adjusting the composition seems to be unique to glasses. Taking advantage of this possibility is a remarkable achievement in materials science and engineering.

One should point out that the data in Table 2.14 have been calculated for the wavelength $\lambda = 589.3 \text{ nm}$ of the electromagnetic wave. The specification of the wavelength is necessary, since the refractive index n and the strain-optical coefficients p_{1i} which enter the figures of merit depend on λ . Usually the dispersion of n is very well known, whereas the dispersion of the strain-optical coefficients has not yet been investigated for many glasses.

Basically, dispersive effects in acousto-optical devices have two different origins. The first one arises from the diffraction. The diffraction angle Θ_m depends on the period of the modulated refractive index, which is given by the wavelength of the acoustic waves (2.138): the larger the wavelength λ of the electromagnetic wave, the larger the diffraction angle Θ_m . Thus, changing the wavelength λ of the

Table 2.14. Figures of merit of selected optical glasses for the wavelength $\lambda = 589.3\text{ nm}$

Type of glass	p_{11}	p_{12}	v_{long} /m/s	v_d /m/s	M_{11} /10 ⁻⁷	M_{12} cm ² s/g	M_{21} /10 ⁻¹⁸	M_{22} s ³ /g	M_{31} /10 ⁻¹²	M_{32} cm s ² /g
FK 3	0.15	0.24	4920	4510	3	7	1	2	0.6	1.5
FK 5	0.14	0.23	5430	5030	2	6	1	1	0.4	1.2
FK 51	0.17	0.20	5440	4730	2	3	1	1	0.4	0.6
PSK 3	0.14	0.23	5760	5370	3	7	0	1	0.4	1.2
PSK 53A	0.17	0.20	5310	4650	5	6	1	1	0.8	1.2
BK 7	0.12	0.22	6030	5700	2	6	0	1	0.4	1.4
K 5	0.13	0.23	5630	5250	2	7	0	1	0.4	1.2
K 7	0.12	0.23	5540	5200	2	7	0	1	0.3	1.2
ZK N7	0.11	0.23	5660	5320	2	7	0	1	0.3	1.2
BaK 1	0.13	0.20	5250	4780	2	6	1	1	0.5	1.1
BaK 2	0.14	0.22	5360	4970	3	7	1	2	0.5	1.2
BaK 4	0.12	0.21	5400	4970	2	6	0	1	0.4	1.1
SK 2	0.14	0.20	5210	4700	3	6	1	1	0.6	1.2
SK 4	0.12	0.18	5360	4800	2	5	0	1	0.4	0.9
SK 5	0.14	0.21	5570	5060	3	6	1	1	0.5	1.1
SK 10	0.13	0.19	5320	4740	3	5	1	1	0.5	1.0
SK 14	0.13	0.19	5540	5000	3	5	1	1	0.4	0.9
SK 15	0.14	0.20	5410	4810	3	6	1	1	0.5	1.1
SK 16	0.16	0.22	5540	4980	4	7	1	1	0.7	1.3
BaLF 4	0.11	0.20	5250	4910	2	6	0	1	0.3	1.1
BaLF 5	0.14	0.23	5080	4690	3	7	1	2	0.5	1.5
SSK N5	0.11	0.17	5510	4880	2	5	0	1	0.4	0.9
SSK N8	0.13	0.21	5490	4990	3	7	1	1	0.5	1.3
LaK N7	0.11	0.16	5470	4850	2	4	0	1	0.4	0.7
LaK 8	0.10	0.16	6300	5510	2	5	0	1	0.3	0.7
LaK 9	0.11	0.17	6370	5600	2	5	0	1	0.3	0.8
LaK 10	0.10	0.16	6150	5390	2	5	0	1	0.3	0.8
LaK N12	0.13	0.17	5270	4620	3	5	1	1	0.6	1.0
LaK N13	0.15	0.19	5180	4530	4	6	1	1	0.8	1.3
LaK 21	0.16	0.22	5510	4920	4	7	1	1	0.7	1.4
LaK N22	0.13	0.18	5470	4900	3	5	1	1	0.5	1.0
LLF 6	0.15	0.25	4990	4730	3	8	1	2	0.6	1.8
BaF 4	0.14	0.21	4720	4330	3	7	1	2	0.7	1.6
BaF N10	0.14	0.20	5510	4860	3	7	1	1	0.6	1.3
BaF 50	0.14	0.19	5510	4950	3	7	1	1	0.6	1.2
BaF 51	0.09	0.16	5670	5090	2	5	0	1	0.2	0.8
BaF 52	0.12	0.19	5170	4700	3	6	1	1	0.5	1.1
LF 5	0.18	0.25	4600	4290	6	11	2	3	1.2	2.3
F 2	0.17	0.23	4300	4010	5	10	2	3	1.3	2.3
F 5	0.15	0.22	4370	4090	4	9	1	3	0.9	2.0
F N11	0.10	0.20	5990	5590	2	7	0	1	0.3	1.2
BaSF 2	0.15	0.20	4490	4120	4	8	1	3	1.0	1.8
BaSF 51	0.12	0.17	4920	4310	3	6	1	1	0.6	1.3
LaF 2	0.11	0.15	5290	4630	3	5	1	1	0.5	0.9
LaF 3	0.11	0.15	5450	4780	2	5	0	1	0.4	0.8
LaF N7	0.13	0.17	4820	4260	4	7	1	2	0.8	1.4
LaF N21	0.07	0.12	6200	5390	1	3	0	0	0.2	0.5
LaF 22A	0.09	0.14	5530	4880	2	4	0	1	0.4	0.9

Table 2.14. (cont.)

Type of glass	p_{11}	p_{12}	v_{long} /m/s	v_d m/s	M_{11} / 10^{-7}	M_{12} $\text{cm}^2\text{s/g}$	M_{21} / 10^{-18}	M_{22} s^3/g	M_{31} / 10^{-12}	M_{32} $\text{cm s}^2/\text{g}$
LaSF 3	0.08	0.14	6130	5360	2	5	0	1	0.3	0.8
LaSF N30	0.10	0.14	6070	5280	2	4	0	1	0.4	0.7
LaSF N31	0.10	0.14	5560	4790	3	5	1	1	0.5	1.0
SF 1	0.22	0.25	3860	3570	12	16	5	6	3.2	4.1
SF 2	0.19	0.25	4070	3780	8	13	3	5	1.9	3.2
SF 4	0.20	0.23	3710	3420	12	15	5	6	3.1	4.1
SF 5	0.18	0.22	4010	3710	7	11	3	4	1.8	2.7
SF 6	0.24	0.25	3580	3270	19	21	8	9	5.4	5.9
SF L6	0.09	0.16	5820	5260	3	8	0	1	0.4	1.4
SF 8	0.19	0.24	3920	3630	9	13	3	5	2.2	3.5
SF 10	0.20	0.24	4160	3860	10	15	3	5	2.5	3.6
SF 11	0.19	0.21	4020	3720	10	13	4	5	2.7	3.3
SF 14	0.20	0.23	4080	3780	11	15	4	5	2.8	3.7
SF 15	0.18	0.22	4160	3850	7	11	3	4	1.9	2.8
SF 53	0.19	0.22	3920	3620	10	13	4	5	2.4	3.3
SF 55	0.20	0.22	3760	3440	11	14	5	6	3.2	3.8
SF L56	0.07	0.15	5810	5280	2	6	0	1	0.3	1.2
SF 56	0.21	0.22	3740	3440	13	16	5	6	3.7	4.1
SF 57	0.23	0.23	3430	3130	20	20	9	9	6.0	6.0
SF 58	0.24	0.23	3290	2960	29	26	14	13	8.5	7.8
SF 59	0.25	0.24	3200	2870	34	30	17	15	10.5	9.7
SF 61	0.21	0.23	3810	3530	12	15	5	6	3.3	4.0
KzFS N4	0.13	0.20	4870	4320	3	7	1	2	0.6	1.5

electromagnetic wave implies a different angle of diffraction. This effect cannot be avoided except by adjusting the wavelength λ_{ac} of the acoustic wave.

The second effect is related to the intensity of the diffracted beams and to the diffraction efficiency characterized by the figure of merit M_{2i} [see (2.139–141)]. Since the factor λ^2 occurs in the denominator of the argument in (2.139) and M_{2i} can show dispersion of the order of several ten percent, e.g. in the visible spectral region, the intensity of the diffracted beam clearly depends on the wavelength λ of the electromagnetic waves. In order to compensate this effect, the acoustic power has to be adjusted. On the other hand, the dispersion of the figure of merit M_{2i} in the numerator may compensate the factor λ^2 in the denominator. Therefore it is worth while investigating the figure of merit M_{2i} as a function of wavelength in different glass systems. If it is proportional to λ^2 in a certain range of wavelength, the intensity of the diffracted beams is independent of λ . There are indications that this occurs for glasses with a large content of lead oxide.

2.6 Nonlinear Optical Properties

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Nonlinear optics deals with the nonlinear changes in optical properties of materials induced by externally-applied light beams. In general, nonlinear optical effects are sufficiently small so that their observation requires the application of the intense light fields made available by lasers. The theoretical analysis of nonlinear optical effects was first reported in the early 1960s [2.254]. In the same time period, the available laser intensities from pulsed lasers reached a sufficient level so that useful nonlinear phenomena such as the generation of a second-harmonic laser beam became readily observable [2.255]. However, nonlinear effects were also responsible for undesirable thread-like damage observed in optical elements exposed to high-intensity laser beams. It was quickly realized that this was due to the self-focusing of a propagating laser beam at high-fluence locations from localized light-induced increases in the refractive index. From this time to the early 1970s, research emphasized the reduction of nonlinear effects in order to maintain the beam quality of high-power glass laser systems. Later, interest in nonlinear refraction broadened when optical bistability and phase conjugation were demonstrated in the late 1970s.

Today, intensified research with nonlinear materials is currently directed at possible application areas such as all-optical switching, modulation, image processing, sensors, advanced targeting, optical limiting, adaptive optics and optical storage/memory systems. In addition to these, more novel applications are still being discovered. There are numerous texts explaining the concepts and theory underlying nonlinear optics [2.256–258]. New technologies and potential commercial markets for nonlinear optical materials have been thoroughly reviewed in [2.259].

A wide variety of materials are known to possess nonlinear properties, including gases, vapours, polymeric media, liquid crystals, biological systems, organic solutions, water, crystals and glasses [2.260]. Inorganic crystals and organic polymers are well known for their large nonlinear effects, but they have several problems which often prevent practical application of these materials: cost, optical quality and laser damage threshold. Solid-state nonlinear materials, on the other hand, are particularly attractive because of their easy handling; and they often possess a higher laser damage threshold. Of the solid-state nonlinear materials, crystals exhibit more pronounced nonlinear behaviour than glassy materials. Indeed, several crystals have already been marketed for harmonic generation and parametric processes. However, glasses have several competitive advantages over crystals such as mechanical stability, easy handling, possible fabrication of large optics, excellent optical homogeneity, quicker production processes and potential lower cost.

Today, glasses are recognized as one class of usable nonlinear material. Numerous papers have appeared discussing their properties, the origin of their nonlinearity and actual device fabrication and performance. We will present a

discussion of glass as a nonlinear material by classifying nonlinear glasses into homogeneous and heterogeneous types. Whereas homogeneous glasses are completely vitrified and isotropic on both a microscopic and a macroscopic scale, heterogeneous glasses contain, on the other hand, additional zones, usually sub-microscopic of different chemical identity such as particles, crystals or other glassy phases. Most of the nonlinear examples discussed here are a result of the third-order susceptibility, $\chi^{(3)}$, since glassy materials are, in general, isotropic and lack non-centrosymmetric structure, normally regarded as essential for the manifestation of $\chi^{(2)}$ processes [2.256–258]. Nevertheless, the best-known $\chi^{(2)}$ process, second-harmonic generation of light, has been observed in glasses as well [2.261]. Interested readers can also consult the several review papers that have appeared in the literature [2.262–267]. A short description of nonlinear terms and of related physical units is given in Appendix A.

2.6.1 Homogeneous Glasses

There have been several attempts to identify a relationship to predict the nonlinear response of a material, including a purely empirical relationship known as *Miller's rule* [2.268] and a semi-empirical approach using a perturbation treatment [2.269]. These early efforts all predict that glasses with high linear refractive indices will also possess a correspondingly high nonlinear refractive index. Today, the equation widely accepted to evaluate nonlinear refractive index is that proposed in [2.270],

$$n_2 = \frac{68(n_d - 1)(n_d^2 + 2)^2}{\nu_d \sqrt{1.52 + \frac{(n_d^2 + 2)(n_d + 1)\nu_d}{6n_d}}}, \quad (2.146)$$

where n_2 is the nonlinear refractive index, n_d is the refractive index at 587.6 nm and ν_d is the Abbe number at 587.6 nm. According to (2.146), a high nonlinear refractive index is expected for glasses with a high linear refractive index and a low Abbe number (high dispersion). Figure 2.57 presents a schematic representation of lines of constant nonlinear refractive index, as calculated with (2.146), on the Abbe diagram. Current empirical laws thus suggest that a high $\chi^{(3)}$ value will be obtained by formulating a glass with heavy elements as the basic constituents.

Actual measurement of nonlinear refractive index values started in the 1970s when high-intensity laser beam sources became available for laboratory use. The initial choice of test samples were active laser materials since self-focusing due to high nonlinear index values was a problem in high-power glass laser systems. A study by *Weber et al.* [2.271] utilized time-resolved interferometry in order to obtain nonlinear index values for optical glasses and crystals at 1064 nm. As a result of their work, the empirical expression above, (2.146), was shown to provide a good estimate of the nonlinear index of optical materials in the long-wavelength limit.

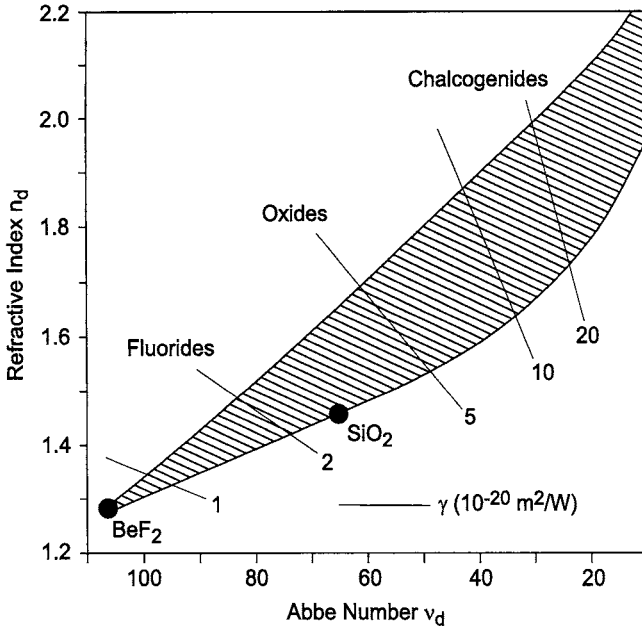


Fig. 2.57. Expected nonresonant nonlinear refractive index value, n_2 , as a function of n_d and ν_d , as predicted by (2.146). The dashed lines correspond to lines of constant γ [2.263]. There are two different units for the nonlinear index n_2 and γ . For their relation to one another see Appendix A

In search of higher nonlinear glass materials, glass developers have formulated experimental compositions, including glasses with high contents of lead and bismuth oxides [2.272, 273], germanates [2.272], chalcogenides [2.272, 274, 275] and glasses with large amounts of titanium and niobium oxides [2.276]. Reported results are summarized in Table 2.15 along with corresponding data for the commercial optical glass with the highest linear index, SF 59, for comparison. The highest $\chi^{(3)}$ value, in the range of 10^{-11} esu, was obtained in an As-S-Se composition [2.274, 275]. This is two orders of magnitude larger than that of conventional silicate-type glasses and is comparable to values obtained with nonlinear organic compounds. The degree of nonlinearity in homogeneous glasses is felt to be sufficient to render them useful once available laser output powers become higher in the future.

One of the leading application pursuits in nonlinear optical research is the demonstration of an optical transistor or optical switch. Another contribution [2.278] has considered this application and defined a figure of merit for a functional material by considering the number of switching operations which can be performed within a thermal lifetime. They then investigated this figure of merit for a number of representative nonlinear materials: GaAs, a GaAs/GaAlAs multiple quantum well structure, the organic molecule PTS, a semiconductor-doped glass and Schott SF 59. They found that because of its extremely small absorp-

Table 2.15. Nonlinearity of high-refractive-index glasses

Glasses	$n_{623 \text{ nm}}$	$n_{1060 \text{ nm}}$	$n_2 \times 10^{-13} \text{ esu}^f$
Silicate SF 59 ^a		1.91	14.8
QR ^{a,b}		2.02	20.5
Germanate VIR-3 ^a		1.84	9.8
QS ^{a,c}		1.94	15.6
Gallate RN ^{a,d}		2.30	68.8
Chalcogenide (As ₂ S ₃) ^e		2.48	258.0
Oxides ^g			
30TiO ₂ -20Na ₂ O-50SiO ₂	1.72		12.8
15NbO _{5/2} -15TiO ₂ -20Na ₂ O-50SiO ₂	1.83		11.8
30NbO _{5/2} -30TiO ₂ -20Na ₂ O-20SiO ₂	1.95		18.8

^a Reference [2.272]^b QR: 60PbO-40SiO₂^c QS: 60PbO-40GeO₂^d RN: 40PbO-35BiO_{1.5}-25GaO_{1.5}^e Reference [2.275]^f Reference [2.324]^g Reference [2.276]

tion and inertness to thermal effects, the SF 59 glass had a larger figure of merit than any of the other nonlinear optical materials, despite the far larger nonlinear coefficients of the competitive semiconductor materials. This demonstrates how the advantages offered by glass can, under certain situations, make it the material of choice for a particular application [2.277, 278].

Consistent with material developments there has also been parallel research in fabricating guided wave structures of fibres and planar waveguides from homogeneous nonlinear glasses [2.279]. This work resulted in the observation of the self-induced confinement of a light beam referred to as a spacial soliton [2.280] as well as two newly discovered problems: two-photon absorption [2.281] and absorption by colour centres which mimics two-photon absorption [2.282]. Materials with a high nonlinear refractive index are necessary to accumulate a phase shift in an all-optical switching device. However, nonlinear absorption could cancel this effect before a switchable level is achieved. In [2.281] this limitation is investigated with a lead-glass fibre consisting of an SF 6 glass core and an SF 56 cladding glass. It was reported that this system did not have usable properties at either 532 nm or 1060 nm, but that the mismatch at 1060 nm was nearly acceptable. Since the two-photon transition probability drops significantly as the photon energy is decreased, we expect that adequate performance should be possible if such a device is operated further into the IR.

Although glassy materials are centrosymmetric and thus normally incapable of exhibiting a $\chi^{(2)}$ -type phenomena, observations of second-harmonic generation (SHG) were first reported in germania-doped optical fibres by *Osterberg* in 1986 [2.261]. A number of models have been proposed to account for this phenomena

which involve some kind of symmetry-breaking organizational process which gives the non-zero $\chi^{(2)}$ necessary for frequency doubling. Two leading theories to explain SHG in glass fibres are the creation of a DC polarization in the fibre through a third-order interaction between the fundamental and second-harmonic light beams [2.283] and the creation of electrostatic fields by a photogalvanic effect [2.284].

2.6.2 Heterogeneous Glasses – Semiconductor-Doped Glasses

Semiconductor-doped glasses are known as colloiddally coloured, longpass filter glasses in the glass industry and they are readily available from glass filter catalogues [2.285]. The colour in these glasses are, in most cases, due to small microcrystallites of semiconductor particles, principally of a mixed $\text{CdSe}_x\text{S}_{1-x}$ variety, which are nucleated and grown by a secondary heat treatment (known as a striking process) of the initially colourless glass. Commercial semiconductor-doped glasses have a yellow, orange, or red appearance with a very steep absorption edge at a wavelength which strongly depends on the thermal history during the secondary heat treatment, as well as on the characteristics and the concentration of the colourants and base glass constituents.

Although considerable research has been completed on the commercial filter glasses as supplied, there has been a number of disagreements in the findings among research groups. In addition to the fact that different companies may prepare these products with different base compositions and manufacturing schedules, these disagreements may also be partially because each filter glass supplier has the freedom, by selecting different glass formulations and by adjusting the striking schedule employed, to supply a given filter type with a variety of different glasses covering a range of base glass and crystalline phase compositions. Furthermore, use of this flexibility also leads to a corresponding variation in available crystal densities and size distributions.

The nonlinearity of these $\text{CdSe}_x\text{S}_{1-x}$ glasses was first noticed some time ago and the first application as a saturable absorber appears to have been in 1964 [2.286]. More active research on these materials began after the nonlinear refractive index had been measured for this type of glass in four-wave mixing experiments in 1983 [2.287]. These results are shown in Table 2.16 along with theoretically calculated values given in [2.288], where the near-resonant effect theory was employed.

Most efforts in this field have concentrated on revealing the mechanisms that cause the high nonlinearity in this type of glass, since the linear refractive indices of these glasses are fairly modest. Apparently the empirical correlation described in Sect. 2.6.1 does not explain such a high nonlinearity in this case. Two different nonlinear mechanisms have been proposed, depending upon the microcrystalline particle sizes: a band-filling effect for particles larger than the effective radius of the exciton (a bound electron-hole pair), created in the semiconductor particles upon absorption of incident radiation, and a quantum confinement effect for particles smaller than this effective radius.

Table 2.16. Nonlinearity of $\text{CdSe}_x\text{S}_{1-x}$ glasses

Glasses	Wavelength /nm	Sulphur fraction x	Measured $\chi(3)^a$ / 10^{-9} esu	Calculated n_2^b / 10^{-8} esu	Estimated n_2^c / 10^{-8} esu
Corning #3484	532.0	0.9	13	32	7.4
Corning #2434	580.0	0.7	5	12	3.2
Schott RG 695	694.3	0.1	3	7	2.1

^a Reference [2.287]^b Reference [2.324]^c Reference [2.288]

Band-Filling Effect

For the short-to-medium wavelength cut-on, commercial samples which are available in a ready-struck condition, the usual particle size is in the range of 100 Å. The observed nonlinearity for this size regime of semiconductor particles has been explained by many research groups as due to band-filling [2.289]. Correspondingly observed phenomena are the saturation of absorption and a blue-shifting of the absorption edge with increasing intensity of the incident probe light.

The response times reported for this case vary over a wide range: microseconds to nanoseconds to femtoseconds. An understanding of this wide range of response domains became possible with the observation given in [2.290, 291] of a photodarkening effect in which initially observed slow nonlinear response times (several tens of nanoseconds) changed into a much quicker (picoseconds) region. With this faster response, they also observed the disappearance of a broad luminescence band at higher wavelengths which they assigned to trapped-carrier recombination. In order to explain this faster response due to the photodarkening effect, a three-level system [2.292] was suggested, which includes a conduction and valence band and an intermediate long-lived trapping level. The photodarkening phenomenon changes the carrier-recombination routes by population saturation of this low-lying trapping state. Later, a four-level system with a second, non-radiative trapping level was also proposed. The extremely slow millisecond regime response times have thus been assigned to recombination of this deep-level trapped carrier, and the femtosecond-order nonlinearity is attributed to intraband carrier decay [2.289].

Quantum Confinement Effect

When the microcrystallite becomes smaller, photo-excited carriers are effectively confined in volumes whose dimensions are all below a critical length. The optical behaviour of such systems is particularly affected by this quantum confinement effect. In particular, with decreasing particle size, the separation between individual energy states in the valence and conduction bands as well as the effective band gap of the semiconductor species increases. In order to exhibit distinct quantum confinement effects, the particle size must be smaller than its char-

acteristic bulk exciton radius, which depends upon the microcrystallite species [2.293–299].

Corresponding to the shifting of energy levels described above, quantum confinement effects are manifested by corresponding energy shifts of luminescence and absorption spectra and the appearance of discrete absorption structures attributable to transitions between discrete quantized levels. The response times in the quantum confined particle region are, as in the band-filling regime above, also affected by photodarkening in room-temperature experiments [2.300]. Apparently the discovery of a photodarkening effect in the $\text{CdSe}_x\text{S}_{1-x}$ -doped glasses had been delayed since many of the initial experiments in the quantum confined region were carried out at lower temperatures.

Efforts have thus been made to develop particle-size control methods. These techniques include striking condition control [2.301] and the doping of microcrystallites directly into porous glasses [2.302, 303]. Direct observation of CdSSe microcrystallites by transmission electron microscopy to determine particle-size distributions and densities has indicated that these crystallites are hexagonal columns [2.304] and that they are spherical [2.305, 306]. The size distribution of the crystallites has also been reported [2.307].

Applications for $\text{CdSe}_x\text{S}_{1-x}$ -Particle-Doped Glasses

Fabrication of semiconductor-doped glass planar waveguides began in the mid-1980s. Ion-exchange is an effective method of increasing the refractive index. This technique has been applied to both bulk and thin-film glass samples. *Gabel et al.* presented the first effective generated four-wave mixing by utilizing the ion exchange technique [2.308]. Others have shown that phenomena observed in bulk semiconductor-doped glasses are also seen in waveguides, i.e. saturable absorption, the photodarkening effect and the resolution of thermal and electronic effects [2.309].

Successful fabrication of guided wave fibre structures has also been reported in [2.310]; however, attempts to demonstrate optical switching were unsuccessful because of saturation of the index change in the band-gap energy region. They recently shifted their attention to the off-resonance regime; i.e., for photon energies well below the band gap. This was motivated by their discovery that certain semiconductor-doped glasses display substantial refractive nonlinearity (up to 40 times greater than for silica) even at a wavelength of $1\text{ }\mu\text{m}$. In order to utilize this off-resonance nonlinearity, the optical fibre must be prepared with sufficiently long pathlengths of a controlled microcrystallite size; a difficult process since the temperature regimes for fibre drawing usually lead to changes in the crystal size distribution.

Other Semiconductor Dopants

Semiconductor dopants other than $\text{CdSe}_x\text{S}_{1-x}$ -mixed crystals have been studied in the search for systems of higher nonlinearity. These dopants include pure CdTe and CdSe, Bi_2S_3 , PbS, HgSe, In_2Se_3 and AgI [2.311].

Researchers investigated the optical nonlinearity of pure CdTe quantum dots in glass, because CdTe has the largest Bohr radius of any semiconductor successfully grown as a quantum dot in glass [2.312]. Therefore, it was anticipated that quantum confinement effects would be observed in larger dots, making CdTe dots especially sensitive to electric fields. One observation of exceptional nonlinearity has been reported by another group who doped phosphate glasses with a concentration of CdSe two orders of magnitude higher than that found in conventional filter glasses [2.313]. Their degenerate four-wave mixing experiments showed the $\chi^{(3)}$ of such a glass to be of the order of 10^{-6} esu near maximum resonance.

An all-optical switching material figure of merit for this type of glass has been presented and material figures of merit calculated for several semiconductors. ZnTe, CdTe and GaAs have been identified as dopants that should exhibit higher nonlinear activity [2.314].

The next-generation nonlinear optical material project in Japan disclosed CuCl and CuF-doped glasses as new materials expected to have a 1000-times larger light sensitivity than existing $\text{CdSe}_x\text{S}_{1-x}$ -doped glasses [2.315]. These systems have thus attracted considerable attention from the nonlinear glass research community.

2.6.3 Heterogeneous Glasses – Metallic-Particle-Doped Glasses

The nonlinearity of silver- and gold-particle-dispersed glasses has been observed for several years. Successful optical phase conjugation utilizing such a system was reported in 1985 [2.316]. Representative nonlinear properties of these metallic-particle-doped glasses are presented in Table 2.17. The nonlinearity in these types of glasses is assigned to the local field effect enhanced near the surface plasma resonance. The observed nonlinear responses are in the picosecond region [2.317–319] even though a slower nonlinearity believed to be of thermal

Table 2.17. Nonlinearity of metal-particle-dispersed glasses

Glasses	Measured $\chi^{(3)}$ / 10^{-9} esu ^a	Calculated $\chi^{(3)}$ / 10^{-9} esu ^b
Gold particle Dispersed (530 nm)	15	0.87
Silver particle Dispersed (400 nm)	2.4	0.18

^a Reference [2.316]

^b Reference [2.317]

origin has also been reported for an Mn-doped glass [2.320]. Although the bulk of reported research has been in silver and gold colloid systems, modeling the effective nonlinearity of spheroidal metal particles embedded in a linear host medium suggests that aluminum can be exploited to obtain a larger enhancement of nonlinearity [2.321].

2.6.4 Heterogeneous Glasses – Other Systems

Theoretical calculations of nonlinear behaviour given in [2.322] have shown that model composites of nanospheres with a metallic core and nonlinear shell or with a nonlinear core and metallic shell can exhibit a 10^8 order of magnitude increase in nonlinear activity for photon energies where the optical field can match the surface-mediated plasmon resonance for this system. Such a nonlinear system has been demonstrated in solution, and there is strong motivation to develop techniques for fabricating such systems in a glass.

There has also been research in the area of organically doped glasses. Here, research has focused on identifying suitable glass hosts that can be prepared with high optical quality. The difficulty is that these glasses, must have the low viscosities required for a uniform blending of the organic dopant at temperatures low enough so that the employed organic molecules are not decomposed, vapourized, or experience combustion. One demonstrated example is lead-tin-fluorophosphate-based glasses, which can be melted at or below 500°C and which have glass transition temperatures as low as 95°C [2.323]. Other methods to prepare a suitable glass host at low temperature, in many cases utilizing technology appropriate for the fabrication of integrated optic devices, are under development [2.262–264].

Today the field of nonlinear optics remains a field of research. The applications of this technology are in the early stage of development. Of particular current interest is the possibility of switching a light beam by the application of a second light beam within a very short time period. Such capability in the form of a commercial optoelectronic device is of extreme importance to the telecommunications and computer industry of the future.

Appendix A. Terms of Nonlinear Optics

If a light beam of low intensity is propagated through an isotropic medium such as glass, there is an induced polarization, P , in the material given by

$$P = \chi E , \quad (2.147)$$

where E is the magnitude of the applied electric field and the constant of proportionality, χ , is called the dielectric susceptibility. Note that for small values of E , the induced polarization depends linearly on the magnitude of the applied electric field. If the electric field strength of the applied light is very high, this linear relationship is no longer valid, and this function then has to be described in a series of powers of E :

$$P = \chi^{(1)} E + \chi^{(2)} E^2 + \chi^{(3)} E^3 + \dots , \quad (2.148)$$

where $\chi^{(2)}$ and $\chi^{(3)}$ are the second- and third-order susceptibilities, respectively.

The refractive index can be expressed as

$$n = n_0 + n_2 E^2 , \quad (2.149)$$

where n_0 is the linear refractive index and n_2 the nonlinear refractive index. Both n_2 and E are often expressed in electro-static units (esu). Another equation also used (applying SI units) is

$$n = n_0 + \gamma I , \quad (2.150)$$

where γ is the nonlinear refractive index (in m^2W^{-1}) and I is the intensity of the light beam (in Wm^{-2}).

For the case of linearly polarized, monochromatic light the relationship between n_2 and $\chi^{(3)}$ is given in (2.151):

$$n_2 (\text{esu}) = \frac{12\pi}{n_0} \chi^{(3)} (\text{esu}) \quad (2.151)$$

and the relation between γ and $\chi^{(3)}$ is given in (2.152):

$$\gamma (\text{m}^2\text{W}^{-1}) = \frac{480\pi^2}{n_0^2 c} \chi^{(3)} (\text{esu}) , \quad (2.152)$$

where c is the velocity of light ($3 \times 10^8 \text{ ms}^{-1}$).

If one has the value of $\chi^{(3)}$ in esu, the calculation in SI is done according to (2.153):

$$\chi^{(3)} (\text{m}^2\text{V}^{-2}) = 1.4 \times 10^{-8} \chi^{(3)} (\text{esu}) . \quad (2.153)$$

The conversion between n_2 (esu) and γ (m^2W^{-1}) is made by (2.154):

$$n_2 (\text{esu}) = \frac{cn_0}{40\pi} \gamma (\text{m}^2\text{W}^{-1}) . \quad (2.154)$$

2.7 Resistance to Laser Radiation

Joseph S. Hayden

Optical glass is often utilized in transmissive or reflective applications in conjunction with laser radiation. In many cases, no change in the optical glass employed in these situations is observed. However, in some cases, generally those involving either high-energy photons or high-intensity light beams, changes are observed in the appearance and properties of the glass. Some of these changes are reversible, whereas others are not only permanent but can be serious enough in nature to render the optical element unusable.

We will discuss here the nature of changes that can take place in optical glass as a result of its use with laser radiation. Emphasis will be placed on a number

of particular applications with specific information provided on the resistance of certain glasses to both reversible and irreversible changes in properties upon application with laser sources. Readers interested in a more complete treatment including, for example, discussions relating to crystalline materials, are referred to the literature [2.325–329].

2.7.1 Bulk Damage Resistance to Laser Radiation

Permanent internal damage in optical glass due to laser radiation takes several forms. For a sufficiently strong applied electric field, dielectric breakdown of the optical material can take place. A related form of damage to be considered is thread-like damage due to self-focusing of the laser beam, which is correlated with the nonlinear refractive index of the sample. As described in Sect. 2.6, the refractive index of glass increases if it is under the influence of a strong electric field. The laser beam intensity profile is generally higher in the centre and decreases towards the outside region of the beam profile. The effective refractive index is altered according to the intensity distribution and this causes a lensing effect. By self-focusing, the cross section of the laser beam is confined so strongly that dielectric breakdown is caused within the glass (forming thin dim threads).

There is a critical power, P_2 , for self-focusing given by

$$P_2 = \frac{3.77c\lambda^2}{32\pi^2n_2}, \quad (2.155)$$

where c is the speed of light, λ the laser wavelength and n_2 the nonlinear refractive index (in esu). If the power of the laser beam is considerably less than P_2 (i.e. 25% of P_2), self-focusing does not influence the measurement of the damage threshold [2.330].

Other forms of damage include that due to absorbing inclusions within the glass which, upon illumination, can heat up to a point that they explode within the glass; leaving behind point-like, individual damage sites. The absorbing inclusions can be undissolved particles of the glass batch, parts of the glass melting unit, or precipitated platinum particles. Another thermal effect occurs when the bulk glass is sufficiently absorbing that the irradiated portion of the optical element increases in temperature, ultimately leading to thermal deformation and even melting.

There have been numerous investigations of laser-induced damage over the last 30 years; for instance, see the Annual Reports of the Laser Program at Lawrence Livermore National Laboratory [2.331] and the series of Symposia on Laser Induced Damage in Optical Materials [2.332]. One consequence of this work is that transparent materials for use with laser beams must not have any absorbing particles larger than a few tenths of microns. Particles a few microns in size can lead to damage sites larger than 1 mm (see Fig. 2.58 [2.333]).

The short-pulse laser damage threshold for a wide range of optical glasses has been experimentally evaluated in [2.334] using a 3 ns duration Nd:YAG laser at 1060 nm and in [2.335] with a 25 ns duration ruby laser at 694 nm. Analysis

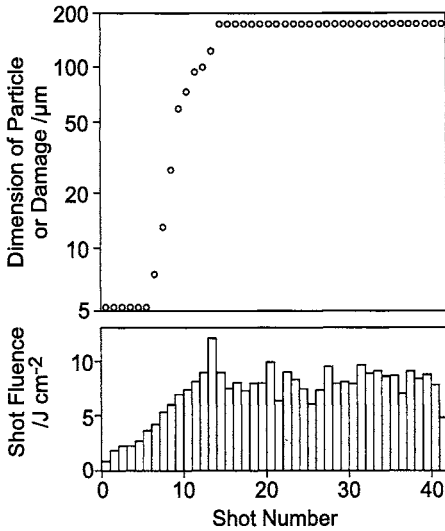


Fig. 2.58. Increasing of the damage dimension of a $4\mu\text{m}$ inclusion in glass by irradiation with laser pulses of increasing energy density [2.333]

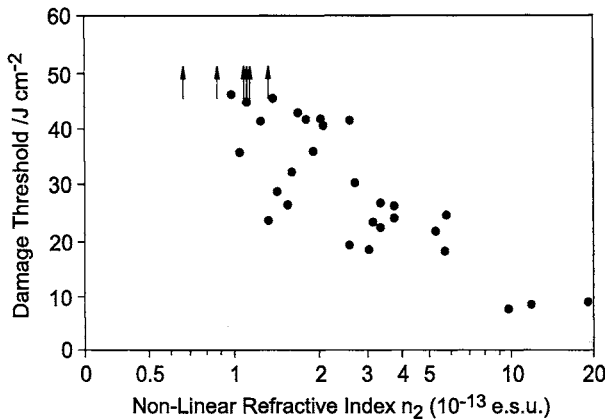


Fig. 2.59. Correlation of the threshold for thread-like damage caused by dielectric breakdown from self-focusing with the nonlinear refractive index [2.334]. Up to energy densities marked by the arrows, no damage could be detected (higher energy densities were not measured)

of these findings verifies the relationship of thread-like damage correlating with the nonlinear refractive indices of the materials tested. Additionally, self-focusing also scales with the thickness of the irradiated optic as given by the well-known B integral [2.329]. A summary of threshold levels for thread-like damage in optical glasses is presented in Fig. 2.59 along with computed values for the nonlinear refractive index (2.146). In critical cases, it is advantageous to apply glass in the form of thin sheets because the damage by self-focusing occurs only for a sufficient accumulated pathlength. The thread-like damage starts first at the exit surface and grows with increasing beam intensity into the sample.

Point-like damage has been observed in nearly all of the twenty-two optical glasses investigated in [2.334], with the damage threshold varying from 7 to more than 47 J/cm^2 for the 3 ns pulse length Nd:YAG laser employed. Optical glasses of the highest optical quality are generally prepared in platinum melting units. Consequently, platinum particles in glass are of special importance in determining the resistance of glass to point-like laser damage. The aggression of platinum by the glass melt depends on the chemical composition of the glass and on the parameters of the melting process. Therefore, it is not possible to quantify a definite threshold energy for point-like damage of an individual glass type. The problem of a damaging particle in glass has been thoroughly modelled [2.336]; and, more recently, also for platinum particles in laser glass [2.337].

In 1986, as a result of manufacturing developments discussed further in Sect. 8.5, phosphate-based laser glasses became available in a form free of all damaging inclusions [2.338]. The bulk damage threshold for these materials is limited only by self-focusing. These glass compositions are already additionally characterized by low values of the nonlinear refractive index. The complete absence of damaging inclusions is normally verified by the deliberate damage testing of the glass with a high-fluence Nd:YAG laser beam [2.339].

2.7.2 Surface Damage Resistance to Laser Irradiation

In practice it is the surface of an optical element that first experiences damage from laser radiation as the test conditions become more severe. Surface preparation plays an important role in the resistance of glass surfaces to short-pulse laser radiation. The lowering of the damage threshold by dust, stains and scratches on a glass surface has been considered in detail by *Bloembergen* [2.340]. So that they do not play a role in the observed damage threshold of a material, such defects must generally be smaller than one-hundredth of the employed laser wavelength.

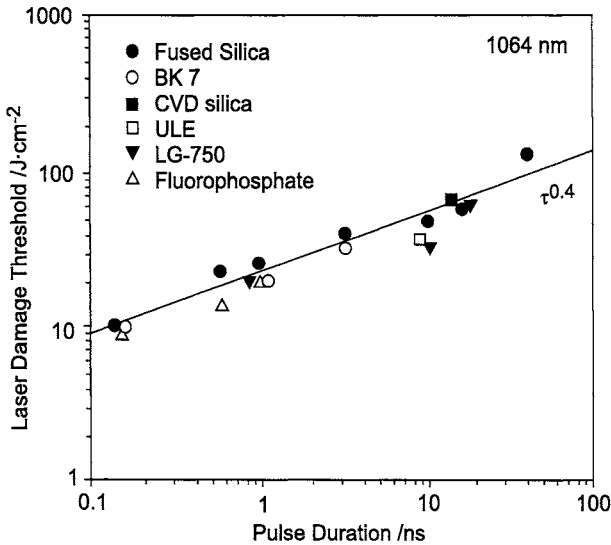
For sufficiently high laser fluences, surface damage becomes clearly visible upon inspection with a differential-interference microscope as circular elevations called pustules with diameters of about 0.01 to 0.1 mm. Other studies [2.334] indicated that the threshold for initial damage, when a Nd:YAG laser of nominal 3 ns pulse length was used, was nearly the same for all optical glasses if the polishing process of the samples was the same. The observed damage threshold depended only on the polishing technique employed and not on the chemical composition of the glass itself. The number of observed pustules increased with increasing energy density in the test beam and their size grew with increasing pulse intensity.

The damage mechanism has been assigned to the following. Residue from the glass polishing material acts as the initial nucleation site for damage, especially when this residue is highly absorbing at the wavelength of interest, leading to localized strain points within the glass surface layer. Table 2.18 presents the findings given in [2.334] as a function of the polishing technique employed. Glass polished with cerium oxide showed a higher damage threshold than that polished with a mixture of ZrO_2 and Fe_2O_3 . The threshold energy density for surface

Table 2.18. Surface damage threshold values in J/cm^2 for BK 7 as a function of various polishing parameters for 3 ns 1060 nm laser pulses [2.334]

Polishing material	Polishing plate		
	Plastic	Pitch	Felt
CeO_2	26	22	21
$0.8 \text{ ZrO}_2 + 0.2 \text{ Fe}_2\text{O}_3$	18	18	20

damage increases with the pulse length of the laser irradiation, τ , as $\tau^{0.4}$; see Fig. 2.60 [2.327]. (With respect to the energy density the threshold is considerably smaller for smaller pulse length than for higher pulse length.) In this figure it should be noted that the damage behaviour is very similar for a wide variety of glass types if they have been prepared with the same high-quality polishing technique [2.331]. The measurements in Fig. 2.60 were done with a pulsed laser operating at a wavelength of 1060 nm; at shorter wavelengths the damage energy thresholds become smaller [2.341].

**Fig. 2.60.** Average threshold of surface damage of bare polished glasses as a function of laser pulse length; laser wavelength 1064 nm [2.327]

It is well established that application of an appropriate coating on the glass surface can reduce the damage threshold value considerably. The development of new coating techniques is still in process. Treatment of coatings by (moderate) laser irradiation increases their damage resistance.

2.7.3 Multiphoton Darkening of Optical Glass

Reversible damage to optical glass as a result of short-pulse-length laser illumination usually takes the form of darkening of the exposed glass. Although it is most commonly observed with high photon energies characteristic of UV light sources, this is often referred to as a solarization. This damage is termed reversible since, in many cases, the discolouration can be at least partially removed by application of a bake cycle at an appropriate elevated temperature. However, in some cases, this discolouration is correlated to defects which are sufficiently stable that the material's initial transparency cannot be recovered.

A discussion of UV-induced solarization and of optical glasses resistant to this and to other forms of high-energy radiation can be found in Sects. 2.3 and 8.7. Here we will restrict our discussion to the laser-induced discolouration in optical glass which is assigned to multiphoton absorption processes.

The first reported solarization attributed to multiphoton absorption in glass was in 1968 [2.342]. Here discolouration was detected in a flint glass exposed to Nd:YAG radiation of 532 nm. Subsequently, others reported measurements of nonlinear absorption and of solarization in glasses exposed to laser light of UV and visible wavelengths at which the glasses had negligible single-photon absorption [2.343, 344]. In general, the magnitude of induced solarization was undetectable for materials with band-gap values more than twice the photon energies of the laser sources employed. However, unlike classic solarization, this form of damage has been found to be both able at room temperature and irreversible with high-temperature treatments [2.345].

White et al. have studied the optical glasses BK 3, BK 7 and BK 10 at 532 nm by a photothermal lensing technique [2.344]. The degree of observed darkening became greater as the band-gap energy for the glasses became less than twice the value of the photon energy available from the employed laser source. The induced absorption was found to increase with accumulated exposure up to a saturation point proportional to the peak intensity of the excitation source. The saturation level of the induced discolouration also correlated with the measured two-photon absorption coefficient of each glass. The reported two-photon absorption values at 532 nm were 0.6, 2.9 and 0.4 cm^{-1} per TW/cm^2 for BK 3, BK 7 and BK 10, respectively. These levels are two orders of magnitude smaller than those for crystalline materials with similar band-gap energies [2.346]. The level of induced single-photon absorption for BK 7 at 532 nm was reported as 0.07 cm^{-1} per GW/cm^2 .

2.8 Environmentally Friendly Optical Glass

Uwe Kolberg, Silke Wolff

"The Schott Group is committed to being a responsible partner of society and the environment". This quotation, which is one of the guidelines issued in "We

at Schott" [2.347], is a summary of the company's social and environmental commitment arising from historical tradition.

Unlike many of its competitors, Schott has never used a number of environmentally hazardous substances, such as HgO, Ti_2O_3 , BeO and BeF_2 . In 1980, the use of ThO_2 was stopped completely and the use of CdO was restricted to coloured optical glass. In the late 1980s, when the public became increasingly sensitive to health problems caused by Pb, we started developing PbO-free glasses as a first step towards the production of environmentally friendly glass. Apart from the health aspect, the reduction and replacement of PbO in the glass composition led to a considerable weight reduction, which is advantageous in several respects for outdoor applications. Since 1995 the use of another substance, As_2O_3 , has been restricted; it is used as a refining agent in trace amounts only. With the intention of replacing PbO and As_2O_3 , we initiated a comprehensive programme that will be completed in the near future. The main target has been to redesign all glass types so as to preserve n_d and ν_d while gaining a Pb-free and As-free composition. Moreover, the partial-dispersion values $P_{g,F}$ and $\Delta P_{g,F}$ have been preserved for several special glasses. There are, however, some special glasses whose optical properties cannot be preserved without the addition of PbO. Of course, these glass types will be available in the future in their known quality and composition. They will be clearly marked as Pb-containing products. The bio-availability of the pure toxic substances in the products that are finally sold is negligibly low. For example, even high amounts of PbO in the composition of a glass would not cause any health problems. But although there is no objective reason, the customers' negative associations regarding Pb or As lead to a strong rejection of products containing these substances.

There are, however, serious problems caused by hazardous compounds that make expensive and time-consuming development necessary. High amounts of fine dust are produced during acquisition of the raw material and during batch preparation. Of course, the dust does not contain a higher percentage of toxic substances than the base material, but it has a very much larger specific surface. This large surface in combination with the fact that dust tends to deposit in the lung tissue in large amounts causes a bio-availability of the toxic compounds that is in the latent-toxic range. High yields of grinding waste result from finishing processes, particularly from grinding and polishing. These must be disposed of at high costs on hazardous waste disposal sites. Experience in this field shows that there is a high risk of wrong handling and a tendency towards illegal disposal because of the high costs. Improving technical measures to protect directly affected persons from the unhealthy effects of dust and waste is theoretically possible, but hard to control. In particular in foreign production, for example in developing countries, safety could not be guaranteed. Schott therefore decided to dispense with these substances and normally uses TiO_2 , Nb_2O_5 , ZrO_2 , WO_3 and others instead of PbO. As_2O_3 as refining agent is replaced by Sb_2O_3 . All glass compositions reformulated in this way are designated by an "N" prefix to the original name. New data sheets for the environmentally

friendly optical glasses will be published soon. For 50% of all glass compositions originally containing As_2O_3 but no PbO , replacing As_2O_3 by Sb_2O_3 does not change the properties. Examples of such glasses are:

BK 7	→	N-BK 7,
SK 16	→	N-SK 16,
LaK 8	→	N-LaK 8,
LaF N28	→	N-LaF 28.

For the compositions originally containing As_2O_3 and PbO (up to 80%), some physical and optical properties are significantly changed upon the replacement of both elements. Besides improvements, these changes also include some disadvantages. Improvements are: better chemical resistance; higher Knoop hardness; higher T_g point; lower density. Disadvantages are: higher tendency of crystallization; reduced transmission in the UV region; changed values for $P_{g,F}$ and $\Delta P_{g,F}$ (except for N-KzFS series).

Examples of such glasses are:

LF 5	→	N-LF 5,
SF 10	→	N-SF 10,
BaF 4	→	N-BaF 4,
KzFS N4	→	N-KzFS 4.

Table 2.19 compares some of the physical and optical properties of SF 10 and KzFS N4 with the corresponding new types N-SF 10 and N-KzFS 4. These data do not show general trends for all reformulated glasses, but they represent two interesting groups. (N-)SF 10 is chosen as an example for the large group of heavy-flint glasses containing high amounts of PbO . Upon PbO replacement, the optical properties tend to change considerably, so preserving them requires great developmental effort. Additionally, other properties, such as internal transmission or density, are unavoidably changed. (N-)KzFS 4 is chosen as an example for Schott's short-flint special glasses, which are known for their negative $\Delta P_{g,F}$. This property has been successfully maintained, but a considerable increase in the batch price has to be accepted.

One parameter that is changed in all redeveloped glasses is the physical density. The higher the former PbO content, the stronger is the decrease in density for the new glasses. Thus the density of crown and dense-crown glasses is only slightly decreased, whereas the density of dense-flint glasses is decreased by a factor of down to 0.7 (see Fig. 2.61). The values for V_A , E_w and T_g commonly increase, but in very different ranges, from some degrees up to 250°C . Glass types with large increases in these parameters, such as N-SF 10 (see Table 2.19), formerly often contained much PbO , which is now directly replaced by compounds with higher melting points and related viscosity properties. For glasses with low amounts of PbO in the K, KF and LLF regions, a complete rearrangement of the glass composition is necessary. This rearrangement inevitably leads to an

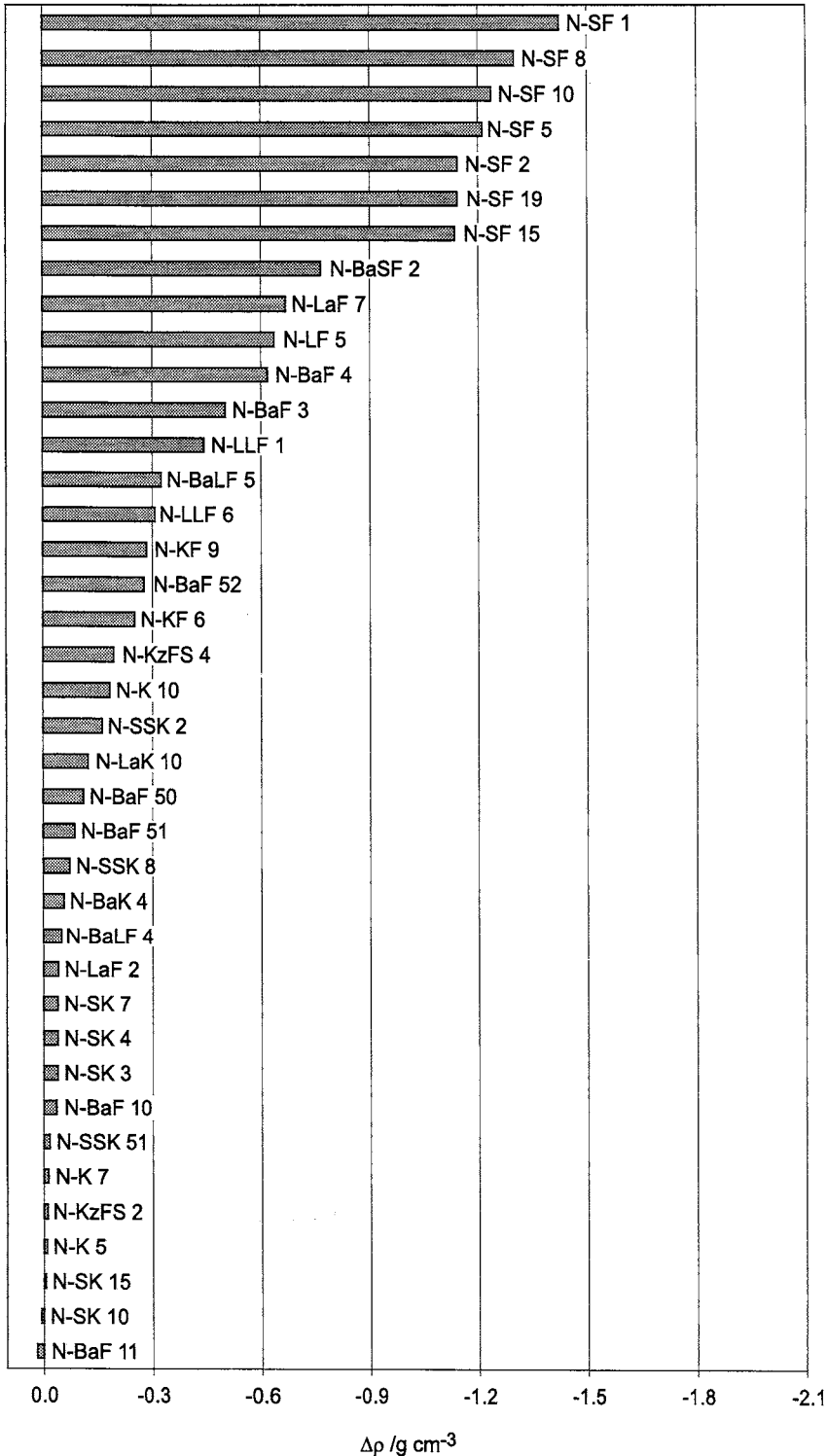


Fig. 2.61. Deviation of the physical density (g cm^{-3}) of environmentally friendly optical glasses in comparison with former glass types

Table 2.19. SF 10 and KzFS N4 compared with the new types N-SF 10 and N-KzFS 4

	SF 10	N-SF 10	KzFS N4	N-KzFS 4
n_d	1.72825	1.72828	1.61340	1.61336
ν_d	28.41	28.53	44.29	44.49
$P_{g,F}$	0.6046	0.6066	0.5607	0.4958
$\Delta P_{g,F}$	+0.0085	+0.0108	-0.0087	-0.0100
$\alpha_{20-300}/10^{-6} \text{ K}^{-1}$	8.40	10.8	5.50	8.20
$\rho/\text{g cm}^{-3}$	4.28	3.05	3.20	3.00
$E_w/^\circ\text{C}$	595	652	594	675
$T_g/^\circ\text{C}$	453	559	492	547
SR	1.0	1.0	52.3	3.4
AR	1.2	1.0	4.3	1.2
FR	0	0	2.0	1.0
CR	1.0	1.0	3.0	1.0
τ_i (400 nm/25 mm)	0.700	0.640	0.974	0.950

increase in SiO_2 , which is a component with a high melting point, and finally results in strongly increased V_A , E_w and T_g values for these glasses.

The acid and alkali resistances are commonly improved because PbO is replaced by compounds with higher chemical resistance. An example for a special change is N-KzFS 4, whose acid resistance rose two classes from a poor chemical resistance up to a quite good one. The alkali resistance and climate resistance similarly improved. In contrast, the internal transmission, $\tau_i(400 \text{ nm}, 25 \text{ mm})$, is commonly reduced. Depending on the former PbO and As_2O_3 content, τ_i decreases by a few per cent for compositions with low amounts, and up to several ten per cent for compositions with high amounts, for example the heavy-flint glasses. Three effects are mainly responsible for this behaviour. First, PbO is replaced by TiO_2 , whose maximum absorption band is shifted from the far-UV (PbO) towards the near-UV range. The second effect is the insufficient reduction of oxidized crucible material, here PtO_2 to metallic Pt , if As_2O_3 is replaced by Sb_2O_3 as the refining agent. The reductive potential of Sb_2O_3 is much lower, so the transmission-decreasing PtO_2 remains in the glasses. The third effect is the equilibrium of the $\text{Fe}^{2+}/\text{Fe}^{3+}$ system. As_2O_3 induces an excess of the reduced form, whereas the use of Sb_2O_3 decreases this excess. Therefore the oxidized form with a higher absorption coefficient in the UV region gains stronger influence.

Although maintaining the n_d and ν_d values is the primary goal of development for glasses, where only minimal deviations from the original values are tolerable, the $P_{g,F}$ and $\Delta P_{g,F}$ values are changed for nearly all of the reformulated glasses. As a rule, $\Delta P_{g,F}$ is increased. This is not tolerable for the N-KzFS glass types, because they are used for chromatic correction, where a high negative value for $\Delta P_{g,F}$ is mandatory. For these glasses, the use of TiO_2 , which is the standard source for increasing the refractive index, must be avoided because it increases $P_{g,F}$ and, in consequence, $\Delta P_{g,F}$. High refractive indices in connection with high negative $\Delta P_{g,F}$ are only obtainable by a specially chosen combination of rare high-refracting oxides, which are expensive and accordingly

raise the glass price. As can be seen from Table 2.19, the goal of maintaining the high negative $\Delta P_{g,F}$ of the old glass composition was not only achieved but exceeded. Unfortunately, this combination of oxides cannot be used for producing KzFS glasses with much higher refractive indices because of crystallization problems. Thus, an attempt to redesign KzFS N5 failed and a completely new concept must be established.

Though the development of PbO-free and As_2O_3 -free glasses is not yet finished, one can see that the position of the new environmentally friendly glasses in the Abbe diagram is equal to those of the former compositions, and most of the other optical properties have been maintained.

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